

INTERNATIONAL STANDARD

**Test methods for electrical materials, printed boards and other interconnection structures and assemblies –
Part 3: Test methods for interconnection structures (printed boards)**



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Part 3: Test methods for interconnection structures (printed boards)**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

TEST METHODS FOR ELECTRICAL MATERIALS, PRINTED BOARDS AND OTHER INTERCONNECTION STRUCTURES AND ASSEMBLIES –

Part 3: Test methods for interconnection structures (printed boards)

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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International Standard IEC 61189-3 has been prepared by IEC technical committee 91: Electronics assembly technology.

This second edition cancels and replaces the first edition, published in 1997, its amendment 1 (1999) and constitutes a technical revision.

The document 91/698/FDIS, circulated to the National Committees as Amendment 2, led to the publication of the new edition.

The major technical changes with regard to the previous edition concern the addition of 25 new tests, as follows:

- 6 V: Visual test methods: 3V01, 3V02 and 3V03;
- 7 D: Dimensional test methods: 3D03;

- 8 C: Chemical test methods: 3C02, 3C13 and 3C14;
- 9 M: Mechanical test methods: 3M01, 3M03, 3M04, 3M07 and 3M09;
- 10 E: Electrical test methods: 3E03, 3E04, 3E05, 3E11, 3E12, 3E13, 3E16, 3E17 and 3E18;
- 11 N: Environmental test methods: 3N03, 3N07 and 3N12;
- 12 X: Miscellaneous test methods: 3X01

This edition also includes the deletion of Annex B: Conversion table, as the referred documents were disbanded in 2005 and do not officially exist. Should any one wish to consult such information, they should refer to the first edition of IEC 61189-3 (1997).

The text of this standard is based on the first edition, its Amendment 1 and the following documents:

FDIS	Report on voting
91/698/FDIS	91/727/RVD

Full information on the voting for the approval of this amendment can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 61189 series, under the general title *Test methods for electrical materials, printed boards and other interconnection structures and assemblies*, can be found on the IEC website.

NOTE Future standards in this series will carry the new general title as cited above. Titles of existing standards in this series will be updated at the time of the next edition.

This standard should be used in conjunction with the following parts:

- Part 1: General test methods and methodology
- Part 2: Test methods for materials for interconnection structures
- Part 3: Test methods for electronic components assembling characteristics
- Part 5: Test methods printed board assemblies and also the following standard
- Part 6: Test methods for materials used in manufacturing electronic assemblies

IEC 60068 (all parts), Environmental testing

The committee has decided that the contents of the base publication and its amendments will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

IEC 61189 relates to test methods for printed boards and printed board assemblies, as well as related materials or component robustness, irrespective of their method of manufacture.

The standard is divided into separate parts, covering information for the designer and the test methodology engineer or technician. Each part has a specific focus; methods are grouped according to their application and numbered sequentially as they are developed and released.

In some instances test methods developed by other TCs (e.g. TC 50) have been reproduced from existing IEC standards in order to provide the reader with a comprehensive set of test methods. When this situation occurs, it will be noted on the specific test method; if the test method is reproduced with minor revision, those paragraphs that are different are identified.

This part of IEC 61189 contains test methods for evaluating printed boards and other forms of interconnection structures. The methods are self-contained, with sufficient detail and description so as to achieve uniformity and reproducibility in the procedures and test methodologies.

The tests shown in this standard are grouped according to the following principles:

- P: preparation/conditioning methods
- V: visual test methods
- D: dimensional test methods
- C: chemical test methods
- M: mechanical test methods
- E: electrical test methods
- N: environmental test methods
- X: miscellaneous test methods

To facilitate reference to the tests, to retain consistency of presentation, and to provide for future expansion, each test is identified by a number (assigned sequentially) added to the prefix (group code) letter showing the group to which the test method belongs.

The test method numbers have no significance with respect to an eventual test sequence; that responsibility rests with the relevant specification that calls for the method being performed. The relevant specification, in most instances, also describes pass/fail criteria.

The letter and number combinations are for reference purposes, to be used by the relevant specification. Thus "3D02" represents the second dimensional test method described in this publication.

In short, for this example, 3 is the part of IEC standard (61189-3), D is the group of methods, and 02 is the test number.

A list of all test methods included in this standard, as well as those under consideration is given in Annex B. This annex will be reissued whenever new tests are introduced.

TEST METHODS FOR ELECTRICAL MATERIALS, PRINTED BOARDS AND OTHER INTERCONNECTION STRUCTURES AND ASSEMBLIES –

Part 3: Test methods for interconnection structures (printed boards)

1 Scope

This part of IEC 61189 is a catalogue of test methods representing methodologies and procedures that can be applied to test materials used for manufacturing interconnection structures (printed boards) and assemblies.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60051 (all parts), *Direct acting indicating analogue electrical measuring instruments and their accessories*

IEC 60068-1: 1988, *Environmental testing – Part 1: General and guidance*

IEC 60068-2-20: 1979, *Environmental testing – Part 2: Tests – Test T: Soldering*
Amendment 2 (1987)

IEC 60068-2-78, *Environmental testing – Part 2-78: Tests – Test Cab: Damp heat, steady state*

IEC 60169–15, *Radio-frequency connectors – Part 15: RF coaxial connectors with inner diameter of outer conductor 4,13 mm (0,163 in) with screw coupling – Characteristic impedance 50 ohms (Type SMA)*

IEC 60454-1:1992, *Specifications for pressure-sensitive adhesive tapes for electrical purposes – Part 1: General requirements*

IEC 60454-3-1:1998, *Pressure-sensitive adhesive tapes for electrical purposes – Part 3: Specifications for individual materials – Sheet 1: PVC film tapes with pressure-sensitive adhesive*

IEC 60584-1, *Thermocouples – Part 1: reference tables*

IEC 60695-11-5, *Fire hazard testing – Part 11-5: Test flames – Needle flame test method – Apparatus, confirmatory test arrangement and guidance*

IEC 61188-1-2:1998, *Printed boards and printed board assemblies – Design and use – Part 1-2: Generic requirements – Controlled impedance*

IEC 61189-1:1997, *Test methods for electrical materials, interconnection structures and assemblies – Part 1: General test methods and methodology*

IEC 61190-1-1, *Attachment materials for electronic assembly – Part 1-1: Requirements for soldering fluxes for high quality interconnections in electronics assembly*

IEC 61190-1-2, *Attachment materials for electronic assembly – Part 1-2: Requirements for*

solder pastes for high quality interconnections in electronic assembly

IEC 62326-4:1996, *Printed boards – Part 4: Rigid multilayer printed boards with interlayer connections – Sectional specification*

IEC 62326-4-1:1996, *Printed boards – Part 4: Rigid multilayer printed boards with interlayer connections – Sectional specification – Section 1: Capability Detail Specification – Performance levels A, B and C*

ISO 4046:1978, *Paper, board, pulp and related terms – Vocabulary (withdrawn)*¹

ISO 9002:1994, *Quality systems – Model for quality assurance in production, installation and servicing (withdrawn)*

ISO 9453:2006, *Soft solder alloys – Chemical compositions and forms*

3 Accuracy, precision and resolution

Errors and uncertainties are inherent in all measurement processes. The information given below enables valid estimates of the amount of error and uncertainty to be taken into account.

Test data serve a number of purposes which include:

- to monitor a process;
- to enhance confidence in quality conformance;
- to arbitrate between customer and supplier.

In any of these circumstances, it is essential that confidence can be placed upon the test data in terms of:

- accuracy: calibration of the test instruments and/or system;
- precision: the repeatability and uncertainty of the measurement;
- resolution: the suitability of the instruments and/or system for the test.

3.1 Accuracy

The regime by which routine calibration of the test equipment is undertaken shall be clearly stated in the quality documentation of the supplier or agency conducting the test, and shall meet the requirements of 4.11 of ISO 9002. The calibration shall be conducted by an agency having accreditation to a national or international measurement standard institute. There should be an uninterrupted chain of calibration to a national or international standard.

Where calibration to a national or international standard is not possible, "round robin" techniques may be used, and documented, to enhance confidence in measurement accuracy.

The calibration interval shall normally be one year. Equipment consistently found to be outside acceptable limits of accuracy shall be subject to shortened calibration intervals. Equipment consistently found to be well within acceptable limits may be subject to relaxed calibration intervals.

A record of the calibration and maintenance history shall be maintained for each instrument. These records should state the uncertainty of the calibration technique (in $\pm$ % deviation) in order that uncertainties of measurement can be aggregated and determined.

¹ ISO 4046 has been withdrawn and replaced by ISO 4046: Parts 1 to 5.

A procedure shall be implemented to resolve any situation where an instrument is found to be outside calibration limits.

3.2 Precision

The uncertainty budget of any measurement technique is made up of both systematic and random uncertainties. All estimates shall be based upon a single confidence level, the minimum being 95 %.

Systematic uncertainties are usually the predominant contributor, and will include all uncertainties not subject to random fluctuation. These include:

- calibration uncertainties;
- errors due to the use of an instrument under conditions which differ from those under which it was calibrated;
- errors in the graduation of a scale of an analogue meter (scale shape error).

Random uncertainties result from numerous sources but can be deduced from repeated measurement of a standard item. Therefore, it is not necessary to isolate the individual contributions. These may include:

- random fluctuations such as those due to the variation of an influence parameter. Typically, changes in atmospheric conditions reduce the repeatability of a measurement;
- uncertainty in discrimination, such as setting a pointer to a fiducial mark, or interpolating between graduations on an analogue scale.

Aggregation of uncertainties: Geometric addition (root-sum-square) of uncertainties may be used in most cases. Interpolation error is normally added separately and may be accepted as being 20 % of the difference between the finest graduations of the scale of the instrument.

$$U_t = \pm \sqrt{(U_s^2 + U_r^2)} + U_i$$

where

U_t is the total uncertainty

U_s is the systematic uncertainty

U_r is the random uncertainty

U_i is the interpolation error

Determination of random uncertainties: Random uncertainty can be determined by repeated measurement of a parameter and subsequent statistical manipulation of the measured data. The technique assumes that the data exhibits a normal (Gaussian) distribution.

$$U_r = t \sigma / \sqrt{n}$$

where

U_r is random uncertainty

n is the sample size

t is the percentage point of the "t" distribution (from 3.5, statistic tables)

F is the standard deviation (F_{n-1})

3.3 Resolution

It is paramount that the test equipment used is capable of sufficient resolution. Measurement

systems used should be capable of resolving 10 % (or better) of the test limit tolerance.

It is accepted that some technologies will place a physical limitation upon resolution (e.g. optical resolution)

3.4 Report

In addition to requirements detailed in the test specification, the report shall detail:

- the test method used;
- the identity of the sample(s);
- the test instrumentation;
- the specified limit(s);
- an estimate of measurement uncertainty, and resultant working limit(s) for the test;
- the detailed test results;
- the test date and operators' signature.

3.5 Student's "t" distribution

Table 1 gives values of the factor "t" for 95 % and 99 % confidence levels, as a function of the number of measurements. It is sufficient to use 95 % limits, as in the case of the worked examples shown in Annex A.

Table 1 – Student's "t" distribution

Sample size	t value 95 %	t value 99 %		Sample size	t value 95 %	t value 99 %
2	12,7	63,7		14	2,16	3,01
3	4,3	9,92		15	2,14	2,98
4	3,18	5,84		16	2,13	2,95
5	2,78	4,6		17	2,12	2,92
6	2,57	4,03		18	2,11	2,9
7	2,45	3,71		19	2,1	2,88
8	2,36	3,5		20	2,09	2,86
9	2,31	3,36		21	2,08	2,83
10	2,26	3,25		22	2,075	2,82
11	2,23	3,17		23	2,07	2,81
12	2,2	3,11		24	2,065	2,8
13	2,18	3,05		25	2,06	2,79

3.6 Suggested uncertainty limits

The following target uncertainties are suggested:

- a) Voltage < 1 kV: $\pm 1,5 \%$
- b) Voltage > 1 kV: $\pm 2,5 \%$
- c) Current < 20 A: $\pm 1,5 \%$
- d) Current > 20 A: $\pm 2,5 \%$

Resistance

- e) Earth and continuity: $\pm 10 \%$
- f) Insulation: $\pm 10 \%$
- g) Frequency: $\pm 0,2 \%$

Time

- h) Interval < 60 s: $\pm 1 \text{ s}$
- i) Interval > 60 s: $\pm 2 \%$
- j) Mass < 10 g: $\pm 0,5 \%$
- k) Mass 10 g to 100 g: $\pm 1 \%$
- l) Mass > 100 g: $\pm 2 \%$
- m) Force: $\pm 2 \%$
- n) Dimension < 25 mm: $\pm 0,5 \%$
- o) Dimension > 25 mm: $\pm 0,1 \text{ mm}$
- p) Temperature < 100 °C: $\pm 1,5 \%$
- q) Temperature > 100 °C: $\pm 3,5 \%$
- r) Humidity 30 to 75 % RH: $\pm 5 \%$ RH

Plating thicknesses

- s) Backscatter method: $\pm 10 \%$
- t) Microsection: $\pm 2 \text{ } \mu\text{m}$
- u) Ionic contamination: $\pm 10 \%$

4 Catalogue of approved test methods

This standard provides specific test methods in complete detail to permit implementation with minimal cross-referencing to other specific procedures. The use of generic conditioning exposures is accomplished in the methods by reference, for example IEC 61189-1 and IEC 60068 and, when applicable, is a mandatory part of the test method standard.

Each method has its own title, number and revision status to accommodate updating and improving the methods as industry requirements change or demand new methodology. The methods are organized in test method groups and individual tests.

5 P: Preparation/conditioning test methods

6 V: Visual test methods

6.1 Test 3V01: Visual examination, 3× magnification

6.1.1 Object

This method describes the procedure for visual examination of materials and finished printed boards, where 3× magnification is required by the relevant sectional specification (SS) or customer detail specification (CDS).

6.1.2 Test specimen

Finished printed board(s), portion of finished printed board, or test coupon(s), as specified by the relevant sectional specification (SS) or customer detail specification (CDS).

6.1.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Optical device capable of providing 3× magnification.
- b) Referee optical device capable of providing 10× magnification.

6.1.4 Procedure

The following steps shall be taken:

- a) The product shall be carefully examined, per the specified requirement, under 3× magnification device.
- b) In the case of a questionable evaluation at 3×, a referee examination may be performed at 10×.

6.1.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of evaluation;
- c) identification and description of specimen;
- d) the magnification used for the examination;
- e) the referee magnification examination, if applicable;
- f) the results of evaluation, including requirement(s) failed, failure modes and degree of failure, in case of failure;
- g) any deviation from this test method;
- h) the name of the person that conducted the test.

6.1.6 Additional information

In addition to stated requirements, any other objective evidence of defective and/or substandard conditions should be noted, such as dirt, oil, corrosion, fingerprints, foreign matter, etc.

6.2 Test 3V02: Visual examination, 10× magnification

6.2.1 Object

This method describes the procedure for visual examination of materials and finished printed boards where 10× magnification is required by the relevant sectional specification (SS) or customer detail specification (CDS).

6.2.2 Test specimen

Finished printed board(s) or test coupon(s), as specified in the relevant specification.

6.2.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Optical device capable of providing 10× magnification.
- b) Referee optical device capable of providing 50× magnification.

6.2.4 Procedure

The following steps shall be taken:

- a) The product shall be carefully examined, per the specified requirement, under 10× magnification device.
- b) In the case of a questionable evaluation at 10×, a referee examination may be performed at 50×.

6.2.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of evaluation;
- c) identification and description of specimen;
- d) the magnification used for the examination;
- e) the referee magnification examination, if applicable;
- f) results of evaluation, including requirement(s) failed, failure modes and degree of failure, in case of failure;
- g) any deviation from this test method;
- h) the name of the person that conducted the test.

6.2.6 Additional information

In addition to stated requirements, any other objective evidence of defective and/or sub-standard conditions should be noted, such as dirt, oil, corrosion, fingerprints, foreign matter, etc.

6.3 Test 3V03: Visual examination, 250× magnification

6.3.1 Object

This method describes the procedure for visual examination of materials and finished printed boards, where 250× magnification is required by the relevant sectional specification (SS) or customer detail specification (CDS).

6.3.2 Test specimen

Finished printed board(s), portion of finished printed board, or test coupon(s), as specified in the relevant specification.

6.3.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Optical device capable of providing 250× magnification.
- b) Referee optical device capable of providing at least double the power magnification.

NOTE Caution is advised not to falsely identify problems at this magnification.

6.3.4 Procedure

The following steps shall be taken:

- a) The product shall be carefully examined, per the specified requirement, magnification device under 250×
- b) In the case of a questionable evaluation at 250×, a referee examination may be performed at any magnification of at least double the power.

6.3.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of evaluation;
- c) identification and description of specimen;
- d) the magnification used for the examination;
- e) the referee magnification examination, if applicable;
- f) the results of evaluation, including requirement(s) failed, failure modes and degree of failure, in case of failure;
- g) any deviation from this test method;
- i) the name of the person that conducted the test.

6.3.6 Additional information

In addition to stated requirements, any other objective evidence of defective and/or substandard conditions should be noted, such as dirt, oil, corrosion, inclusions, foreign matter, etc.

6.4 Test 3V04: General visual (under consideration)

7 D: Dimensional test methods

7.1 Test 3D01: Optical method (under consideration)

7.2 Test 3D02: Conductor width and spacing

7.2.1 Object

The purpose of this test method is to provide a procedure for determining the conductor width and spacing of a printed board.

7.2.2 Test specimen

The specimen shall be a suitable printed board having conductor patterns for test.

Where the use of test coupons, as specified in IEC 62326-4-1, is agreed between the user and the supplier, the measurement shall be carried out on specimen F.

7.2.3 Test apparatus and materials

The following test apparatus and materials shall be used:

An illuminated eyepiece or microscope or projector having an ocular micrometer with a resolution of 0,01 mm or better shall be used.

7.2.4 Procedure

The following steps shall be taken:

The conductor width and spacing between conductors shall be measured at random points, including central and corner areas, according to the relevant specification, and viewed vertically from above. The measured value shall be recorded to the nearest 0,01 mm. Edge defects such as indentations, projections, and slivers shall be excluded from measurement.

7.2.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of the test;
- c) the identification and description of the specimen;
- d) the conductor width and spacing measured;
- e) the layer number;
- f) the number of measurements;
- g) the maximum and minimum observed conductor widths and spaces;
- h) the average conductor width and spaces;
- i) any deviation from this test method;
- j) the name of the person conducting the test.

7.2.6 Additional information

None.

7.3 Test 3D03: Automated optical inspection (AOI)

7.3.1 Object

AOI is an integrated system for the automated visual inspection and verification of printed boards, printed board inner layers, and printed board artworks. Software and hardware combinations are available which enable accurate inspection at a high throughput.

AOI is undertaken to ensure that a printed board complies with the customer's requirements. AOI is capable of detecting many defect types (as listed in 7.3.6), and may be used as a screening process (100 %), or in accordance with a sampling plan.

The benchmark for AOI verification may be physical ("golden board" or artwork) or theoretical (design rules or CAD data).

The principal of operation of AOI equipment is the interpretation/comparison of data from the item under test with the benchmark. The equipment either operated on a "reflected light" principle, or uses the fluorescence of substrates subjected to laser light.

7.3.2 Test specimen

The test specimens shall comprise of production artwork, printed boards or inner layers.

7.3.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Automated equipment capable of viewing an area equal to 500 mm × 700 mm using raster scanning techniques. The resolution of the machine shall be 4× greater than the feature size being determined.

- b) Light sensors and recording mechanisms shall be capable of determining feature ambiguities as programmed into the equipment to facilitate identification of non-conforming feature sizes of printed boards, artwork, etc.

7.3.4 Procedure

The test procedure differs for new and repeat jobs:

7.3.4.1 Procedure for new jobs

The AOI equipment requires undertaking a learning stage followed by an inspection stage.

7.3.4.1.1 Learning stage

During the learning stage, the AOI equipment acquires the parameters required for the inspection stage. These typically include:

- inspection areas
- non-inspection areas
- minimum conductor width
- minimum conductor spacing

The AOI uses the data to create a data file for reference during the inspection stage. Data may be collected from a "golden board" or from CAD data.

A database should be maintained in order to expedite repeat jobs/orders.

7.3.4.1.2 Inspection stage

During the inspection stage, the AOI equipment uses the learned data to perform the inspection of the test item. Detected defects may be marked, or logged as (X, Y) coordinates.

Detected defects should then be verified by the operator.

7.3.4.2 Repeat jobs

Learned data from previous jobs shall be loaded from the database, and the procedure detailed in 7.3.4.1.1 shall then be followed.

7.3.5 Report

The report should include:

- a) test number and revision
- b) date of test
- c) description and identification of the test items
- d) details of verified defects
- e) a statistical analysis of the defect data (optional);
- f) identification of AOI apparatus if AOI is subcontracted;
- g) the name of the person that conducted the test.

7.3.6 Additional information

AOI apparatus should be capable of detecting:

- a) design rule defects
 - open circuit
 - conductor width violation
 - conductor spacing violation
 - copper splash
 - pin holes
- b) reference defects
 - short circuits
 - missing conductors
 - missing spacing
 - pad violation (incorrect number of conductors entering a pad)
 - errors in clearances in power and ground planes
 - multiple line width defects (MLW)
 - short circuit between split planes
- c) outer layer defects
 - annular ring defects
 - break out
 - missing holes
 - conductor dish down
- d) general
 - dimension error measurements

7.4 Test 3D04: Dimensional examination, general (under consideration)

8 C: Chemical test methods

8.1 Test 3C01: Flammability, rigid printed board metal removal (under consideration)

8.2 Test method 3C02: Flammability; glow wire test, rigid printed boards

8.2.1 Object

The object of the test is to determine the effect upon a printed board of exposure to a glowing wire under specific conditions.

The intensity of the ignition source is similar to that of an accidentally overheating or glowing single electronic component.

Timings measured by this test are an indication of the ability of the printed board to self-extinguish. Correlation with other properties of the material(s), such as Oxygen Index, is not possible.

Materials suitable for testing in accordance with this technique include rigid printed boards.

8.2.2 Test specimen

The test specimen shall be production board, or a test board that is representative of the production board in terms of:

- a) Base material(s)

- b) Surface coating(s)
- c) Type (e.g. multilayer, single-sided etc.)
- d) Size ²
- e) Design
- f) Surface area
- g) Thickness
- h) Metal distribution

A minimum of five specimens shall be used.

8.2.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) The glow-wire shall consist of a loop of nickel/chromium (80 % / 20 %) wire having a diameter of 4 mm (see Figure 1). Care must be taken when forming the loop to avoid fine cracking at the tip.
- b) A sheathed fine-wire thermocouple, having an overall diameter of 0,5 mm and wires of NiCr and NiAl with the welded point located inside the sheath, is used for measuring the temperature of the glow-wire.
- c) The sheath consists of a metal which must be resistant to a temperature of at least 960 °C. The thermocouple is arranged in a pocket hole, 0,6 mm in diameter, drilled in the tip of the glow wire, as shown in “detail Z” of Figure 1. The thermo-voltages shall comply with IEC 60584-1; the characteristics given in that publication are practically linear. The cold connection shall be kept in melting ice unless a reliable reference temperature is obtained by other means.
- d) The instrument for measuring the thermo-voltage should be accurate to 1 % (i.e. Class 0,5 according to IEC 60051).
- e) The glow-wire is electrically heated. The current necessary for heating the specified tip to a temperature of 960 °C is between 120 A and 150 A.
- f) The test apparatus shall be so designed that the glow-wire is kept in a horizontal plane and that it applies a force of between 0,8N and 1,2N to the specimen. The force shall be maintained at this value whilst the glow-wire or test specimen is moved horizontally in relation to each other over a distance of at least 7mm.
- g) A wooden board shall be placed underneath the specimen. The board shall be covered with tissue paper which complies with 6.86 of ISO 4046.
- h) An example of the test apparatus is shown in Figure 2.

8.2.4 Procedure

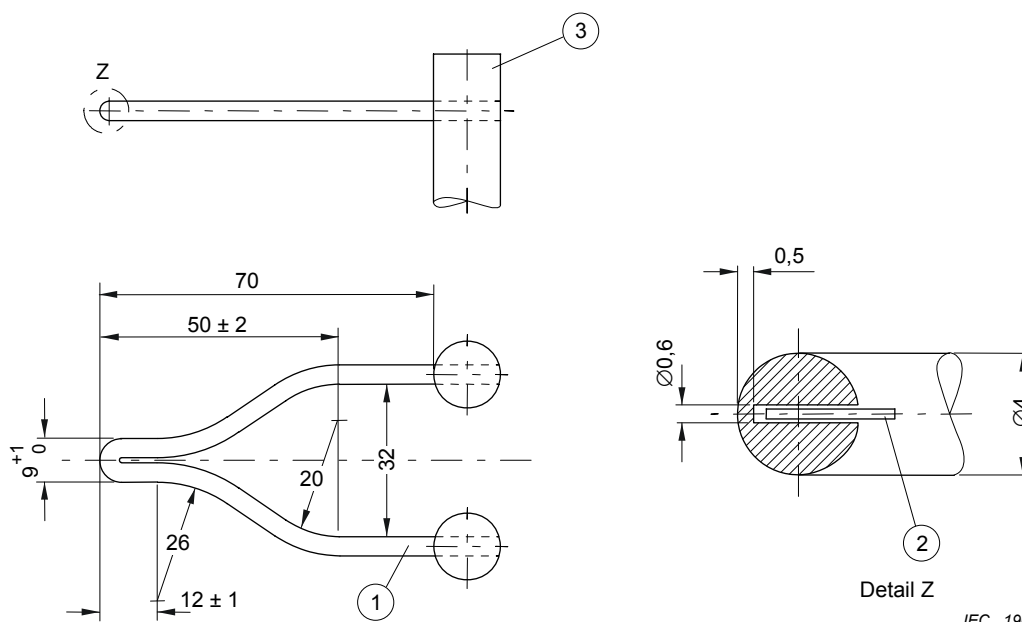
The following steps shall be taken:

- a) The test specimens shall be pre-conditioned for 24 h at (124 ± 2) °C in an air-circulating oven. The test specimens shall then be allowed to stabilize for 4 h at room temperature in a desiccator over anhydrous calcium chloride.

² Test boards of 150mm x 150mm may be considered large enough to represent larger production boards. Smaller production boards should be tested in their actual size.

- b) The relevant specification shall detail the orientation of the test specimen, if other than vertical.
- c) The thermocouple shall be calibrated at a temperature of 960 °C. The standard calibration technique shall be to place silver foil (99,8 % pure, 2 mm², 0,6 mm thick) on the upper face of the tip of the glow-wire. The glow-wire is heated by increasing current, and a temperature of 960 °C has been achieved when the silver foil melts.
- d) Calibration should be repeated after five measuring runs in order to compensate for alterations in the thermocouple and in the connections.
- e) Care should be taken to ensure that the thermocouple can follow the movement of the tip of the glow-wire caused by thermal elongation.
- f) The test specimen shall be held in a manner that minimizes heat loss into the test fixture.
- g) The tip of the glow-wire shall be applied to a part of the test specimen that is likely to be subjected to thermal stresses in normal use and/or which features the greatest combination of base material and surface coating(s). The tip shall be applied at least 15 mm below the upper edge of the test specimen. The glow wire shall be electrically heated to one of the preferred temperatures as given in the detail specification. This temperature shall be measured by means of the calibrated thermocouple. Care must be taken to ensure that, before starting the test, the temperature and heating current are stable for a minimum period of 60 s, and that heat radiation does not influence the specimen during this period or during the calibration (e.g. by providing an adequate separation or by using a screen).
- h) The tip of the glow-wire shall be brought into contact with the test specimen for a period of (30 ± 1) s, or a period as given in the appropriate detail specification. The heating current shall be maintained during this period. After this period, the glow-wire and test specimen shall be slowly separated, avoiding any further heating of the test specimen and any movement of air which might prejudice the result of the test.
- i) Before each test or calibration, it is necessary to remove any residue from the tip of the glow wire (by means of brushing).

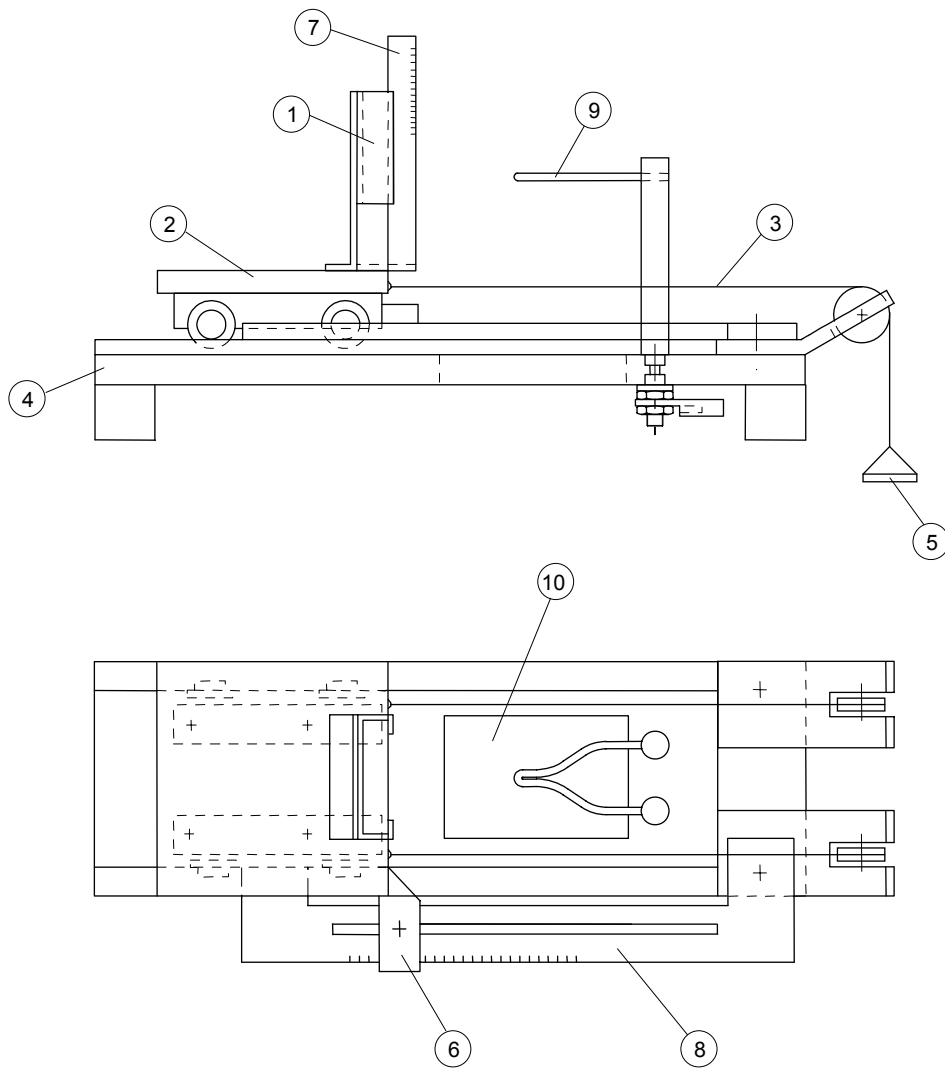
Dimensions in millimetres



Key:

- 1 glow-wire hard soldered at 3
- 2 thermocouple
- 3 stud

Figure 1 – Glow wire



Key:

- | | |
|---------------------|--|
| 1 positioning clamp | 6 stop |
| 2 carriage | 7 scale for measure of flame |
| 3 tensioning cord | 8 scale for penetration |
| 4 base plate | 9 glow-wire (Figure 1) |
| 5 weight | 10 break-through in base plate for particles falling from the specimen |

Figure 2 – Test apparatus

8.2.5 Report

In addition to the general requirements for reporting, the report shall include:

- the number of specimens, if other than five;
- the duration between the beginning of the tip application and the time at which the test specimen or the tissue paper beneath ignites;
- the duration between the beginning of the tip application and the time at which flames extinguish, during or after the period of glow-wire application;
- the maximum height of any flame. The height of the flame is the vertical distance measured between the upper edge of the glow-wire when applied to the test specimen and the visible tip of the flame. Any high flame produced at ignition, lasting approximately 1 s, shall be

ignored;

- e) whether the test specimen flames or glows and whether it, or the tissue paper, has been completely consumed by the flames;
- f) whether flaming or glowing extinguishes within 30 s from the removal of the glow-wire tip from the test specimen, and whether this applies to the test specimen and/or the tissue paper.

8.2.6 Additional information

- a) There are obvious hazards associated within flammability testing. Training of test operators, and familiarity with laboratory safety procedures is paramount. All fire effluent should be considered to be toxic, for the purposes of safety if not in fact.
- b) Small scale flammability tests such as that described herein give an indicator of the behaviour of the material(s) tested. Fire integrity of equipments in which printed boards are used can only be assessed by equipment level testing.
- c) Test methods in development by IEC Technical Committees include the use of a Cone Calorimeter for the determination of fire integrity of materials.

8.3 Test 3C03: Flammability; needle flame test, rigid printed boards

8.3.1 Object

The object of the test is to determine the effect upon a printed board of exposure to a glowing wire under specific conditions.

The intensity of the ignition source is similar to that of an accidentally overheating or glowing of a single electronic component.

Timings measured by this test are an indication of the ability of the printed board to self-extinguish. There is no correlation with other properties of the material(s), such as the oxygen index.

8.3.2 Test specimen

The test specimen shall be a production board or a test board that is representative of the production board in terms of:

- base material(s);
- surface coating(s);
- type (for example multilayer, single-sided, etc.);
- size;
- design;
- surface area;
- thickness;
- metal distribution.

Test boards of 150 mm × 150 mm may be considered large enough to represent larger production boards. Smaller production boards should be tested in their actual size.

A minimum of five specimens shall be tested.

8.3.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A room or compartment, in which the test is conducted, having dimensions adequate to ensure that the test is carried out in a substantially draught-free atmosphere, but which allows a sufficient supply of air for normal combustion. Subdued light is advantageous.
- b) A burner to produce the test flame, which is a tube with a length of at least 35 mm with an internal diameter of $(0,5 \pm 0,1)$ mm and an external diameter not exceeding 0,9 mm. The burner shall be held in a movable fixture.
- c) A burner shall be supplied with butane gas having a minimum purity of 95 %. No air may be admitted to the burner tube. Propane gas may be used, but butane gas shall be the reference.
The burner shall be adjusted whilst in a vertical axis, in order to produce a flame having a height of (12 ± 1) mm. See Figure 3.
- d) A wooden board to be placed underneath the specimen. The board shall be covered with tissue paper which complies with 6.86 of ISO 4046. The distance between the lower edge of the specimen and the tissue paper shall be (200 ± 5) mm.
- e) A hand-operated timing device with a resolution of $\pm 0,5$ s or better.
- f) An air circulating oven capable of maintaining (125 ± 5) °C.
- g) A desiccator capable of maintaining 20 % R.H. or less.

8.3.4 Procedure

The following steps shall be taken:

- a) The test specimens shall be pre-conditioned for 24 h at (125 ± 5) °C in an air-circulating oven. The test specimens shall then be allowed to stabilize for 4 h at room temperature in a desiccator over anhydrous calcium chloride.
- b) The relevant specification shall detail the position of the test specimen and the point of application of the flame (for example surface, edge). The attitude of the specimen (for example horizontal or vertical) should mimic the intended mode of operation in the assembled equipment.
- c) Where surface application is used, the point of application of the flame shall, specimen size permitting, not be less than 10 mm from the nearest edge, in order to minimize edge effects.
- d) Where edge application is used, the flame shall not, specimen size permitting, be less than 10 mm from the nearest corner.
- e) The burner shall be mounted at an angle of about 45°, so that any drops from the test specimen can fall freely onto the underlying tissue paper.
- f) If the intended operational attitude of the printed board is not known or is variable, the test specimens shall be positioned as follows.
- g) Edge application: The lower edge shall be horizontal and the specimen shall be inclined at approximately 80°. The flame shall be applied to the lower side of the test specimen.
- h) The burner shall be ignited away from the test specimen, and the height of the flame shall be adjusted to (12 ± 2) mm. The burner shall then be positioned as described, so that the test specimen penetrates the flame by approximately 2 mm. A vertical distance of between 8 mm and 10 mm from the tip of the burner and the surface or edge to be tested is adequate for this purpose. In the case of application to a vertical surface, a horizontal distance of approximately 5 mm is necessary.
- i) The detail specification shall state the severity to be used.

- j) Test flame application times as detailed in Clause 7 of IEC 60695-11-5 are preferred. These severities are: 5 s – 10 s – 20 s – 30 s – 60 s – 120 s.

8.3.5 Report

The report shall include:

- a) the test number and revision;
- b) the date of the test;
- c) the identification and description of the specimen;
- d) the number of specimens, if other than five;
- e) the position of the test specimens;
- f) the point of application of the test flame;
- g) the duration of application of the test flame;
- h) for each specimen whether flames and/or burning or glowing particles fall from the test specimen spreading fire to surrounding parts of the test specimen or to the tissue paper beneath the test specimen;
- i) for each specimen whether there is flame or glowing at the end of the application of the test flame;
- j) for each specimen whether the duration of burning is less than 30 s;
- k) any deviation from this test method;
- l) the name of the person conducting the test.

8.3.6 Additional information

There are obvious hazards associated with flammability testing. Training of test operators and familiarity with laboratory safety procedures is of paramount importance.

The smoothness of the specimen edges can be critical to the performance of the sample. A polished finish is recommended. A rough finish (for example blanked) will significantly degrade performance due to the increase in surface area available to the flame.

Small-scale flammability tests, such as the one described herein, are an indicator of the behaviour of the material(s) tested. Fire integrity of equipment in which printed boards are used can only be assessed by equipment level testing.

Detail requirements to be stated in specifications which reference this test method should be based on Clause 12 of IEC 60695-11-5.

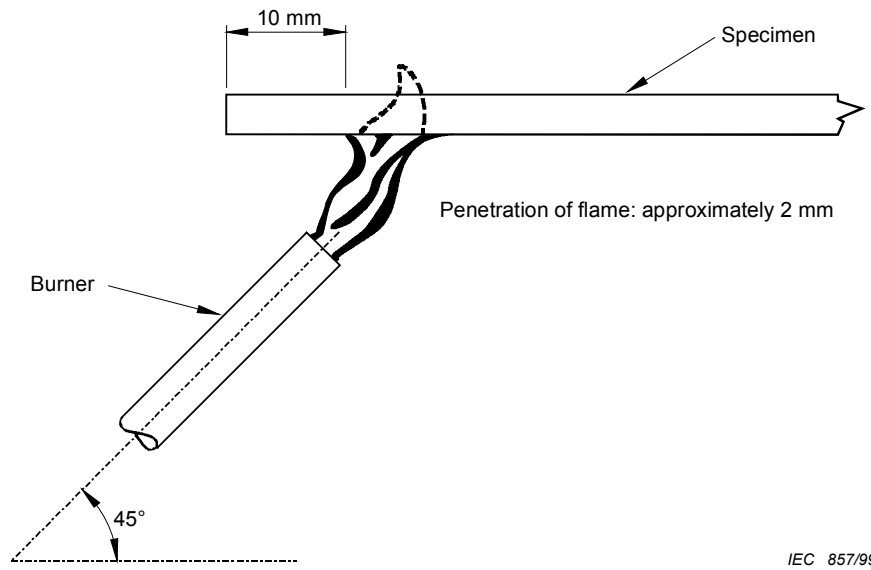


Figure 3a – Horizontal specimen – Flame applied to surface

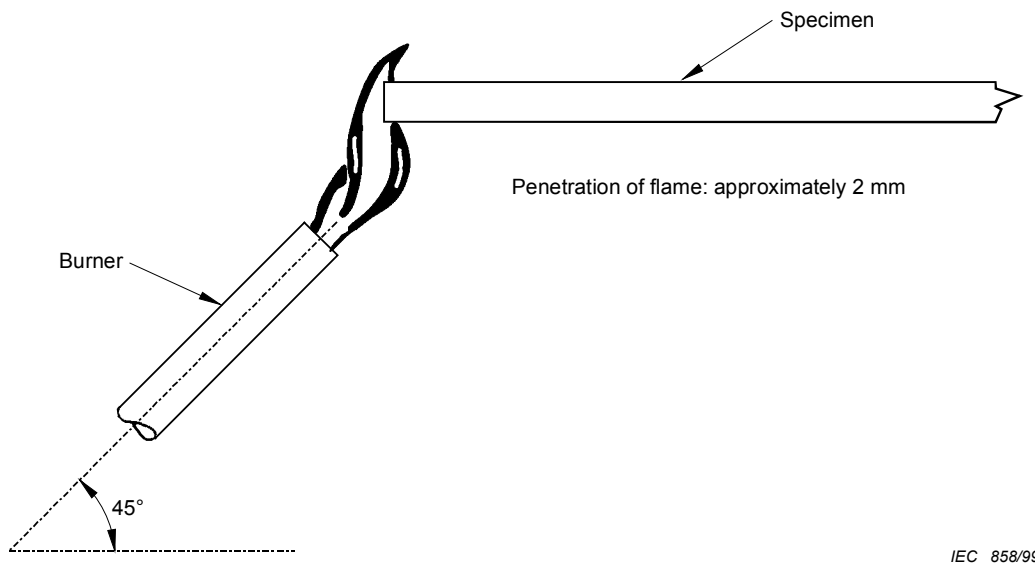


Figure 3b – Horizontal specimen – Flame applied to edge

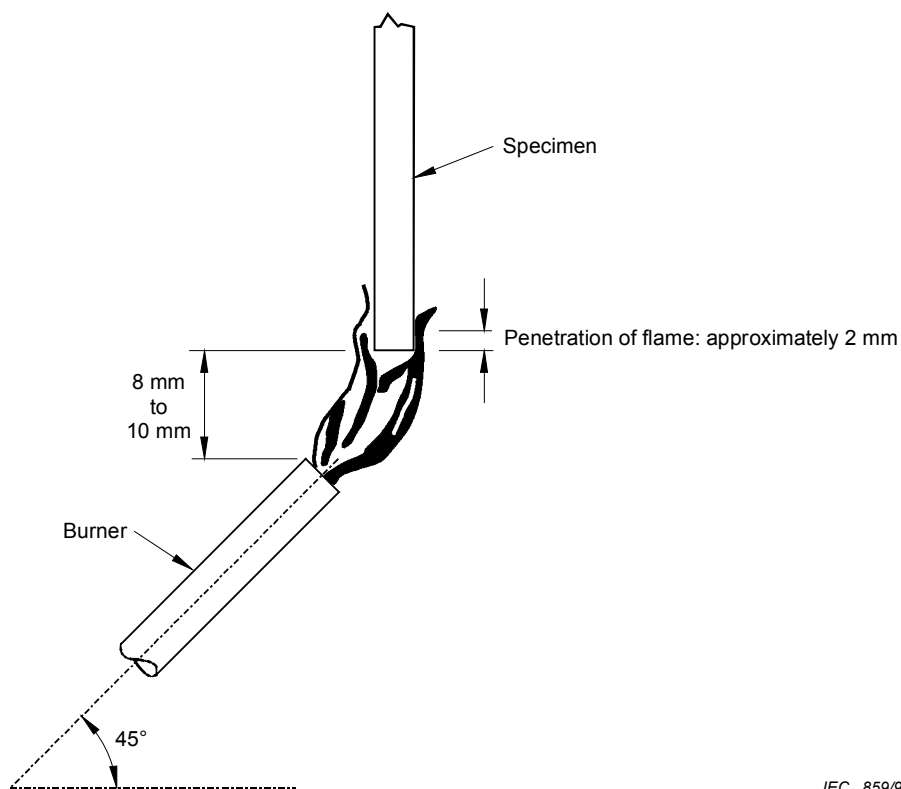


Figure 3c – Vertical specimen – Lower edge horizontal – Flame applied to edge

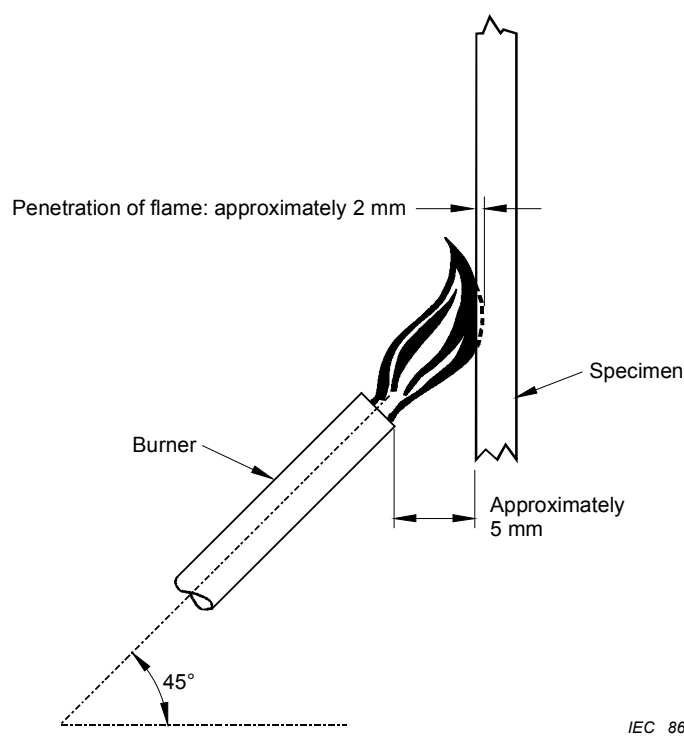


Figure 3d – Vertical specimen – Lower edge horizontal – Flame applied to surface

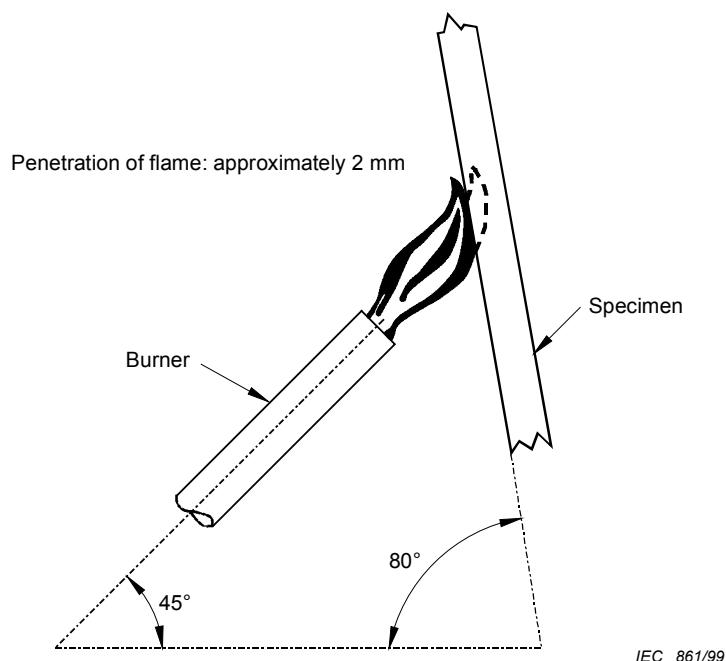


Figure 3e – Needle burner test – Side views of test board and burner

Figure 3 – Needle burner test

- 8.4 Test 3C04: Solvent and flux resistance** (under consideration)
- 8.5 Test 3C05: Electrolytic corrosion, rigid and thin film** (under consideration)
- 8.6 Test 3C06: Flammability, glow-wire test, rigid printed boards** (under consideration)
- 8.7 Test 3C07: Flammability, needle flame, rigid printed boards** (under consideration)
- 8.8 Test 3C08: Vertical burning** (under consideration)
- 8.9 Test 3C09: Water absorption** (under consideration)
- 8.10 Test 3C10: Surface organic contaminants (in-house)** (under consideration)
- 8.11 Test 3C11: Resistivity of solvent extract (ionic contaminants)** (under consideration)
- 8.12 Test 3C12: Surface organic contaminants (infrared)** (under consideration)
- 8.13 Test 3C13: Ionic analysis of printed circuit boards, ion chromatography method**

8.13.1 Object

This test procedure is designed to measure the level of anionic contaminants on the surface of printed boards by ion chromatography.

8.13.2 Test specimen

Three specimens of printed board (PWB) for extraction

8.13.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) an Ion chromatograph apparatus. The system consists of a pump and an anion column,

and a conductivity detector. A system which is operating properly should be capable of providing an accuracy of 50^{-9} or better. The equipment and chemistry should be set up and standardized per manufacturers' instructions;

- b) hot water bath capable of maintaining $(80 \pm 5) ^\circ\text{C}$;
- c) use a clean heat sealable bag with less than 250^{-9} extractable contaminants. (Specify cleanliness level or part number of manufacturers.);
- d) cleanroom vinyl gloves. ($<3^{-6}$ of Cl);
- e) hi-purity deionized water ($18,3 \text{ M}\Omega\text{-cm}$ grade and chloride levels of less than 50^{-9});
- f) hi-purity chemicals necessary for ion chromatography;
- g) isopropanol, electronic grade.

8.13.4 Procedure

The following steps shall be taken:

8.13.4.1 Extraction

- a) Record area of PWB. General rule on surface area is (length x width x 2).
- b) Use clean gloves when handling the specimens to be tested, and then place each specimen in the extraction bag.
- c) Prepare 75 % / 25 % isopropanol/ H_2O solutions for the extraction.
- d) Add (100-250) ml of the extraction solution to the extraction bag (enough to cover the printed board).
- e) Heat seal the extraction bag and place it in the $80 ^\circ\text{C}$ water bath for one hour (cut a vent hole in the bag).
- f) Measure solution volume after extraction.
- g) Prepare unprocessed PWB as control.

8.13.4.2 Standard and specimen analysis

- a) Inject solution into ion chromatograph (IC) unit and calculate the concentration of from known reference data.
- b) Values of chemicals obtained from the IC are expressed in parts per billion ($^{-9}$).
- c) Standards should be used per instruction of the manufacturer. (Chloride levels of 100^{-9} are recommended).
- d) A calculation to take into account for surface area and evaporation must be done so as to compare all different sizes of circuit boards.

$$\mu\text{g}/\text{cm}^2 = \frac{(\text{ppb value from IC}/1000) \times (\text{final volume} / \text{original volume})}{\text{Surface area}(\text{cm}^2)}$$

8.13.5 Additional information

None.

8.13.6 Reference documents

IPC-TP-1043 "Cleaning and Cleanliness Test Program, Phase III, Water Soluble Fluxes, Part 1: B-Z4, interactions of Water Soluble Fluxes with Metal/Substrates. October, 1992."

IPC-TP-1044 "Cleaning and Cleanliness Test Program, Phase III, Water Soluble Fluxes, Part 2: B-36, Comparison to Phase 1 Rosin Benchmark", September 1992.

8.14 Test 3C14: Flux induced corrosion (copper mirror method)

8.14.1 Object

This test method is designed to determine the effect the flux has (if any) to remove the copper film that is vacuum deposited on a glass slide.

8.14.2 Test specimen

A minimum of 100 ml of liquid flux, a representative container of solder paste, reflowed solder-paste flux, extracted solder preform flux or extracted flux from a flux-cored solder wire.

8.14.3 Test apparatus and materials/reagents

- a) 0,5l of control standard rosin flux to be agreed upon between user and supplier;
- b) 0,5l of reagent grade (99 % pure) isopropanol;
- c) a vacuum deposition system or the means to procure glass test panels having a copper mirror coating as described in this method;
- d) 0,5l of reagent grade 0,5 % solution of ethylene diamine tetra acetic acid (EDTA);
- e) 0,5l of reagent grade ethanol or methanol;
- f) 100ml medicine bottle with dropper;
- g) test cabinet capable of achieving $(23 \pm 2) ^\circ\text{C}$ and $(50 \pm 5) \%$ relative humidity;
- h) glass slides;
- i) a relative humidity gauge having an accuracy of $\pm 2 \%$ or better, shall be used to continuously monitor the test environment. The gauge should be calibrated periodically.

8.14.4 Procedure

The following steps shall be taken:

8.14.4.1 Preparation

8.14.4.1.1 Preparation of control standard flux

Dissolve 35 g the rosin into 100 ml of reagent grade 99 % isopropanol and stir thoroughly.

8.14.4.1.2 Preparation of temperature/humidity chamber

When acid or salt solutions are used the chamber shall be conditioned for a minimum of 48 h prior to exposing the copper mirror specimens, to assure compliance with the $(50 \pm 5) \%$ relative humidity requirement.

8.14.4.1.3 Preparation of copper mirror test panels

- a) Apply by vacuum deposition, a film of copper metal on one surface of a flat sheet of clear, polished glass.
- b) Apply a uniform thickness of approximately 50 nm and assure that the finished mirror permits $(10 \pm 5) \%$ transmission of normal incident light of nominal wave length of 500 nm. This may be determined using a suitable photoelectric spectrophotometer. Commercially available copper mirrors meeting the above specifications are acceptable. (See 8.14.6.2.)
- c) Prevent oxidation of the copper mirror by storing in a closed container which has been flushed with nitrogen.

- d) Immediately before testing, immerse the copper mirror in a 5 g/l solution of EDTA for copper oxide removal. Mirrors stored in a non-oxidizing environment, do not require cleaning with the EDTA solution prior to testing. The cleaning step must be used if test results are in dispute.
- e) Rinse thoroughly in running water, immerse in clean ethanol or methanol and dry with clean, oil free air.
- f) Carefully examine the mirror before testing. There must be no oxide.

8.14.4.2 Test

- a) Place the copper mirror test panel on a flat surface (see Figure 4), mirror side up, and protect from dust and dirt at all times.
- b) Place one drop of test flux or extract to be tested (approximately 0,05 ml) on each copper mirror test panel. The number of panels per test is no less than 3. Do not allow the dropper to touch the test panel. The number of tests shall be specified in the relevant specification
- c) Solder-paste shall be applied directly to the mirror without scratching the copper mirror, with a volume approximating a 0,5 mm thickness and 8 mm diameter. (It has been determined that significant variations from this quantity have little effect for most fluxes.)
- d) Immediately also place one drop of the control standard flux adjacent to the test flux. Do not allow drops to touch.
- e) Place test panels in a horizontal position in the dust free cabinet at $(23 \pm 2) ^\circ\text{C}$ and $(50 \pm 5) \%$ relative humidity for $(24 \pm 0,5) \text{ h}$.
- f) At the end of the 24 h period, remove the test panels from the chamber and remove the test flux and control standard flux from the panels by immersing the panels into clean isopropanol.

8.14.4.3 Evaluation

- Carefully examine each test panel for possible copper removal or discoloration.
- If there is any complete removal of the copper film as evidenced by the background showing through the glass, the test flux has failed the L category. Complete removal of the copper only around the perimeter of the drop defines the flux as “M”. Complete removal of the copper places the flux in the “H” category.
- If the control flux fails, repeat the entire test, using new copper mirror test panels.
- Discoloration of the copper film due to a superficial reaction or only a partial reduction of the copper film thickness is not considered a failure.
- A number of chemicals can cause failure of copper mirror: free halides, stronger organic and inorganic acids and free amines.

8.14.5 Notes

8.14.5.1 Safety

Observe all appropriate precautions described in the material safety data sheets (MSDS) for chemicals involved in this test method.

8.14.5.2 Sources for prepared copper mirrors

- EMF Corp., 239 Cherry St., Ithaca, NY 14850, 800-456-7070
- H.L. Clausing, Inc., 8038 Monticello Ave., Skokie, IL 60076, 847-676-0330

8.14.6 Additional information

None.

8.14.7 Reference document

ASTM E104 Maintaining constant relative humidity by means of aqueous solutions

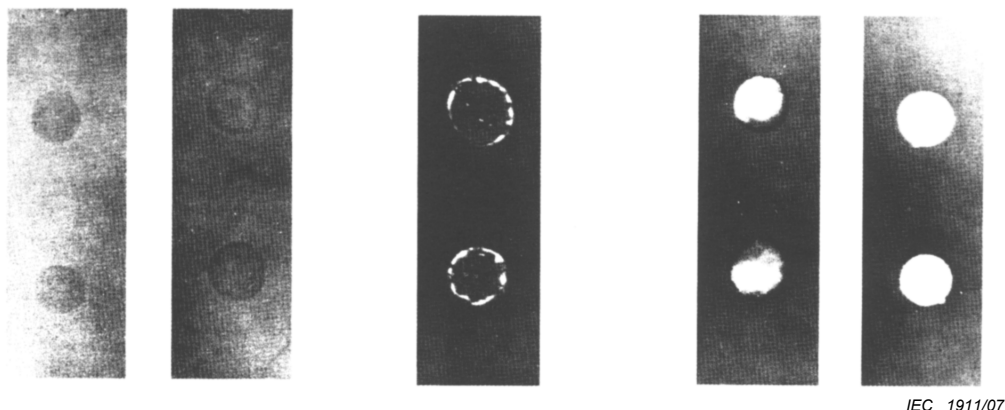


Figure 4 – Flux type classification by copper mirror test

9 M: Mechanical test methods

9.1 Test Method 3M01: Peel strength, standard atmospheric condition

9.1.1 Object

This test method is to determine the peel strength of conductors in the "as received" condition, and is applicable only to external foils that are used in foil lamination processes, and only as specified in customer specification, and/or purchase contract. (Note: Other peel strength tests are performed at the laminated printed board material level.)

9.1.2 Test specimen

The test specimens are individual test specimens (ITS) (coupons). Each coupon shall have 4 strips, parallel to the X-axis and Y-axis as shown in Figure 5 of the applicable printed circuit board. One coupon is necessary for each side of the printed circuit board laminated with foil. The coupons are located on a production board and shall be produced together with the applicable printed circuit board produced by the same conditions.

9.1.3 Test apparatus and test materials

The following test apparatus and materials shall be used:

- Tensile tester: A standard calibrated tensile strength tester, equipped with a low load cell, capable of measuring 45 mN, and at least 450 mm long light load wire or chain and clamp, and which includes its weight in the calculation. The clamp jaws must cover the foil width of each peel strip tab. Any equipment or apparatus having the described accuracy, precision, and reproducibility may be used.
- A suitable hold-down clamping system, equivalent in performance to that shown in Figure 5a.
- For qualification testing, a recording system must be incorporated into the test apparatus.

9.1.4 Procedure

The following steps shall be taken:

- a) Specimens (coupons) shall be placed on the production panel, such that the peel strips on each cut specimen are in both directions, and such that the copper can be completely removed (etched) from the opposite side of the test specimens.
- b) Pull two strips per specimen (coupon) for conformance testing; Pull four strips per specimen (coupon) for qualification.
- c) Cut the specimens from the panel, parallel to the glass fibres. The outside 25 mm border shall be excluded. Specimens may be cut to facilitate individual testing. Thin specimens must be provided with support by bonding them to a rigid aluminium, or similar base, or may be tested with the aid of a "keyhole" fixture (Figure 5d). Bonding to a rigid base shall be used as a referee evaluation.
- d) Assign an identification number to each specimen, and record relative panel location.
- e) Peel the foil back at the tab end.
- f) Clamp the end of each individual copper foil tab. The wire for connecting the clamp to the tensile tester must be free to pull vertically within $\pm 5^\circ$ angle.
- g) Fasten specimen (coupon) with hold-down fixture, such that an unencumbered vertical pull can be exerted. The tab end of the copper foil strip should be in a vertical position, ready for testing.
- h) Adjust tester to compensate for the weight of the wire or clamp.
- i) Start tester and apply force in the vertical direction at the rate of 50 mm/min, for $(30 \begin{smallmatrix} -0 \\ +10 \end{smallmatrix})$ s, until a 25 mm peel is completed, or until metal or foil breaks or tears. If the full width of the strip does not peel, the results shall be discarded, and another strip shall be tested.
- j) Observe and record the average load (Figures 5b and 5c) for each specimen; include assigned identification number.

9.1.5 Report

The report shall include:

- a) test method number and revision
- b) the date of evaluation;
- c) identification and description of specimen;
- d) actual average load for each specimen/test; including relative identification specified by the user.
- e) any deviation from this test method.
- f) the results of evaluation, including requirement(s) failed, failure modes and degree of failure, in case of failure;
- g) the name of the person that conducted the test.

9.1.6 Additional information

- a) Copper foil breaks may be caused by either a superior bond, or by brittle copper. Superior bond is indicated when the value at the break is above specification; in this case the actual load at the break should be reported for each specimen. The report shall indicate that the value is greater than the required specification.
- b) The condition of the underlying base material may also be evaluated after the peel strength test is completed.

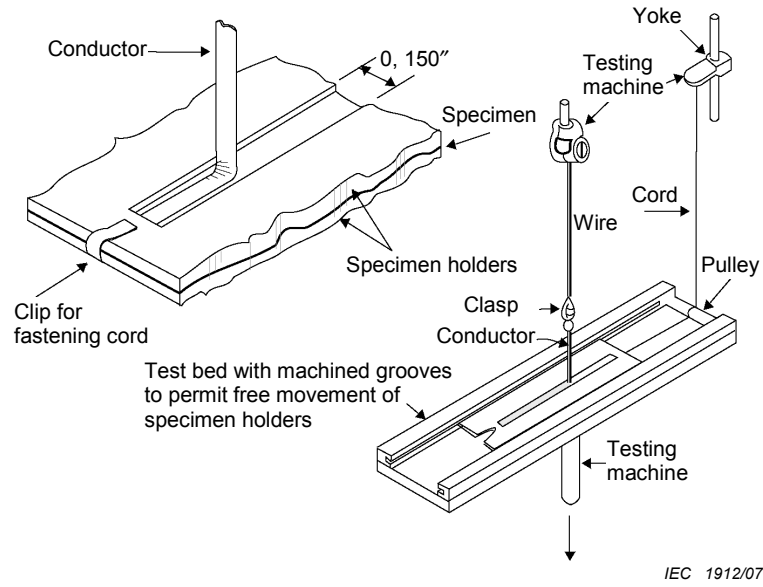


Figure 5a – Hold down clamping system

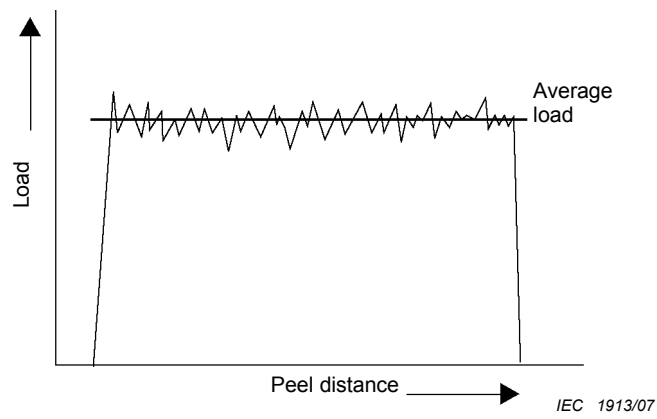


Figure 5b – Single load mode

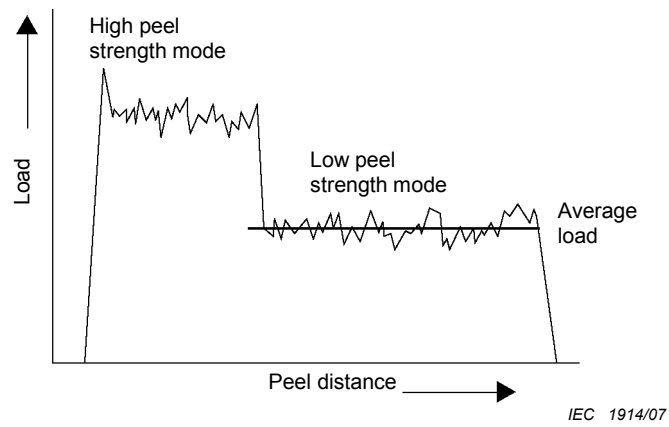


Figure 5c – Multiple load mode

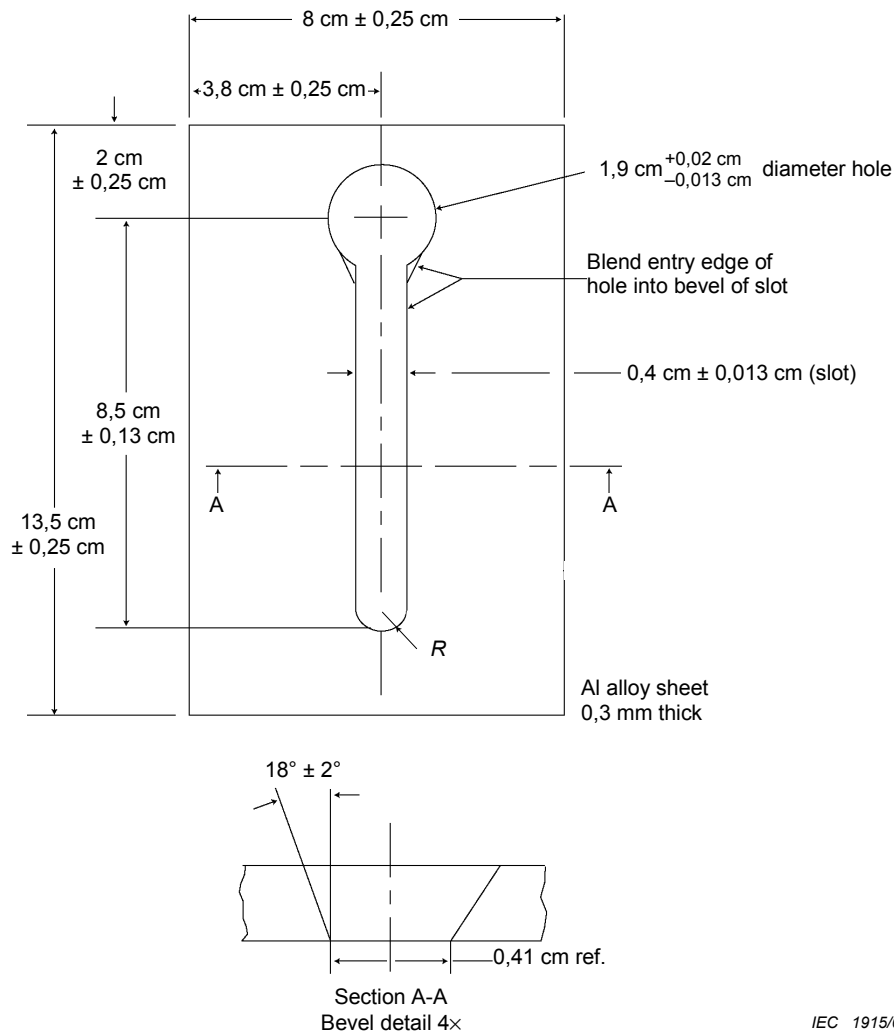


Figure 5d – Keyhole hold down fixture

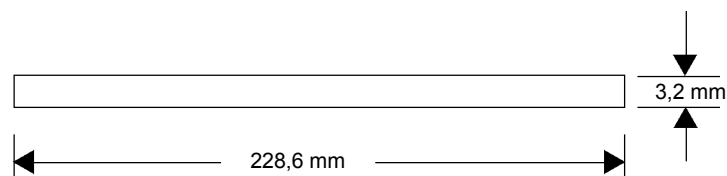


Figure 5 – Copper foil for peel test

9.2 Test 3M02: Peel strength, elevated temperature (under consideration)

9.3 Test 3M03: Pull out strength, plated through-holes, with or without lands

9.3.1 Object

This test method is to test the bond strength, or terminal pull strength of plating in plated through holes through-out manufacturing, as well as reworks and/or repairs. The five test cycles are intended to simulate initial soldering, component placement in original manufacturing, and subsequent component replacement for purpose of equipment repair.

9.3.2 Test specimen

A standard test specimen (coupon) or a printed board with plated through holes which can be used for component mounting. The diameter of the holes to be tested shall be no smaller than would normally be used for through-hole component mounting. Via holes, or other small holes, which would normally be considered only as vias, shall not be used for this test.

NOTE This is a destructive test.

The number of holes to be tested shall be specified in the relevant specification.

9.3.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) 60 W regulated soldering iron..
- b) Tin plated copper wire. The diameter of the test wires shall be (0,25 to 0,70) mm smaller than the test holes.
- c) Sn60Pb40 or Sn63Pb37 solder as specified in ISO 9453.
- d) Force tester: Vertical pull force tester, capable of operating at a speed of 50 mm/min, and measuring up to 100 N.

9.3.4 Procedure

The following steps shall be taken:

- a) If conductors are connected to the terminal area and/or the plated through hole, cut conductor(s) not exceeding 6 mm away from the terminal area, taking care not to disturb the bond at the terminal area or in the plated through hole.
- b) Insert test wires of sufficient length to connect to gripping mechanism of bond strength tester into plated through test holes, and solder by machine or by hand, as applicable. The wires shall not be clinched.
- c) A wire shall be unsoldered and re-soldered in the same hole(s) by hand, five times following initial soldering. During each cycle, the wire shall be completely removed from the hole. If the same wire is used, it shall be allowed to cool to ambient temperature before being replaced in the test hole, and re-soldered. A new wire may be used for each re-soldering.
 - Operate the 60 W soldering iron at a voltage sufficient to produce a tip temperature of 232 °C to 260 °C.
 - Apply the iron to the test wire and solder or unsolder normally. The iron may touch the land surrounding the hole, however, undue pressure and duration should be avoided to minimize damage to the plated through hole under test. The iron should be applied only as long as necessary to perform the soldering and unsoldering operation. (Note: It is recommended that the soldering and unsoldering portion of this test be performed only by personnel trained and/or certified in soldering and soldering rework.)
- d) Following the fifth cycle, the test specimen shall be clamped in the jaws of the bond tester.
- e) The wire on the “pattern side” of the board shall be pulled at a speed of 50 mm/min with a load giving the force in accordance with the following formula:

$$\frac{4L}{\pi(D_2^2 - D_1^2)} \geq 14 \text{ N/mm}^2$$

where

D₁ is the hole diameter;

D_2 is the terminal diameter;

L is the load.

Apply load perpendicular to the major surface of the terminal area until the required load is reached, or when a failure occurs.

9.3.5 Evaluation

It is a failure when

- a) a plated hole is loosened with soldered wire attached, whether the terminal area is loosened or not,
- b) a terminal area around a hole is loosened.

Breaking of a wire, or wire pull out is not a failure; however, a wire pulled out shall be re-soldered and retested (pulled).

9.3.6 Report

The report shall include:

- a) test method number and revision;
- b) any deviation from this test method;
- c) identification and description of specimen;
- d) hole size(s) and wire size(s) used in the test;
- e) pull out strength;
- f) results of evaluation, including failure modes and degree of failure, in case of failure;
- g) the date of evaluation;
- h) name of the person that conducted the test.

9.3.7 Additional information

It should also be noted that test results can vary greatly, based upon the skill of the operator performing the solder and unsolder operations required in the test.

9.4 Test 3M04: Flatness of laminates and printed boards (bow and twist)

9.4.1 Object

Four procedures are presented to determine the flatness of finished rigid printed boards, including single sided, double sided, and multilayer and the rigid segments of rigid flex printed boards.

9.4.2 Definitions

9.4.2.1 Bow

Bow is illustrated in Figure 6.

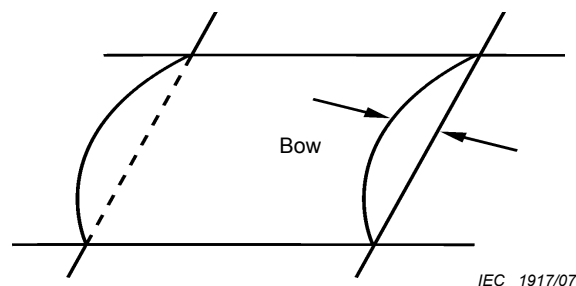


Figure 6 – Bow

9.4.2.2 Twist

Twist is illustrated in Figure 7.

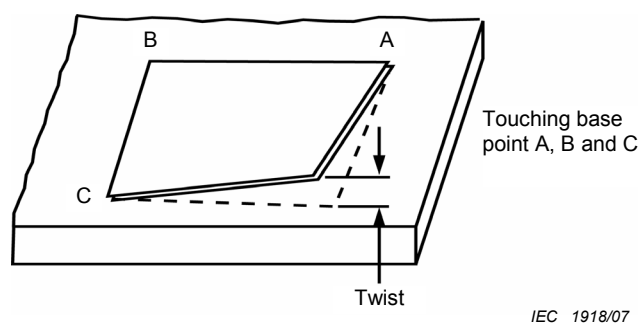


Figure 7 – Twist

9.4.3 Test specimen

- finished boards (single sided, double sided, multilayer, rigid flex boards)

9.4.4 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Inspection grade surface plate with a flatness tolerance $\pm 0,010$ mm.
- b) Standard metrology height dial indicator gauges.
- c) Thickness measurement feeler gauges.
- d) Standard pin gauges.
- e) Levelling jacks.
- f) Gauge blocks.
- g) Shims of suitable thickness.
- h) Linear measuring devices with an accuracy of $\pm 0,025$ mm.
- i) Micrometer.

9.4.5 Procedure

The following steps shall be taken:

9.4.5.1 Procedure number 1, Bow (see Figure 6)

- a) Place the specimen to be measured on the precision surface plate with its convex side

facing upward. For each edge, apply sufficient pressure on both corners of the specimen to ensure contact with the surface. Take a reading with the height dial indicator at the maximum vertical displacement of the edge including the thickness of the specimen (see Figure 8).

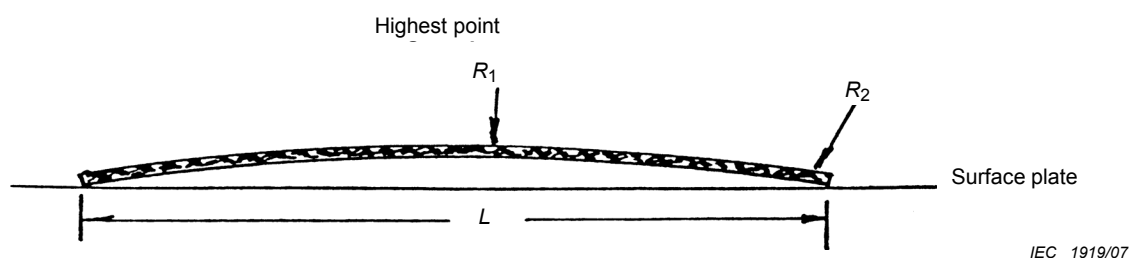


Figure 8 – Test set-up for bow measurement

- b) Repeat this procedure until all four edges of the specimen have been measured. It may be necessary to turn the specimen over to accomplish this procedure. Identify the edge with the greatest deviation from the surface plate; This deviation is to be recorded as R_1 .
- c) Take a reading with the height dial indicator or equivalent measuring device at the corner of the specimen contacting the surface plate, or determine R_2 by measuring the thickness of the specimen with a micrometer.
- d) Apply sufficient pressure so that the entire edge contacts the surface plate. Measure the length of the edge and denote as " L ".
- e) Calculate bow for this edge as follows:

$$\text{Percent bow} = \frac{R_1 - R_2}{L} \times 100$$

The largest bow of all edges is denoted as the bow of the specimen.

9.4.5.2 Procedure number 2, Twist

- a) Place the specimen to be measured on the plate with three corners touching the surface; apply sufficient pressure to ensure that three corners are in contact with the surface plate.
- b) Insert suitable shims under the raised corner so that it is just supported. When the correct shim thickness is used, the three other corners will be in contact with the surface without applying pressure to any corner (see Figure 9).

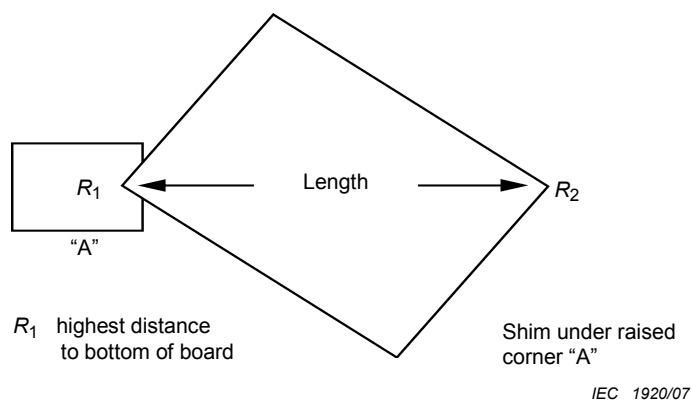


Figure 9 – Specimen set-up for twist measurement

- c) Without exerting any undue pressure on the specimen, take a reading with the height dial indicator, at the maximum vertical displacement (denoted as R_1 in Figure 9), and record the reading.

- d) The thickness R_2 of the specimen shall be measured with a micrometer.

NOTE For fabricated boards, both readings should be made on base material.

- e) Measure the diagonal (length as shown in Figure 9) of the specimen (for rectangular boards) and record the reading. For non-rectangular boards, measure from the corners exhibiting displacement diagonally to the corner on the opposite end of the board.

- f) Calculation:

Deduct R_2 reading from R_1 reading. (This value is divided by 2 in the formula defined because of the method of measurement doubles the vertical deflection).

Divide the measured deviation by the recorded length, and multiply by 100. The result of this calculation is the percent twist, as follows:

$$\text{Per cent twist} = \frac{R_1 - R_2}{2 \times \text{length}} \times 100$$

9.4.5.3 Procedure number 3, Twist referee test

- Place the specimen to be measured on the plate surface, with the two lower opposite corners touching the plate surface, or on raised parallel surfaces of equal height from the plate surface (see Figure 10a).
- Support the board in other two corners with levelling jacks, or other appropriate devices, ensuring that the two raised corners are of equal height from the surface plate. This may be checked by using the height dial indicator (see Figure 10b).
- Using the height dial indicator, measure the highest raised portion of the board, and record the reading as R_1 (see Figure 10c).
- Without disturbing the specimen, take reading with the height dial indicator at one of the corners contacting the surface, and record as R_2 (see Figure 10c).
- Measure the diagonal of the specimen (for rectangular boards) and record the reading. For non-rectangular boards, measure from the corner exhibiting maximum displacement diagonally to the corner on the opposite end of the board.
- Calculation:

Deduct the measurement R_2 from measurement R_1 . This difference is denoted as twist.

Divide the measured deviation by the recorded length, and multiply by 100. The result of this calculation is the per cent of twist:

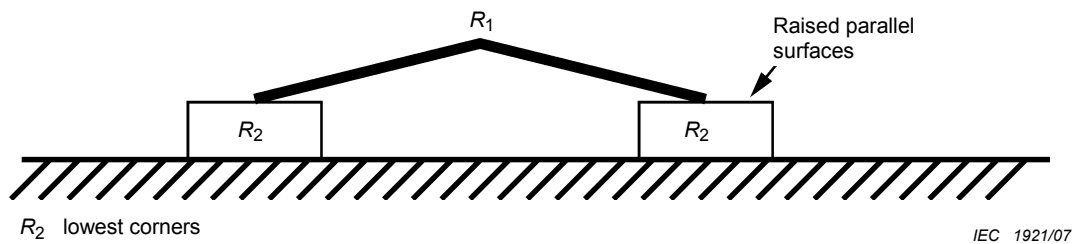


Figure 10a – Specimen set-up for referee test for twist, raised parallel surfaces

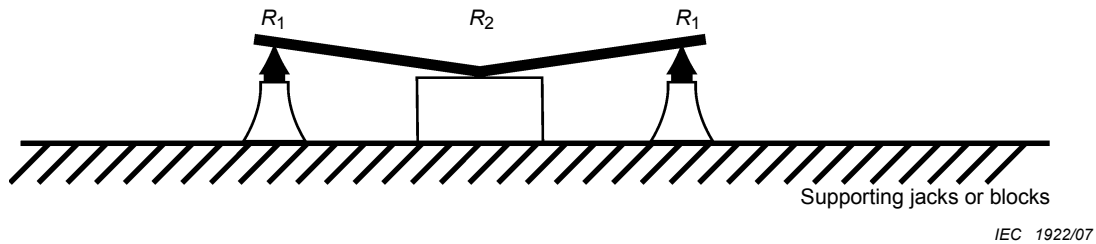


Figure 10b – Specimen set-up for referee test for twist, supporting jacks or blocks

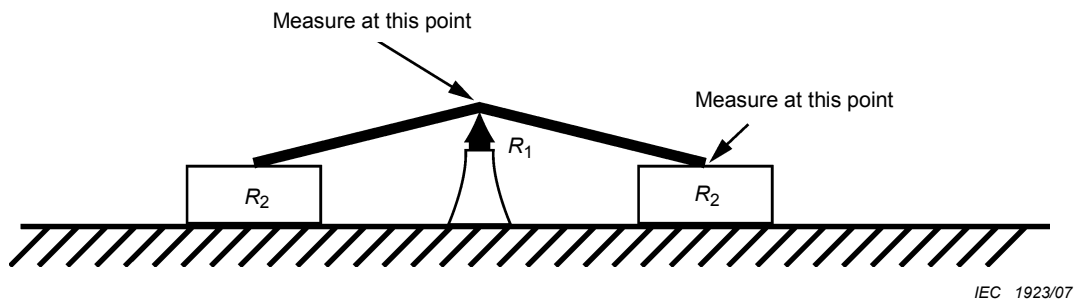


Figure 10c – Specimen set-up for referee test for twist measurements

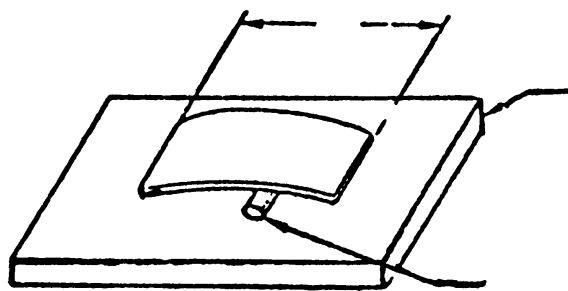
Figure 10 – Specimen set-up for referee test

9.4.5.4 Procedure number 4, Production testing (bow and twist)

Having previously determined the maximum allowable deviation of a board from a flat plane, the following GO-NO GO procedure may be used.

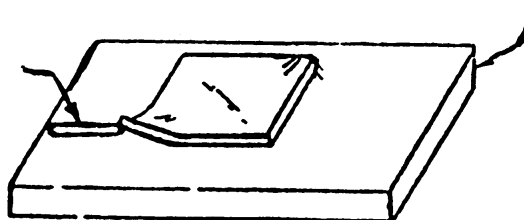
With both corners of an edge touching the plate, and using a suitable known dimensional standard, such as a pin gauge or feeler gauge conforming to the maximum allowable deviation, attempt to insert the gauge between the raised portion of the board and the datum surface (see Figure 11). Attempt to insert a gauge appropriate to the allowable surface dimension between the raised corner of the board and the datum surface (see Figure 12).

If the gauge does not enter the gap, the board will have met the bow and twist acceptability criteria.



IEC 1924/07

Figure 11 – Bow measurement



IEC 1925/07

Figure 12 – Twist measurement

9.4.6 Report

The report shall include:

- a) the test method number and revision;
- b) the date of the test;
- c) the procedure number(s) used for the test(s);
- d) the description identification of the specimen;
- e) actual calculations, where applicable;
- f) the results of evaluation, including failure modes and degree of failure, in case of failure;
- g) any deviation from this test method;
- h) the name of the person that conducted the test.

9.4.7 Additional information

Forms of distortion other than those defined in this test method (such as multiple convolutions) cannot be evaluated accurately by these test methods and should therefore receive special attention.

9.5 Test 3M05: Peel strength, flexible printed boards

9.5.1 Object

To determine the peel strength of a 1 mm and/or 3 mm wide printed conductor on a flexible printed board.

9.5.2 Test specimen

- The test shall be carried out on a minimum of four straight conductors of suitable length and uniform width taken from specified parts of a production board, of a test coupon, or of a composite test coupon.
- The conductor length should preferably be not less than 75 mm.
- Conductors less than 0,8 mm wide shall not be tested.
- Where plated conductors are present on the board, some of them shall be tested.

In the case where the sheets of copper-clad base film are to be tested instead of the production boards, the sample unit of the material shall have a minimum of four etched conductor patterns of 1 mm and/or 3 mm width and 230 mm length, two in the machine direction and two in the transverse direction. For a double-clad laminate, a separate sample unit shall be prepared and tested for each side. The copper foil on the untested side may remain to provide stability and to prevent tenting of the specimen from the rotary drum during testing. In the case of materials clad with copper foil of nominal mass per unit area less than 305 g/m², the foil mass per unit area may be increased to not more than 335 g/m² by any procedure in which ductile copper is deposited.

9.5.3 Test apparatus and materials

The following test apparatus and material shall be used:

- a) a tensile testing machine that achieves a peel rate of (50 ± 5) mm/min;
- b) a clamp to grip the detached end of the foil over its entire width;
- c) an adequate digital type recorder which is able to record the peel load at a rate of not less than three points per second. Otherwise, an adequate analogue type recorder which is able to record and display the peel load profile;
- d) free wheeling rotary drum (Figure 1a), to keep the direction of pull at $(90 \pm 5)^\circ$ to the plane of the printed board;
- e) adequate production processes shall be used to prepare the test patterns on a sheet of copper-clad panel;
- f) a suitable straight rule or optical devices, reading to 0,01 mm, for measuring the width of the conductors to be tested.

9.5.4 Procedure

The following steps shall be taken:

9.5.4.1 Conditioning

As a referee condition specimens for 24 h at standard laboratory conditions $(23 \pm 2)^\circ\text{C}$ and $(45 \pm 5)\%$ RH. Stabilization time may be reduced if statistically sound evidence has been generated on the specific product line to support the shorter stabilization time. For normal testing, standard atmospheric conditions shall be used.

9.5.4.2 Measurement

Measure and record the width of the tested conductor to the nearest 0,01 mm. One end of the conductor shall be detached from the base material for a distance of about 10 mm, and sufficient for the apparatus used. The test board shall be supported in a suitable way, for example by clamping between two flat rigid plates with a cut-out in the upper plate for the conductor to be peeled, or by attaching to a rotating drum as shown in Figure 13. The detached end of the conductor shall be gripped over its entire width or in the jaws of the clamp on the tensile testing machine, and a steadily increasing pull shall be applied in a direction $(90 \pm 5)^\circ$ to the plane of the printed board until the conductor peels off at a rate of (50 ± 5) mm/min, the force required to do this being measured. A length of at least 25 mm shall be peeled at this rate from each of four conductors.

The peel load shall be continuously recorded in the recorder, and the average values of the loads during each 1 s shall be calculated automatically, and be printed out on a form, or a profile of loads shall be drawn on a chart.

9.5.4.3 Evaluation

A minimum of 50 mm must be peeled; the pull loads corresponding to either the first 1 s or the first 5 mm distance shall be disregarded.

Where a digital recorder is used, compute the peel strength for each conductor using the following formula:

$$\text{peel strength (N/mm of width)} = \frac{\text{average load (N) as in 9.5.4.2}}{\text{conductor width (mm)}}$$

Where an analogue recorder is used, average the chart recordings for the specimen over the minimum stable region of the chart.

The average of the four test values shall be calculated and recorded as the peel strength.

9.5.5 Report

The report shall include:

- a) the test number and revision index;
- b) the test date;
- c) the identification of the material tested;
- d) conductors being tested;
- e) minimum peel strength;
- f) any deviation from this test method.

9.5.6 Additional information

The force required to bend the test conductor will affect the measured peel strength. The magnitude of this effect will increase as the conductor thickness increases.

In the event that excessive tenting of the peel specimens occur, suitable support material may be applied to the back of the test specimen. A reference support material may be unclad epoxy glass material.

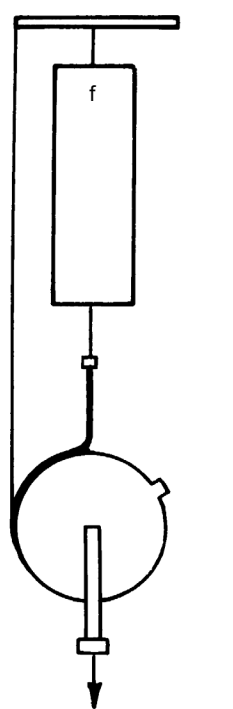


Figure 13a

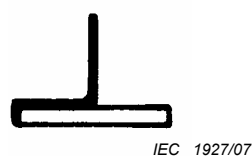


Figure 13b

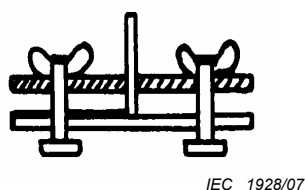


Figure 13c

Figure 13a – Arrangement for flexible specimens

The specimen is held against the surface of a freely rotating drum*. The pull is exerted between the drum spindle and the test strip, and the drum rotates to maintain the required angle of detachment. A cord attached to the circumference and to a fixed point vertically above may be used to ensure smooth rotation.

Figure 13b – Mounting of flexible specimens

Specimens may be cemented on the back to rigid material and tested.

Figure 13c – Support fixture for flexible specimens

The specimen may be held in a fixture to prevent distortion. The peeled strip emerges through a slot in the fixture just wide enough to allow it to pass without restriction.

Figure 13 – Measuring equipment for peel strength of flexible printed boards

9.6 Test 3M06: Flexural fatigue, flexible printed board (under consideration)

9.7 Test method 3M07: Pull-off strength, lands with plain holes

9.7.1 Object

This test method is to assess the quality of adhesion of lands to the base material under the stress of repeated soldering operations. The pull-off strength is measured as the force, normal to the surface of the printed board required to separate the land from the base material. This test gives an indication of an approximate value of the pull-off strength after soldering operation.

* The force caused by friction of the rotating parts indicated by the measuring device shall not exceed 50 mN. It shall also not exceed 10 % of the total force measured during the peel strength test.

9.7.2 Test specimen

Tests shall be carried out on circular lands that have been isolated from the attached conductors as determined by the applicable specification.

The following land, hole and wire dimensions are shown in Table 2.

Table 2 – Preferred land, hole and wire dimensions

Land diameter mm	Hole diameter mm	Wire diameter mm
4,0 ± 0,1	1,3 ± 0,1	0,9 – 1,0
2,0 ± 0,1	0,8 ± 0,1	0,6 – 07

These lands and wires shall be soldered within 3 s by the initial machine or hand soldering.

Other land, wire and hole dimensions may be specified in the relevant specification.

9.7.3 Test apparatus and materials

The following test apparatus shall be used:

- a soldering iron, capable of producing a tip temperature of 232 °C to 260 °C;
- vertical pull force tester capable of operating at a speed of 50 mm/min and measuring up to 50 N load;
- suitable wire of sufficient length to connect to gripping mechanism of bond strength tester;
- the solder to be Sn60Pb40 or Sn63Pb37 as specified in ISO 9453.

9.7.4 Procedure

The following steps shall be taken:

- The wire shall be inserted into the hole located approximately in the centre of the land. And it shall be soldered by hand. The wire shall not be clinched.
- After soldering a specimen must be cool at room temperature for 5 min.
- The iron shall be applied to the leads, not the foil, and shall be applied only as long as is necessary to perform the soldering operation.
- The printed wiring board shall be clamped in horizontal position in the jaws of the bond tester. The wire is connected to the gripping of the tester.
- A force shall then be applied to the wire by means of its tester, pulling at the rate of 50 mm/min the wire at right angle to the printed wiring board. The pulling shall be continued up to the force, that is required to separate the land from the base material.
- The smallest of any of the forces required to detach 5 lands from the base material shall be taken as the pull- off strength of the board under the test.
- Subject wires to four cycles of unsoldering and soldering by hand after the initial machine or hand soldering. During the four cycles the wires shall be completely removed during each unsoldering operation and replaced before each soldering operation.
- The iron shall be applied to the leads, not the foil, and shall be applied only as long as it is necessary to perform the unsoldering or soldering operation.
- After soldering the specimen shall be cooled at room temperature for 5 min each time.
- Following the fourth cycles, the printed board shall be clamped horizontally in the jaws of the bond tester.

- k) A force shall then be applied by means of its tester, pulling at the rate of 50 mm/min the wire at right angles to the printed wiring board. The pulling shall be continued up to the force that is required to separate the land from the base material.
- l) The smallest of any of the forces required to detach 5 lands from the base material shall be taken as the pull-off strength of the board under the test.
- m) Breaking of a wire or a wire pull-out shall not be considered as a failure but the wire shall be soldered and pulled again.

9.7.5 Report

The report shall include:

- a) the test method number and revision;
- b) the test date;
- c) the location of lands to be tested;
- d) soldering method;
- e) the result (each value and the minimum value) of measurement after 1 cycle;
- f) the result (each value and the minimum value) of measurement after 2 cycles;
- g) the name of the person that conducted the test.

9.7.6 Additional information

None.

9.8 Test 3M08: Hardness (resistance to abrasion) of organic surface coatings of printed circuit boards

9.8.1 Object

This test method details the procedure for the determination of the hardness of permanent organic surface coatings used on printed boards.

Organic materials suitable for testing in accordance with this procedure include:

- permanent polymeric coatings;
- conductive inks;
- marking legend inks.

Hardness is determined by comparison with a standard set of pencils with increasing hardness.

9.8.2 Test specimen

The test specimen shall consist of a piece of metal-clad base material $(100 \pm 10) \text{ mm} \times (100 \pm 10) \text{ mm}$, with the organic surface coating applied and cured as recommended by the supplier.

Alternatively, a coated portion of a production board may be used as agreed between the customer and the supplier.

A minimum of three specimens (separate boards) shall be tested.

9.8.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Test device (pencil holder, see Figure 14).

- b) Standard set of pencils of hardness from 4B to 8H.

4B, 3B, 2B, B, HB, F, H, 2H, 3H, 4H, 5H, 6H, 7H, 8H

Soft ————— Hard

9.8.4 Procedure

The following steps shall be taken:

- Place the specimen on a firm horizontal surface.
- The tips of all pencils shall be sharpened as shown in Figure 14.
- Starting with the hardest pencil in the carriage, push the carriage in the direction as illustrated in Figure 14 with a uniform forward force in a 20 mm to 30 mm stroke, whereby, because of the weight of the specimen holder of (750 ± 25) g, a downward directed force of $(7,65 \pm 0,25)$ N will result at the pencil tip. Proceed with the next softest pencil in succession until a pencil is used which will not cut into or gouge the coating material. Used pencils shall be re-sharpened after every test.
- The hardness is an attribute, rather than a variable. No statement of uncertainty is required.

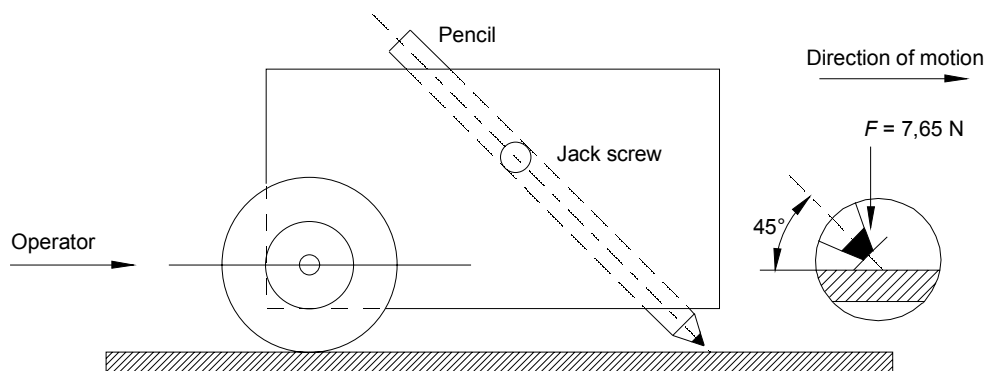
9.8.5 Report

The report shall include:

- the test method and revision;
- the date of the test;
- the identification and description of the specimen(s);
- the achieved hardness; i.e. the hardness of the pencil which failed to cut into or gouge the organic surface coating;
- the supplier and description of the pencil set;
- any deviation from this test method;
- the name of the person conducting the test.

9.8.6 Additional information

It is accepted that the results achieved using this test method are subjective in nature. To limit this subjectivity, it is recommended that only professional grade draftsman's pencils are used. Where disputes between testing agencies result, an agreed single source of pencil should be used for the test.



IEC 862/99

Figure 14 – Pencil holder

9.9 Test 3M09: Degree of cure of organic permanent surface coatings on printed circuit boards

9.9.1 Object

This test method defines the procedure for determining the degree of cure of an organic permanent surface coating used on a printed board.

This technique is qualitative.

This test method is suitable for determining the degree of cure of permanent polymeric solder resistive coating, conductive inks and legend marking inks.

9.9.2 Test specimen

The test specimen may comprise any suitable area of a production board or capability qualifying component (CQC), as agreed between customer and supplier.

A minimum of three separate boards shall be tested.

9.9.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A solvent as agreed upon between user and supplier.
- b) A lint-free cloth or pad.
- c) A stop-watch or stop-clock.

9.9.4 Procedure

The following steps shall be taken:

- a) The material(s) under test shall be processed in accordance with the supplier's instructions.
- b) Three drops of the solvent shall be applied to the same location on the specimen and allowed to stand for 60 s.
- c) The specimen shall be rubbed vigorously with a lint-free cloth or pad.
- d) Excess liquid shall be removed.
- e) Examine the surface if portions of the coating were transferred to the cloth or pad or if the surface of the coating was softened or became tacky.

9.9.5 Report

In addition to the general requirements for reporting, the report shall include:

- a) the test method number and revision;
- b) the date of the testing;
- c) the identification and description of the specimen(s);
- d) the type and grade of solvent used;
- e) if the specimen (other than free surface debris) is transferred to the cloth or pad, or if the surface is softened or becomes tacky;
- f) any deviation from this test method
- g) the name of the person that conducted the test.

9.9.6 Additional information

Care should be exercised when using solvents, in accordance with local statutes and procedures.

Instrumentation is available which provides a quantitative assessment of degree of cure for the suggested test materials. The technique described herein is designed to provide an expedient qualitative measure, suitable for use in both production and laboratory environments.

10 E: Electrical test methods

10.1 Test 3E01: Circuit isolation (under consideration)

10.2 Test 3E02: Circuit continuity (under consideration)

10.3 Test method 3E03: Insulation resistance, surface layers

10.3.1 Object

The purpose of this test is to determine the insulation resistance between specified parts of a conductive pattern on the surface of a printed board or of a layer of a multilayer printed board before lamination.

10.3.2 Test specimen

- a) The insulation resistance shall be measured between any two electrically separate points of a conductive pattern on the surface of a production board, a test coupon or a composite test board.
- b) When the use of test coupons specified in IEC 62326-4-1 is agreed, the measurement shall be carried out on "Specimen E".

10.3.3 Test apparatus and materials

The following test apparatus shall be used:

- a megohm meter, capable of applying (10 ± 1) V, (100 ± 15) V and (500 ± 50) V, and of reading from $1 \times 10^6 \Omega$ to $1 \times 10^{13} \Omega$ with an accuracy of 5 % at its highest scale setting.

10.3.4 Procedure

The specimen shall be preconditioned by Test 1P01 of IEC 61189-1.

The insulation resistance shall be measured with the megohm meter. The test voltage shall be

(10 ± 1) V, or

(100 ± 15) V, or

(500 ± 50) V,

as specified in the relevant specification. The test voltage shall be applied for 1 min before measurement.

10.3.5 Report

The report shall include:

- a) the test method number and revision;
- b) the test date;
- c) parts of the pattern to be measured;

- d) test voltage;
- e) the result of measurement (average and minimum values);
- f) any deviation from this test method;
- g) the name of the person that conducted the test.

10.3.6 Additional information

None.

10.4 Test 3E04: Insulation resistance, internal layers

10.4.1 Object

The purpose of this test is to determine the insulation resistance between specified parts of a conductive pattern on an internal layer of a multilayer printed board.

10.4.2 Test specimen

The insulation resistance shall be measured between any two electrically separate points of a conductive pattern on an internal layer of a production board, a test coupon or a composite test board.

When the use of test coupons specified in IEC 62326-4 is agreed, the measurement shall be carried out on "Specimen J."

10.4.3 Test apparatus and materials

The following test apparatus shall be used:

- a megohm meter, capable of applying (10 ± 1) V, (100 ± 15) V and (500 ± 50) V, and of reading from $1 \times 10^6 \Omega$ to $1 \times 10^{13} \Omega$ with an accuracy of 5 % at its highest scale setting.

10.4.4 Procedure

The specimen shall be preconditioned by Test 1P01 of IEC 61189-1.

The insulation resistance shall be measured with the megohm meter. The test voltage shall be:

- (10 ± 1) V, or
- (100 ± 15) V, or
- (500 ± 50) V,

as specified in the relevant specification. The test voltage shall be applied for 1 min before measurement.

10.4.5 Report

The report shall include:

- a) the test method number and revision;
- b) the test date;
- c) parts of the pattern to be measured;
- d) test voltage;
- e) the result of measurement (average and minimum values);
- f) any deviation from this test method;
- g) the name of the person that conducted the test.

10.4.6 Additional information

None.

10.5 Test 3E05: Insulation resistance, between layers

10.5.1 Object

The purpose of this test is to determine the insulation resistance between specified parts of a conductive pattern between layers of a multilayer printed board.

10.5.2 Test specimen

The insulation resistance shall be measured between any two electrically separate points of a conductive pattern between layers of a production board, a test coupon or a composite test board.

When the use of test coupons specified in IEC 62326-4-1 is agreed, the measurement shall be carried out on "Specimen M."

10.5.3 Test apparatus and materials

The following test apparatus shall be used:

- a megohm meter, capable of applying (10 ± 1) V, (100 ± 15) V and (500 ± 50) V, and of reading from $1 \times 10^6 \Omega$ to $1 \times 10^{13} \Omega$ with an accuracy of 5 % at its highest scale setting.

10.5.4 Procedure

The specimen shall be preconditioned by Test 1P01 of IEC 61189-1.

The insulation resistance shall be measured with the megohm meter. The test voltage shall be:

(10 ± 1) V, or

(100 ± 15) V, or

(500 ± 50) V,

as specified in the relevant specification. The test voltage shall be applied for 1 min before measurement.

10.5.5 Report

The report shall include:

- a) the test method number and revision;
- b) the test date;
- c) parts of the pattern to be measured;
- d) test voltage;
- e) result of measurement (average and minimum values);
- f) any deviation from this standard test method;
- g) name of the person that conducted the test.

10.5.6 Additional information

None.

10.6 Test 3E06: Frequency drift (under consideration)**10.7 Test 3E07: Circuit impedance** (under consideration)**10.8 Test 3E08: Plated-through hole (pth) resistance change, thermal cycling****10.8.1 Object**

The purpose of this test method is to provide a procedure for determining the change in resistance of plated-through holes that may occur when the holes are subjected to thermal cycling. An alternative test method is 3E16: Plated-through hole (pth) and interconnection resistance change, thermal shock.

10.8.2 Test specimen

The test specimens will have the following considerations:

- a) The specimen shall be a suitable printed board, a test coupon or a composite test coupon having a number of plated-through holes connected in series. Composite test patterns are shown in Figure 15.
- b) Where the use of test coupons, as specified in IEC 62326-4-1, is agreed upon between the user and the supplier, the test shall be carried out on specimen A or B.
- c) The specimen shall preferably not be tin-lead plated. If it is, the plating shall be chemically removed prior to testing using a remover specified in 7.8.3 d), but care shall be taken to avoid any detrimental effect on the copper.

10.8.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A circulating air chamber capable of maintaining a uniform temperature of $(85 \pm 2) ^\circ\text{C}$.
- b) A thermal cycling chamber capable of cycling between $-70 ^\circ\text{C}$ and $+150 ^\circ\text{C}$ with a transfer time of 1 min or less.
- c) A milliohm meter with four-point probe with a measuring range of $1 \text{ M}\Omega$ to 1Ω .
- d) A tin-lead removal solution consisting of
 - 330 ml nitric acid 60 % (density, $1,36 \text{ g/cm}^3$ at $20 ^\circ\text{C}$),
 - 3 ml fluoroboric acid 40 % (density, $1,32 \text{ g/cm}^3$ at $20 ^\circ\text{C}$).
 - 670 ml deionized water.

10.8.4 Procedure

The following steps shall be taken:

The specimen shall be conditioned by drying in the circulating air chamber for 1 h at $(85 \pm 2) ^\circ\text{C}$ and cooled down to room temperature for testing.

The resistance of the plated-through holes connected in series shall be measured at a constant current of $(100 \pm 5) \text{ mA}$ using the four-point probe in the standard atmospheric conditions for measurement and test. The measured resistance (R_0) shall be recorded as the initial resistance.

The resistance shall be monitored continuously during the test. The specimen shall be connected to the recording device, for example by a suitable edge board connector.

The thermal cycling shall be carried out choosing one of the following conditions according to

the procurement document. The total number of cycles shall be specified.

Condition A (1 cycle):

–65 °C	-----	30 min.
Transfer	-----	3 $\frac{0}{-1}$ min.
+125 °C	-----	30 min.
Transfer	-----	3 $\frac{0}{-1}$ min.

Condition B (1 cycle):

–65 °C	-----	10 min.
Transfer	-----	1 min. (Max.)
+125 °C	-----	10 min.
Transfer	-----	1 min. (Max.)

The resistance value (or the corresponding voltage drop) is plotted on a graph against the number of cycles.

The increase of resistance in per cent (D) between R_0 and R_N (the measured resistance at an ambient temperature after N cycles) shall be calculated as follows:

$$D = \frac{R_N - R_0}{R_0} \times 100$$

where

R_N is the measured resistance after N thermal cycles;

R_0 is the measured resistance after zero cycles;

D is the resistance increase (in per cent).

The maximum permissible increase in per cent of the resistance between R_0 and R_N shall be specified in the relevant specification.

The increase in resistance shall also be measured at +125 °C and calculated according to the following formula:

$$D_{125} = \frac{R_{125N} - R_0}{R_0} \times 100$$

where

R_{125N} is the measured resistance at 125 °C after N thermal cycles;

R_0 is the measured resistance after zero cycles;

D_{125} is the resistance increase (in per cent) at 125 °C.

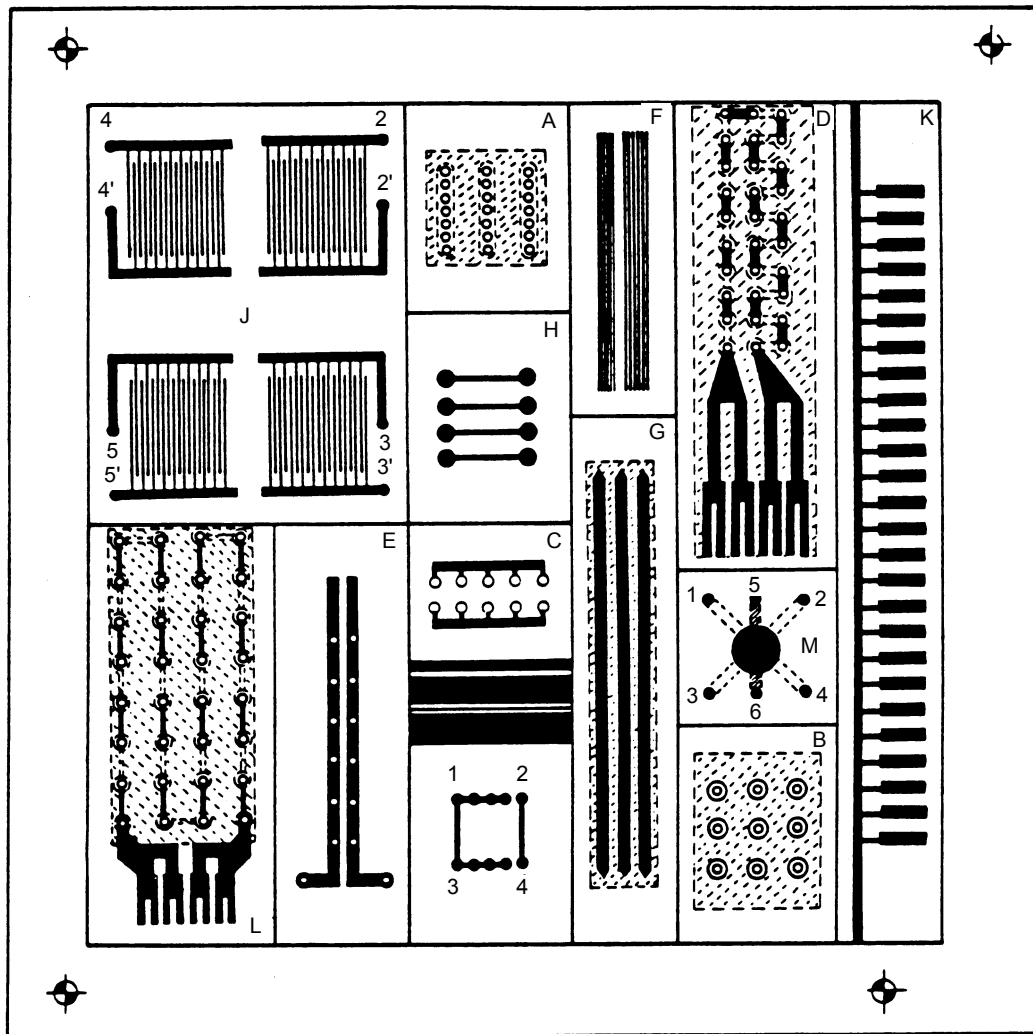
10.8.5 Report

The report shall include:

- a) the test method number and revision;
- b) the identification of the material tested;
- c) the thermal cycling condition (A or B);
- d) the number of cycles;
- e) the maximum increase in per cent of resistance measured;
- f) the maximum increase in per cent of resistance measured at 125 °C;
- g) the date of the test;
- h) any deviation from the test method;
- i) the name of the person conducting the test.

10.8.6 Additional information

When using the tin-lead removal solution, it shall be handled with care, avoiding contact with eyes and skin by wearing protective glasses and chemically resistant gloves respectively.



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Figure 15a is given to identify the locations of the specimens in the composite test pattern; it does not show the pattern of layer L1.

Cross-hatched areas shown on specimens A, B, D, G, L are X-layer patterns.

Figure 15a – Location of test specimens

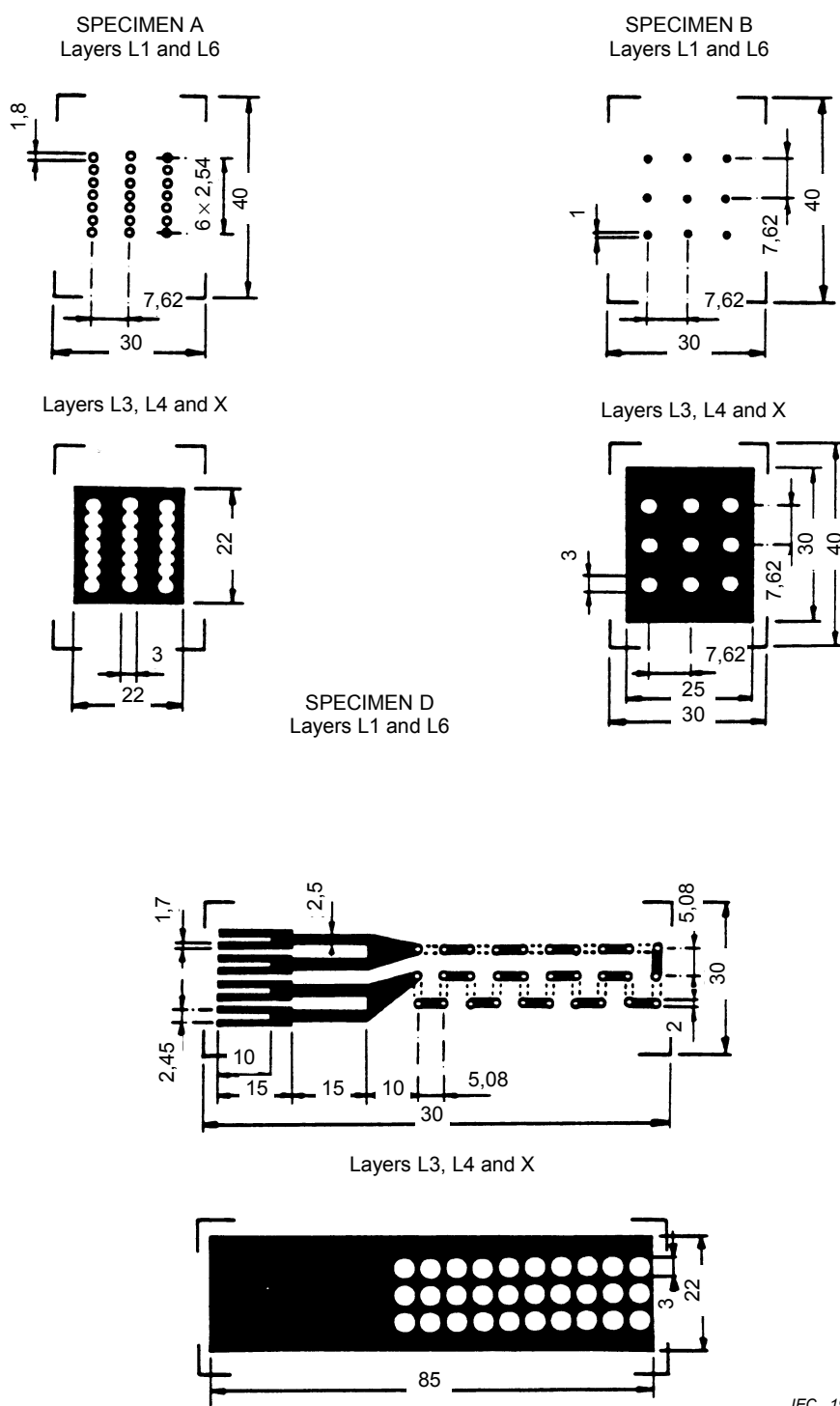


Figure 15b – Location of test specimens

Figure 15 – Composite test pattern

10.9 Test 3E09: Voltage proof, surface layers

10.9.1 Object

To determine the ability of printed boards to withstand specified test voltages without any disruptive discharges as evidenced by flashover (surface discharge), sparkover (air discharge) or breakdown (puncture discharge). The discharge may be visually observed or indicated by the test equipment in an appropriate manner.

10.9.2 Test specimen

The test shall be carried out on specified parts of a pattern on the surface of a printed board. When parts on a surface layer of a multilayer printed board are specified for testing, care shall be taken to avoid the influence of other parts or layers.

The specimen shall be handled carefully in order to avoid any contamination, for example fingerprints, dust, etc.

10.9.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) The following test apparatus and materials shall be used: a d.c. voltage supply or an a.c. peak voltage supply of approximately sinusoidal waveform and a frequency of 40 Hz to 60 Hz.
- b) The equipment shall be capable of supplying the necessary high voltage. It shall indicate the occurrence of disruptive discharge and/or specified leakage current in case of the failure not being visible.

10.9.4 Procedure

The following steps shall be taken:

10.9.4.1 Conditioning

The specimen shall be stored under standard atmospheric conditions for measurements and test for 24 h.

10.9.4.2 Testing

Apply the voltage between the specified test points. Increase the voltage gradually from zero during 5 s up to the specified value. Maintain the value for 1 min.

10.9.5 Report

The report shall include:

- a) test method number and revision;
- b) the identification of the test material;
- c) the points of application;
- d) the test voltage;
- e) the maximum leakage current;
- f) the date of the test;
- g) any deviation from this test method;
- h) the name of the person that conducted the test.

10.9.6 Additional information

None.

10.10 Test 3E10: Voltage proof between layers

10.10.1 Object

To determine the ability of a printed board to withstand specified test voltages without any disruptive discharges as indicated by the test equipment.

10.10.2 Test specimen

The test shall be carried out on specified parts of a pattern on adjacent layers of a printed board.

The specimen shall be handled carefully in order to avoid any contamination, for example fingerprints, dust, etc.

10.10.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a d.c. voltage supply or an a.c. peak voltage supply of approximately sinusoidal waveform and a frequency of 40 Hz to 60 Hz.

The equipment shall be capable of supplying the necessary high voltage. It shall indicate the occurrence of disruptive discharge and/or specified leakage current in case of the failure not being visible.

10.10.4 Procedure

The following steps shall be taken:

10.10.4.1 Conditioning

The specimen shall be stored under standard atmospheric conditions for measurement and tests for 24 h.

10.10.4.2 Testing

Apply the voltage between the specified test points. Increase the voltage gradually during 5 s up to the specified value. Maintain the value for 1 min.

10.10.5 Report

The report shall include:

- a) the test method number and revision;
- b) the identification of the test material;
- c) the points of application;
- d) the test voltage;
- e) the maximum leakage current;
- f) the date of the test;
- g) any deviation from this test method;
- h) the name of the person conducting the test.

10.10.6 Additional information

None.

10.11 Test 3E11: Insulation resistance, multilayer printed wiring board (within a layer)

10.11.1 Object

This test is a method to determine the electrical resistance of the dielectric material between conductors (including plated through holes) within a specific layer of a multilayer printed wiring board during and after cycling of temperature change.

NOTE For the purposes of this standard, the terms “wiring board” and “circuit board” carry the same meaning.

10.11.2 Test specimen

Test coupon “J” from the test pattern described in Figure 15a, test method 3E08, shall be used..

10.11.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A test chamber capable of maintaining (65 – 10) °C and 95 % relative humidity.
- b) A megohm meter capable of delivering 100 V dc and of reading from 1 Ω to 3×10^{11} Ω.

10.11.4 Procedure

The following steps shall be taken:

10.11.4.1 Preparation

- a) Mount specimens by normal mounting means in the normal mounting position of the production boards being represented by the test coupons.
- b) Prior to cycling, leave the specimens in the ambient conditions for 30 min and measure and record the results as the initial resistance measurements.
- c) Subject each specimen to 10 cycles of 30 min at the test condition and 30 min in ambient condition.

10.11.4.2 Test

- a) Take measurements after the fifth and tenth cycles. Electrification must be 100 % ± 10 % for 1 min.
- b) Measurements should also be taken 15 min after removal from the chamber.
- c) Check points are 2&2', 3&3', 4&4' and 5&5' of the test coupons.

10.11.5 Report

The report shall include:

- a) the test method number and revision;
- b) the test date;
- c) parts of the pattern to be measured;
- d) test voltage;
- e) the result of measurement (average and minimum values);
- f) any deviation from this standard test method;

g) the name of the person that conducted the test.

10.11.6 Additional information

Use d.c. voltage for high performance requirements, and use a.c. voltage for general performance requirements.

10.12 Test 3E12: Resistance of conductors

10.12.1 Object

The purpose of this test method is to provide a procedure for determining the resistance of conductors of a printed board with plated through holes.

10.12.2 Test specimen

The specimen shall be a suitable printed board having conductor patterns for test. These conductor patterns shall be as long and narrow as possible. A minimum of three conductors shall be tested per specimen.

10.12.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a four-terminal milliohm meter with an inaccuracy of 1 % or better.

10.12.4 Procedure

The following steps shall be taken:

- a) The resistance of a specified conductor shall be measured using the milliohm meter by contacting measuring probes to the plated through holes connected to the both ends of the conductor.
- b) The measuring error shall not be greater than 5 %. The current shall be kept small enough to avoid heating the specimen appreciably.
- c) Calculate the conductance of the conductor from width, thickness, length and resistance measured of the conductor.

10.12.5 Report

The report shall include:

- a) the test method number and revision;
- b) conductor width and length measured;
- c) layer number;
- d) number of conductors measured;
- e) mean, maximum and minimum conductance of conductors in accordance with the relevant specification;
- f) date of test;
- g) any deviation from this method;
- h) the name of the person that conducted the test.

10.12.6 Additional information

None.

10.13 Test 3E13: Resistance test, plated through holes

10.13.1 Object

To determine the change in resistance of plated-through holes after being subjected to repeated thermal cycling conditions.

10.13.2 Test specimen

The test specimen shall be made using identical laminate to that of the finished product, and shall conform to all details of Figure 16.

10.13.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Circulating air chamber – capable of maintaining a uniform temperature of 135 °C to 150 °C.
- b) mΩ meter – Incorporating use of four-point probe measurement technique, with a range of at least 2 mΩ to 300 mΩ.
- c) Thermal cycling chamber – Either an air-to-air system or a liquid-to-liquid system, capable of cycling between –75 °C and +200 °C and a transfer time of 1 min or less.
- d) Desiccator.

10.13.4 Procedure

The following steps shall be taken:

10.13.4.1 Pre-treatment

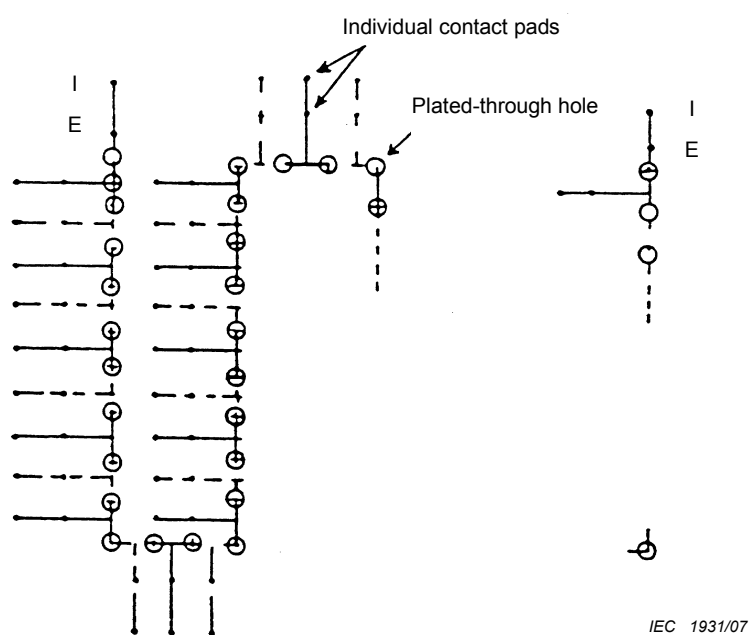
Specimens shall be conditioned by drying in an oven for 4 h at 80 °C to 120 °C and cooled to room temperature in a desiccator.

10.13.4.2 Resistance measurement

Using the four point milliohm meter, connect the current probes to the “I” contact pads at each end of the daisy chain pattern. Connect the voltage measurement probes to the “e” contact pads at the end of the chain. Record the total chain resistance in the appropriate “Initial resistance” record to a chart, as shown in Table 3.

Table 3 – Resistance values

Entry	Number of cycles	Total (mΩ)
1	Initial	
2	Cycle 10	
3	Cycle 20	
4	Cycle 30	
5	Cycle 40	

**Specifications:**

- Daisy chain to consist of 100 holes, 1,0 mm(0,040") diameter.
- Holes to be on 2,54 mm (0,100") centres.
- Midway between each hole pair, conductor traces are to be provided that connect on two, isolated contact pads, also on 2,54 mm (0,100").
- Conductors traces between hole pairs are to alternate from one hole side of the board to the other in order to generate a continuous chain.
- Recommended pad size around plated-through hole to be 1,9 mm (0,075") diameter.

Figure 16 – Test specimen artwork**10.13.4.3 Recommended temperature cycle**

Recommended temperature cycle for the measurement:

–65 °C 10 min;

Transfer time 1 min (max.):

+125 °C 10 min;

Transfer time 1 min (max.):

(Repeat).

10.13.4.4 Actual procedure

- a) Load parts into chamber and allow to cycle for 10 cycles.
- b) After the temperature cycle test, leave the chamber to return to the ambient temperature, and then remove the specimens from the chamber.
- c) Apply voltage probes to pads immediately adjacent to the holes.
- d) Perform resistance measurement with milliohm meter and record value.
- e) Repeat resistance measurement of 10.13.4.1, recording total chain resistance on chart shown in Table 8.

10.13.4.5 Plated through hole resistance measurement

In the event it is desired that the resistance of an individual hole pair measured, use the same milliohm meter as follows:

Using four contact pads provided in artwork as in Figure 16, select two pads on each side of a pair of holes.

Apply current probes to pads furthest from the holes.

10.13.5 Report

The report shall include:

- a) the test method number and revision;
- b) resistance measured;
- c) date of test;
- d) any deviation from this method;
- e) the name of the person that conducted the test.
- f) the name of the person that conducted the test.

10.13.6 Additional information

None.

10.14 Test 3E14: Current proof, plated through hole (under consideration)

10.15 Test 3E15: Current proof, conductors (under consideration)

10.16 Test 3E16: Plated-through hole and interconnection resistance change, thermal shock

10.16.1 Object

The purpose of this test method is to provide a procedure for determining the change in resistance of plated-through holes, including their connections with the internal layers, when the holes are subjected to a thermal shock. The test is also applicable to holes from which the interconnection is achieved with other technologies than the plated-through hole process.

10.16.2 Test specimen

- a) The specimen shall be a suitable printed board, a test coupon or a composite test coupon or a composite test coupon having a number of plating-through holes connected in series.
- b) Where the use of test coupons as specified in IEC 62326-4-1 is agreed, the test shall be

carried out on “specimen D” or on “specimen L” as specified in IEC 62326-4 (see Figures 26, 27 and 28).

- c) The specimen shall not be coated with tin-lead. If it is, the plating shall be chemically removed prior to testing using the stripping solution specified in 10.16.4 e, but care shall be taken to avoid any detrimental effect on the copper.

10.16.3 Test apparatus and materials

The following test apparatus and materials shall be used:

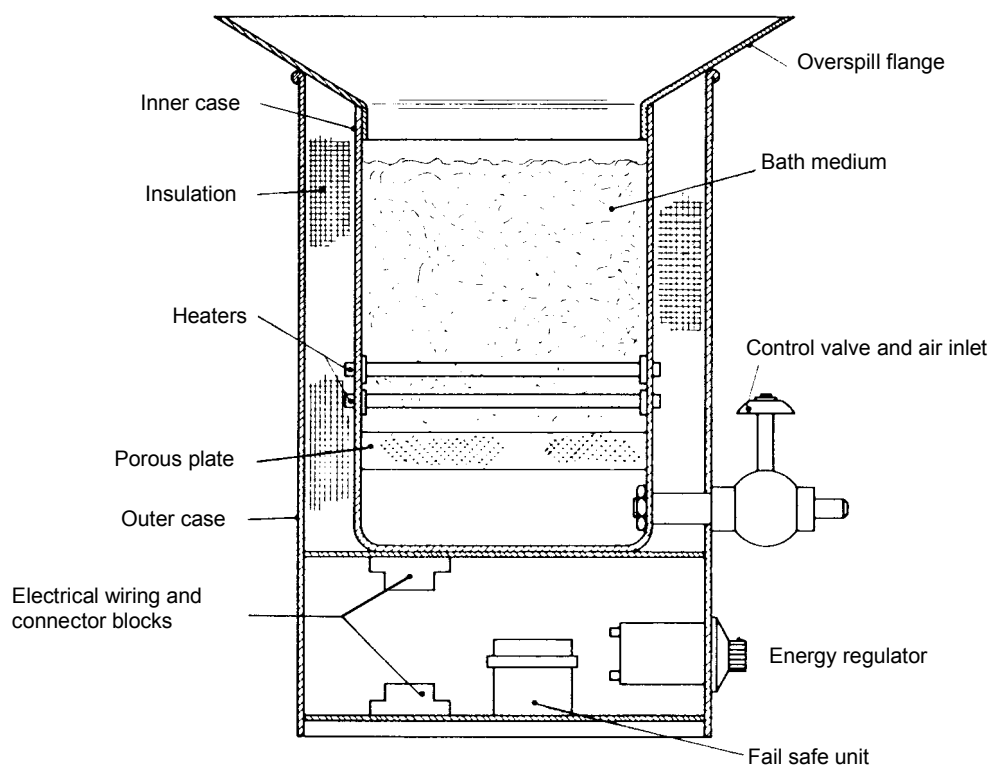
- a) Circulating air chamber capable of maintaining a uniform temperature of $(85 \pm 2) ^\circ\text{C}$.
- b) Fluidized sand bath of suitable design for example, as presented in Figure 17 kept at $(260^{+5}_{-0}) ^\circ\text{C}$ throughout the test.
- c) Milliohm meter with four-point probe capable of measuring at least $1 \text{ m}\Omega$ to 1Ω .
- d) Tin-lead removal solution consisting of
 - 330 ml nitric acid (density $1,36 \text{ g/cm}^3$ at $20 ^\circ\text{C}$),
 - 3 ml fluoboric acid 40 % (density $1,32 \text{ g/cm}^3$ at $20 ^\circ\text{C}$),
 - 670 ml deionized water.

10.16.4 Procedure

The following steps shall be taken:

- a) The specimen shall be conditioned by drying in the circulating air chamber for one hour at $(85 \pm 2) ^\circ\text{C}$ and cooled to room temperature.
- b) The resistance (or the corresponding voltage drop) of the plated-through holes connected in series shall be measured at a constant measuring current of $(100 \pm 5) \text{ mA}$ using the four-point $\text{m}\Omega$ meter. The specimen shall be connected to the recording device, for example, by a suitable edge-board connector. Before the first immersion into the sand bath, the measured resistance (R_0) shall be recorded as the initial resistance.
- c) The resistance shall be monitored continuously during the test. The specimen shall be inserted in a holder that includes the specimen totally and has a heat capacity so low that the temperature of the bath is not brought below $260 ^\circ\text{C}$.
- d) The specimen shall be immersed for the time given in the relevant specification which will also specify the total number of cycles (to take into account the type of base material, and the printed board application) “C”. After removal from the bath, the specimen shall be allowed to cool down to between $15 ^\circ\text{C}$ and $35 ^\circ\text{C}$.
- e) The resistance value (or the corresponding voltage drop) is plotted on a time scale against the number of immersions. The increase of resistance in per cent “D” between R_0 and R_η (the measured resistance at a defined temperature after N cycles) shall be calculated as follows:

$$D = \frac{R_\eta - R_0}{R_0} \times 100 \%$$



IEC 2680/05

Principles of operation

Small solid particles can be readily “fluidized” by means of a suitable gas (air) stream. A diagrammatic cross-section of a fluidized sand bath is shown above. Clean, dry air at a constant pressure of about 2 N/cm² from a pump or from an air line is supplied via a control valve to a chamber beneath the diffuser (porous plate). This porous plate ensures a uniform flow of air across the full section of the container and acts also as a support plate for the solid sand bed.

As the control valve is slowly opened, the solid sand bed remains undisturbed and the air finds its way between the particles; under such conditions the pressure drop is proportional to the rate of flow of air. As the valve is opened further, the air drag on the particles will cause them to separate and the whole mass of the bed can be seen to have expanded. The bed now behaves as a fluid and is said to be “fluidized”. Further opening of the valve is not accompanied by an increase in pressure drop, which remains constant at a value corresponding to the head of the column of particles, but the bed becomes more turbulent and will have the appearance of boiling liquid. The best heat transfer and most uniform temperatures are obtained when the bath is in this “boiling” state.

Figure 17 – Fluidized sand bath

10.16.5 Report

The report shall include:

- a) the test method number and revision;
- b) the identification of the tested material;
- c) the number of immersions;
- d) the results
 - the maximum increase in per cent of resistance between the first and last immersion at 25 °C,
 - the maximum increase in per cent of resistance between the first and the last 260 °C immersion,
 - the maximum increase in per cent of resistance during any one 260 °C immersion;
- e) date of test;
- f) the name of the person that conducted the test.

10.16.6 Additional information

When using the tin-lead removal solution, it shall be handled with care, avoiding contact with eyes and skin by wearing protective glasses and chemically resistant gloves respectively.

10.17 Test 3E17: Production determination of characteristic impedance by TDR

10.17.1 Object

This test method details a technique for measuring the characteristic impedance (Z_0) of conductor configurations on unpopulated printed boards which are intended to perform as transmission lines.

The only practical technique for the accurate measurement of Z_0 is time domain reflectometry (TDR). A high level technique for suitably trained electronics engineers using expensive test equipment is detailed in Test 3E07 of this publication series.

This test method details an alternative technique more suited to printed board fabricators which still retains adequate accuracy. This technique has been benchmarked against the 3E07 method. Correlation was within the order of repeatability for the measurement.

This test method is suitable for both “stripline” and “microstrip” configurations, but is for single needed TDR measurements only. Differential impedance measurement techniques are not detailed herein.

Time domain reflectometry uses a fast step output pulse generator combined with a high bandwidth sampling oscilloscope. The step pulse is applied to the sampled printed board conductor (transmission line) and the reflected voltage is monitored by the sampling oscilloscope. Changes in Z_0 cause reflections in voltage which can be used to calculate the impedance of the transmission line at any given distance from the TDR. Preferably, an average measure over a selected effective length should be determined.

The use, limitations and applicability of TDR test methodology is fully discussed in Clause 9 of IEC 61188-1-2 which should be read before attempting impedance measurements.

10.17.2 Test specimen

- a) The test specimen shall be as detailed in Figure 35, or may comprise suitable conductor configurations on a production board. A minimum “line” length of 150 mm is essential. The

layout given in Figure 35 can be customized to produce a variety of stripline and microstrip configurations.

- b) Test specimens for IECQ approval are well defined. Specimens should not be subjected to special preconditioning or storage conditions, unless specifically requested by the customer.
- c) Some TDR instruments are prone to damage from electrostatic discharges. This damage can be subtle and insidious; difficult to detect and costly to remedy, and responsible for incorrect readings. Suitable precautions should therefore be undertaken when handling the specimens.

10.17.3 Test apparatus and materials

To improve measurement repeatability it is desirable that the instrumentation is automated as far as possible.

Combinations of measurement instrumentation and computer hardware and software are available which carry out suitable automatic system set-up, calibration and measurement routines. Manual equipment is acceptable though the set-up time should be added, and cost and risk of operator error may result.

TDR instrumentation for measuring suitable standard printed board conductors of 150 mm long test coupons, should consist of a combination of a pulse step generator and a sampling measurement system with the following minimum specifications:

10.17.3.1 Pulse step generator

- a) internal 50 Ω , 1 % termination resistance;
- b) 200 ps maximum rise time reflected into the instrument input sampling port;
- c) minimum step amplitude of 200 mV when connected to a 50 Ω load, or greater than 400 mV step amplitude into an open circuit termination;
- d) pulse step aberrations that are: less than 3 % between 10 ns and 35 ps before the step transition; within the range of +10 % and –5 % from 35 ps before the step transition of 500 ps after the step transition; and less than 3 % between 500 ps and 10 ns after the step transition;
- e) offset drift of the step of less than 0,25 % of step pulse amplitude over one calibration period;
- f) amplitude drift of the step of less than 0,25 % of step pulse amplitude over one calibration period.

10.17.3.2 Sampling measurement system

- a) bandwidth of at least 1,75 GHz when measuring 150 mm length test conductors using a 200 ps step pulse generator;
- b) horizontal time base accuracy of at least 25 ps + 0,1 % of the horizontal time scale interval;
- c) horizontal time resolution of at least 25 ps and a display in distance units of millimetres;
- d) vertical voltage interval of at least twice the step voltage;
- e) vertical display of reflection coefficient in milliohms, or impedance in ohms;
- f) vertical resolution of at least 0,1 % of the step voltage amplitude;
- g) less than 0,1 % measurement drift over one calibration interval.

10.17.3.3 Cables

Stable and accurate controlled impedance (50 Ω) high frequency low loss coaxial cable should

be used at all times. The outer conductor shield should be either solid or double screened to minimize radiated losses. The inner conductor should be solid silver coated copper to minimize skin effect and high frequency series resistance loss. Many types of dielectric material are suitable and again these should be chosen to improve stability and reduce losses. The cable assembly should be rugged to prevent wear and operational damage.

10.17.3.4 Connectors

Printed board straight mountable socket and straight cable end surface mount assembly (SMA) of 50 Ω screw-coupled plug coaxial connectors (18 GHz) shall be used as described in IEC 60169-15.

10.17.3.5 Probes

Probes for connecting to the test conductor and associated ground point shall be designed to minimize aberrations and impedance discontinuities. Printed board mounting surface mount assembly connectors are commonly used in standard pitch testing applications, but printed board microstrip probe assemblies can accommodate custom signal to ground pitch spacings and often provide for a more consistent test connection.

10.17.4 Procedure

The following steps shall be taken:

10.17.4.1 Calibration

- a) Where manual measurements are undertaken, the equipment shall be calibrated prior to use, and in accordance with the equipment supplier's instructions. Recalibrations should be undertaken whenever the test interconnections, other than the probe to specimen interface, are disturbed.
- b) Software driven systems may include a self-calibration cycle.

10.17.4.2 Measurement

Figure 18 shows a possible test set-up. In some cases, the reference impedance standard may be within the test instrument.

The method of operation will depend upon equipment choice, and suppliers' instructions must be strictly followed. It is not possible to give individual details in a generic test method.

General handling considerations are given in 10.17.2 c).

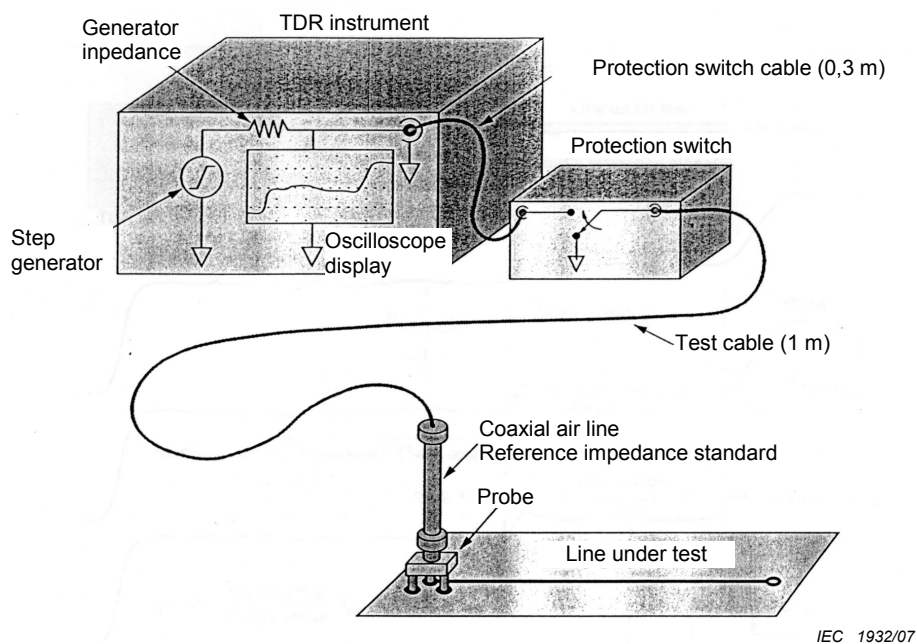


Figure 18 – Possible equipment configuration

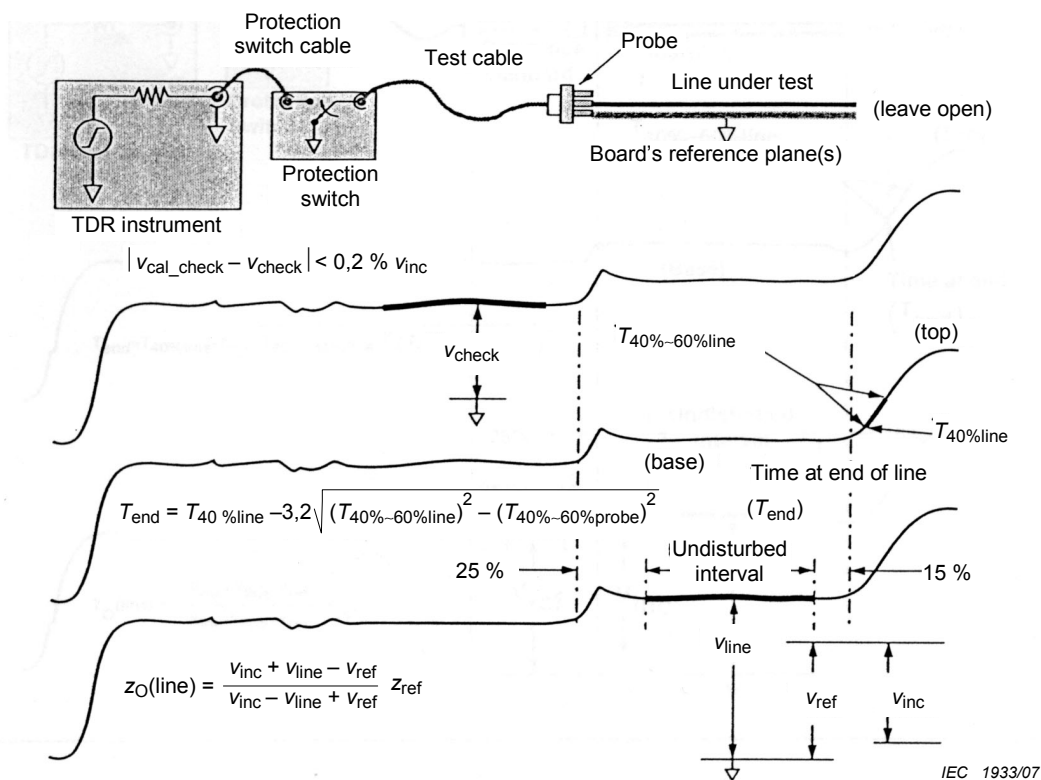


Figure 19 – Schematic showing undisturbed interval

10.17.4.3 Calculation

- The Z_0 value for the conductor configuration under test is based upon the mean voltage value in the test waveform along the undisturbed interval of the test line as shown in Figure 19 or Figure 20.

$$b) Z_0 = Z_{\text{ref}} (V_{\text{inc}} + V_{\text{line}} - V_{\text{ref}}) / (V_{\text{inc}} - V_{\text{line}} + V_{\text{ref}})$$

where

Z_{ref} is the characteristic impedance of the reference impedance standard. This shall normally be 50 Ω ;

V_{line} is the measured mean voltage along the undisturbed region of the line under test;

V_{inc} is the voltage of the incident wave front. This is obtained by disconnecting the cable from the reference impedance standard, and then replacing it with the probe to connector termination as described here. From the resultant waveform it is possible to measure the amplitude of the voltage step between the two forms of termination. The time range for this measurement, as shown in Figure 21, starts at the beginning of the reference impedance standard and extends for two times the reference standard's delay. The voltage amplitude, or V_{inc} , is the difference of the voltage modes of the two forms of termination.

- c) Automated systems may be capable of performing the above calculation, with a visual or printed output in a tabulated and/or graphical format.

10.17.4.4 Specimen handling considerations

10.17.4.4.1 Probe to specimen interconnection

Experiments have shown that the variation in measured impedance due to probe and connector twist, the change of the angle or the change of the depth are individually in the order of 0,2 % and can therefore be ignored. However, variation in probe pressure can cause significantly the changes of high and unstable readings. The probing mechanism should therefore take these, and the following, considerations in account.

10.17.4.4.2 Reversed signal and ground connections

Grossly inaccurate results will be produced if signal and ground connections are reversed. Typically, errors may be in the magnitude of tens of ohms and should be immediately apparent to the experienced operator. This type of error can be eliminated by the use of the standard test specimen given in Figure 22.

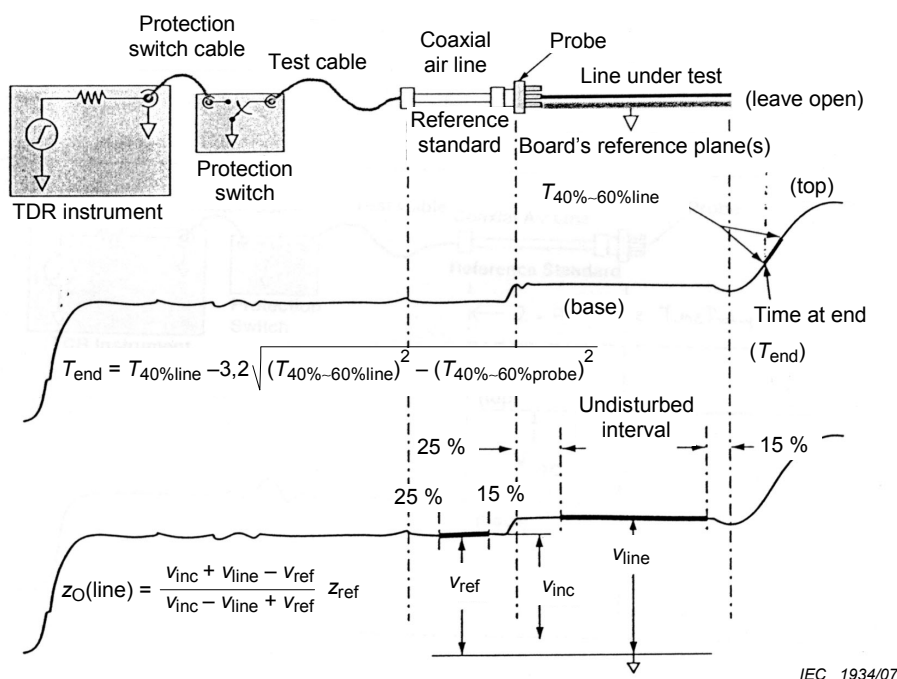


Figure 20 – Test wave form example

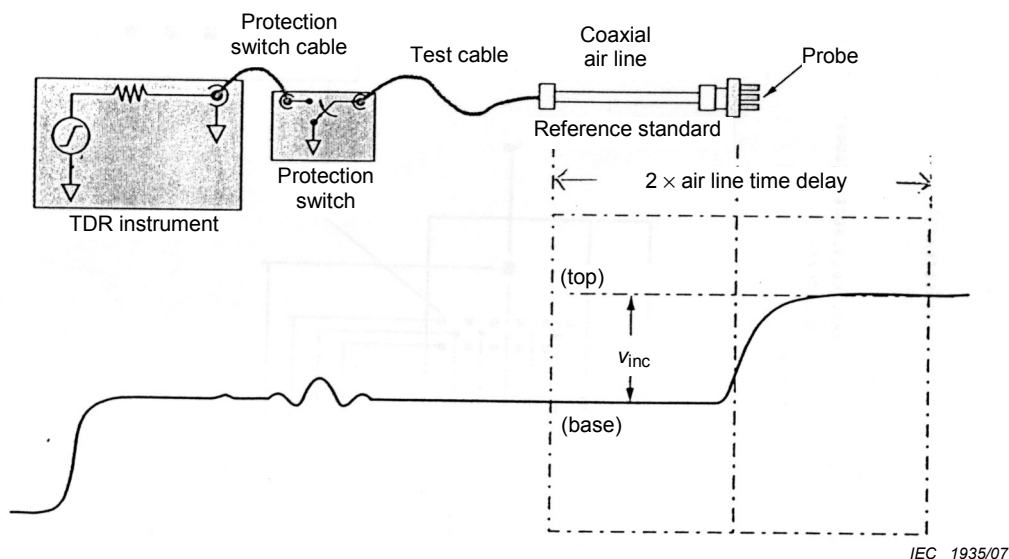


Figure 21 – Incident wave voltage showing (2 X) air line delay

10.17.5 Grounding of stripline ground/power planes

The reference plane for a single or dual stripline configuration should be that plane which is most adjacent to the conductor(s) under test. The choice of other planes within the printed board will produce variation in the results. It is recommended that all ground connections on the test coupon should be connected to the ground plane.

10.17.6 Influence of unwanted contact with the test coupon

- In the case of a microstrip structure, finger contact with the test coupon surface by the operator can result in significant variations in the measured impedance. This impedance

discontinuity can be clearly seen in the form of a distinct dip in the displayed impedance trace. Finger contact creates a localized increase in the capacitive loading. This causes a corresponding localized reduction in the impedance value.

- b) Similarly, all impedance structures are susceptible to large variations in measured impedance if the test point is touched by the operator during test. Microstrip structures will be affected more than stripline configurations.
- c) The impedance of a microstrip structure with conductors on the lower outer surface of the specimen-printed board under test may be influenced by the dielectric properties of the material upon which they are placed. It is recommended that suitable fixtures be used to eliminate this effect by keeping the position of measuring elects stable.

The more complex equipment available is susceptible to electrostatic damage, so the operator is obliged to use suitable matting. However, supporting the specimen in air is preferred where equipment constraints permit.

10.17.7 Variation in ambient temperature

A variation of 5 °C in ambient temperature will, in the case of a specimen constructed using glass epoxide substrates, result in a variation in the order of 0,6 %.

10.17.8 Report

The report shall include the following:

- a) test method number and revision;
- b) identification and description of specimens,
- c) number of specimens tested;
- d) identification of test equipment;
- e) calibration data or status;
- f) any deviation from this method;
- g) the name of the person that conducted the test;
- h) test results, given in ohms.

Graphical or tabulated results may be included, provided that the impedance values are clear, or are separately summarized.

10.17.9 Additional information

- a) Complex manual TDR systems are set-up dominant, and require a highly skilled operator. Such systems undoubtedly provide excellent results at a high cost. However, the high cost relates to the total capability of the instrumentation. In many cases, the measurement of characteristic impedance may be considered to be a small fraction of the total capability of the instrument. As such, they are used by specialist laboratories and manufacturers.
- b) Lower cost systems are available that are adequate for the measurement of impedance structures in printed boards. Although a lower operator skill level is required, the limitations of the equipment should be understood. These systems are acceptable as a process control and batch release tool, and as decision of batch release.
- c) The target measurement uncertainty for the measurement method described in this text, when conducted in accordance with this test method, is 1,5 %.

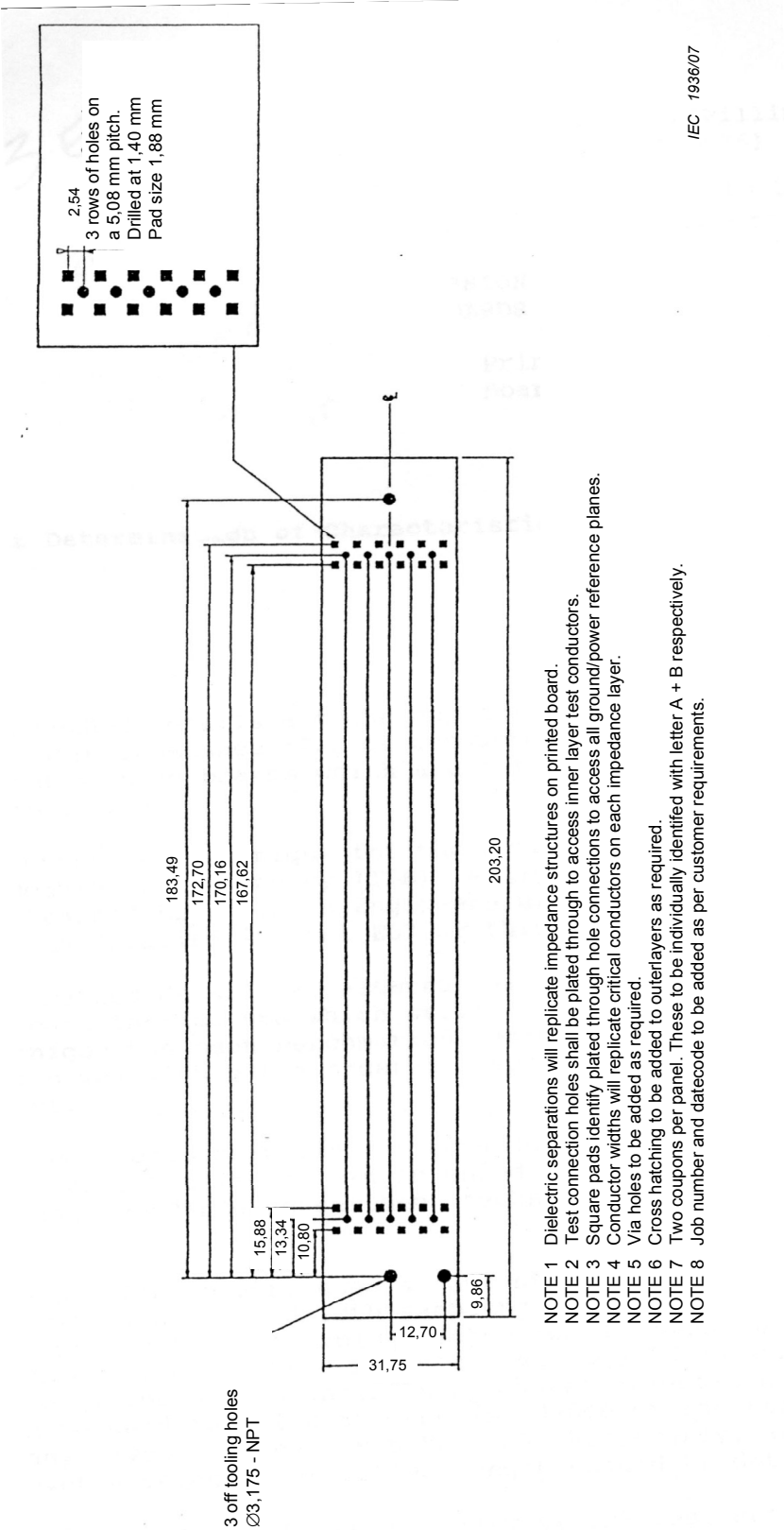


Figure 22 – Details of test specimen

10.18 Test method 3E18: Contact resistance of printed board keypad contacts

10.18.1 Object

This test method details the procedure for measuring the electrical contact resistance between the separate halves of keypad contacts on printed boards bridged via a designated conductive test probe.

The surface finish of the keypad will usually be a conductive ink, although the same procedure may be used for precious metal plated finishes.

10.18.2 Test specimen

- Either one of the keypad patterns shown in Figure 24 shall be used.
- Alternative patterns may be used, as agreed between customer and supplier.
- A minimum of five measurements shall be conducted.

10.18.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- Gold plated cylindrical bridging probe with a flat mating surface, having a diameter of between 3 mm and 5 mm. This may be substituted by a carbon loaded elastomeric pill as agreed between customer and supplier.
- A measurement system, as shown in Figure 23 for its wiring diagram (preferably a resistance bridge), with a range of 0 Ω to 1 000 Ω . A four wire measurement system is preferred. The technique used shall have an uncertainty of measurement of no greater than $\pm 5\%$.

10.18.4 Procedure

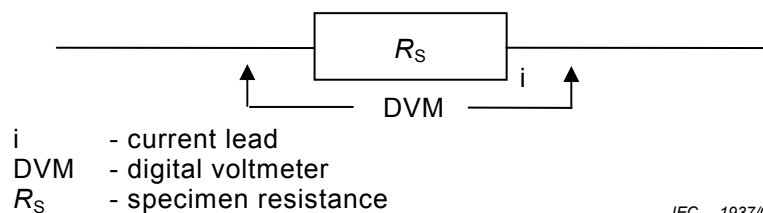
The following steps shall be taken:

The keypad shall be connected to the resistance bridge in a manner that reduces probe contact resistance (i.e. resistance other than that attributable to the contact system).

The electrical test parameters shall be agreed between the customer and supplier, e.g.:

- current (a.c., or d.c. current);
- voltage (open circuit);
- frequency (if other than d.c.).

The gold plated or elastomeric bridging probe, shall be applied with a force of $(3,5 \pm 0,5)$ N over each keypad, and the resistance measured.



IEC 1937/07

Figure 23 – Circuit diagram for measurement of contact resistance

10.18.5 Report

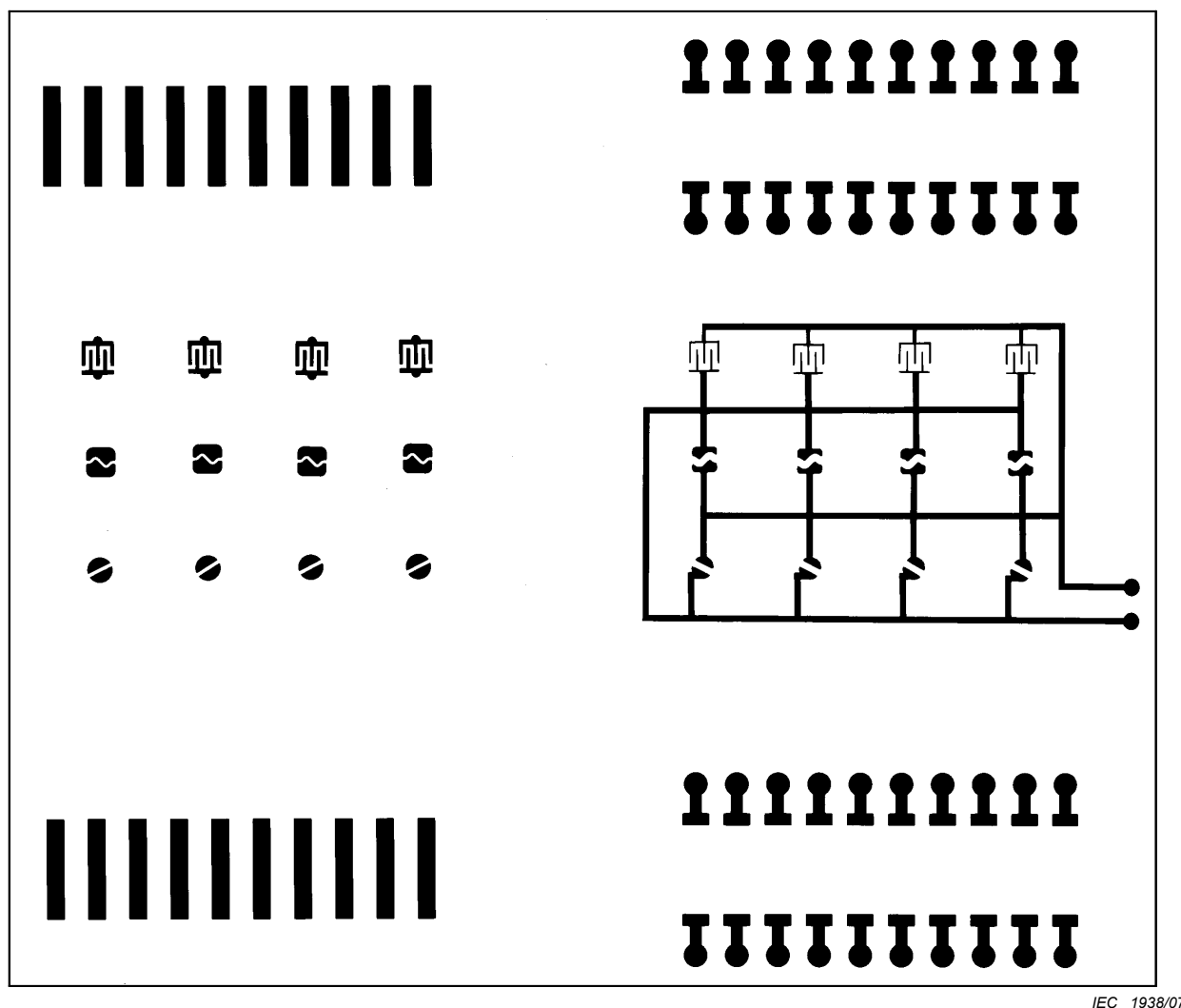
In addition to the general requirements for reporting, the report shall include:

- a) the test method number and revision;
- b) the date of the test;
- c) identification and description of the specimen(s);
- d) the specimen size;
- e) the measured values;
- f) the material of the test probe;
- g) the electrical test parameters;
- h) any deviation from this method
- i) name of the person that conducted the test.

10.18.6 Additional information

Proprietary equipments are available which will measure resistance in the required range, and with the required accuracy.

A four-wire resistance bridge may be easily constructed using a (digital) voltmeter and a constant current power supply. The use of a four-wire technique limits probe contact resistance to that length of conductor between the voltmeter probes.



NOTE These patterns are used to determine the area resistances and contact resistance. No keypad design recommendations are implied. Designs are for test purposes only.

Figure 24 – Keypad contact patterns

11 N: Environmental test methods

11.1 Test 3N01: Thermal shock, immersion, oil bath at 260 °C

11.1.1 Object

The purpose of this test is to determine the ability of a printed board with plated through holes (PTH) test specimen to withstand a specified thermal shock by immersion in a heated fluid. This test is also applicable to holes from which interconnection is achieved with technologies other than the plated through hole process.

11.1.2 Test specimen

Specimens shall be removed from the panel or test coupon by sawing, routing or an equivalent method. Allow sufficient clearance to prevent damage to the area to be tested.

Remove the specimen in such a manner that at least three of the smallest possible adjacent

size plated through holes can be viewed in a finished microsection.

As a referee, the specimen shall be preconditioned in an air circulation chamber at 125 °C for a time designated in the relevant specification, then allowed to cool down under standard atmospheric conditions until its temperature is below 35 °C. The recovery time shall not exceed 8 h.

11.1.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A heating container capable of heating fluid and keeping the temperature at 260^{+5}_0 °C.
- b) A silicone oil or an equivalent fluid.
- c) A thermometer for measuring the temperature of the fluid at a depth of 25 mm below the surface.
- d) A timer capable of resolving to ± 1 s.
- e) A holder with low heat capacity such that the temperature of the fluid is not brought below 260 °C.
- f) A circulating air chamber, capable of maintaining uniform temperature at (125 ± 5) °C.

11.1.4 Procedure

The following steps shall be taken:

The specimen shall be placed in the holder.

The specimen shall be held in a horizontal position and immersed to a depth of 25 mm in the heated fluid.

The specimen shall remain immersed in the fluid for the time given in the relevant specification.

The specimen shall be removed from the oil bath and allow to cool down to ambient temperature.

After cooling, the specimen shall be cleaned with propane 2-OL (isopropyl alcohol) and air dried.

The specimen shall then be microsectioned in accordance with test 3X09 of this standard and examined for conformance.

11.1.5 Report

The report shall include:

- a) test method and revision index;
- b) date of analysis;
- c) identification of the material tested;
- d) immersion time;
- e) results of the microsectional evaluation including but not limited to delamination, inter-connection, separation, through-hole cracks, lifted lands, and resin recession;
- f) any deviation from this test method.

11.1.6 Additional information

A suitable fluid shall have a self-ignition temperature above 300 °C, a decomposition temperature above 270 °C and conduction and oxidation resistance comparable to those of

methylphenyl-polysiloxane.

This test is also applicable as a precondition step prior to the evaluation of plated through hole resistance test method 3E16 of this standard.

Although reliability concerns are normally focused on small holes, large holes may still be a reliability concern even though the smallest holes have passed this test.

11.2 Test 3N02: Thermal shock, floating, solder bath

11.2.1 Object

To determine whether the construction integrity of a printed board has the ability to withstand a thermal shock delivered through exposure to a molten solder bath. This method is intended to simulate the assembly process, wave or drag soldering, where the heat exposure is mainly from one side.

11.2.2 Test specimen

Specimens shall be removed from the printed board, the panel, or test coupon by sawing, routing, or equivalent method. Allow sufficient clearance to prevent damage to the area to be tested.

Remove the specimen in such a manner that at least three of the smallest size through-holes can be viewed in a finished microsection.

11.2.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A solder pot of suitable design, electrically heated, thermostatically controlled and containing at least 1 kg of Sn60, Sn62, or Sn63. The test temperature shall be $(260 \pm 5) ^\circ\text{C}$.
- b) A thermometer for measuring the temperature of the solder at a depth of 25 mm below the surfaces.
- c) A timer with a resolution of ± 1 s or better.
- d) A low- or non-activated water soluble flux.

11.2.4 Procedure

The following steps shall be taken:

Coat the surface and the plated through-holes of the specimen with flux.

Remove the oxide from the solder, and float the specimen on the solder in a manner that only one side is in contact with the solder. The specimens shall be floated for the time specified in the relevant specification.

Remove the specimens from the solder pot and allow to cool to ambient temperature.

After cooling, the specimens shall be cleaned with isopropyl alcohol and air dried.

The specimens shall then be microsectioned in accordance with test method 3X09 and examined for conformance.

11.2.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of the test;
- c) the identification and description of the specimen(s);
- d) the results of the microsectional evaluation;
- e) detailed description of test specimen including, but not limited to preparation, size, quantity, and drying time/temperature;
- f) solder composition, and flux type;
- g) any deviation from this test method;
- h) the name of the person conducting the test.

11.2.6 Additional information

Evaluation criteria include, but are not limited to:

- delamination;
- interconnection separation;
- through-hole cracks;
- lifted lands;
- resin recession.

11.3 Test method 3N03: Thermal shock, hand soldering, through hole mounting

11.3.1 Object

This test method is designed to simulate the procedures for component removal and replacement, in order to determine the effects of rework on the quality and integrity of the plated through hole and conductors on rigid or flexible printed boards. The five test cycles are intended to simulate initial soldering, component replacement in original manufacturing and subsequent component replacement for purpose of equipment repair.

11.3.2 Test specimen

A standard test specimen (coupon) or a printed board with plated through holes and terminal areas which can be used for component mounting. The diameter of the holes to be tested shall be no smaller than would normally be used for through-hole component mounting. Via holes, or other small holes, which would normally be considered only as vias, shall not be used for this test.

NOTE This is a destructive test.

11.3.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) 60 W regulated soldering iron.
- b) Tin-plated copper wire, the diameter of the test wires shall be 0,25 mm to 0,70 mm smaller than the test holes.
- c) Soldering flux, conforming to IEC 61190-1-1.
- d) Solder pastes conforming to IEC 61190-1-2.

- e) Rosin flux, conforming to IEC 61190-1-1.
- f) Metallographic laboratory facilities, conforming to Test 3X09.
- g) Metallographic microscope, capable of up to 400× magnification.

11.3.4 Procedure

The following steps shall be taken:

- a) Condition the specimen at 23 °C, 40 % to 60 % relative humidity for 24 h, prior to tests.
- b) Insert test wires into plated through test holes, and solder by machine or by hand, as applicable. The wires shall not be clinched.
- c) A wire shall be unsoldered and re-soldered in the same holes by hand, five times following initial soldering. During each cycle, the wire shall be completely removed from the hole. If the same wire is used, it shall be allowed to cool to ambient temperature before being replaced in the test hole, and re-soldered. A new wire may be used for each re-soldering cycle.
- d) Operate the 60 W soldering iron at a voltage sufficient to produce a tip temperature of 232 °C to 260 °C.
- e) Apply the iron to the test wire and solder or unsolder normally. The iron may touch the land surrounding the hole, however, undue pressure and duration should be avoided to minimize damage to the plated through hole under test. The iron should be applied only as long as necessary to perform the soldering and unsoldering operation.

NOTE It is recommended that the soldering and unsoldering portion of this test be performed only by personnel trained and/or certified in soldering and soldering rework.

- f) Following the fifth cycle, prepare a microsection specimen as per test 3X09 and prepare for metallographic evaluation.

11.3.5 Report

The report shall include:

- a) test method number and revision;
- b) any deviation from this method;
- c) hole size(s) and wire size(s) used in the test;
- d) information regarding type(s) and measurements of plating;
- e) results of evaluation, including failure modes and degree of failure, in case of failure;
- f) the date of the test;
- g) the name of the person that conducted the test.

11.3.6 Additional information

This test is intended to simulate a strategy of repair and maintenance "in the field", away from the normal manufacturing environment. If a company or organization does not have a component level repair and maintenance strategy for delivered equipment, this may be an inappropriate test. This test was originally designed for situations such as an emergency military operation repair ("in the field"), but should not be considered for common module level field repair strategy. It should also be noted that test results can vary greatly, based upon the skill of the operator performing the solder and unsolder operations required in the test.

11.4 Test 3N04: Thermal shock, solder immersion at 260 °C

11.4.1 Object

The purpose of this test is to determine the ability of a printed board with plated through holes

(PTH) test specimen to withstand repeated thermal shocks by immersion in a bath of molten solder. This test is also applicable to holes from which interconnection is achieved with other technologies than the plated through hole process.

11.4.2 Test specimen

Specimens shall be removed from the panel or test coupon by sawing, routing or an equivalent method. Allow sufficient clearance to prevent damage to the area to be tested.

The specimen shall be removed in such a manner that at least three of the smallest possible adjacent size plated through holes can be viewed in a finished microsection.

As a referee the specimen shall be preconditioned in an air circulation chamber at 125 °C for a time designated in the relevant specification, then allowed to cool down under standard atmospheric conditions until its temperature is below 35 °C. The recovery time shall not exceed 8 h.

11.4.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A solder pot of suitable design, electrically heated, thermostatically controlled and filled to a depth of at least 75 mm. Test temperature shall be 260^{+5}_0 °C throughout the test.
- b) A thermometer for measuring the temperature of the solder at a depth of 25 mm below the surface.
- c) A timer capable of resolving to ± 1 s.
- d) A low- or non-activated water soluble flux or a resin of 0,2 % maximum chloride content.
- e) A suitable fixture (Figure 25).
- f) A circulating air chamber capable of maintaining uniform temperature at (125 ± 5) °C.

11.4.4 Procedure

The following steps shall be taken:

- a) The specimen and the wire shall be coated with the flux.
- b) The specimen and the wire shall be assembled into the fixture to maintain proper board and wire position.
- c) Any oxide shall be removed from the surface of the solder, and the specimen immersed to a depth of 25 mm in the molten solder. The immersion time shall be $(4 \pm 0,5)$ s.
- d) The specimen shall be removed from the solder pot and allowed to cool down to ambient temperature.
- e) A second immersion for $(4 \pm 0,5)$ s shall simulate the thermal shock of unsoldering the wire. After cooling, a third immersion will simulate the resoldering of the wire.
- f) The specimen shall be cleaned if required.
- g) The specimen shall then be microsectioned in accordance with test 3X09 of this standard and examined for conformance.

11.4.5 Report

The report shall include:

- a) test number and revision index;
- b) date of analysis;
- c) identification of the material tested;
- d) immersion time;
- e) results of the microsectional evaluation including but not limited to delamination, interconnection, separation, through-hole cracks, lifted lands, and resin recession;
- f) any deviation from this test method.

11.4.6 Additional information

The three immersions are the first soldering cycle. If more than one cycle has been performed, two immersions shall be added for each additional cycle. The total number of cycles shall be as specified in the relevant specification.

Although reliability concerns are normally focused on small holes, large holes may still be a reliability concern even though the smallest holes have passed this test.

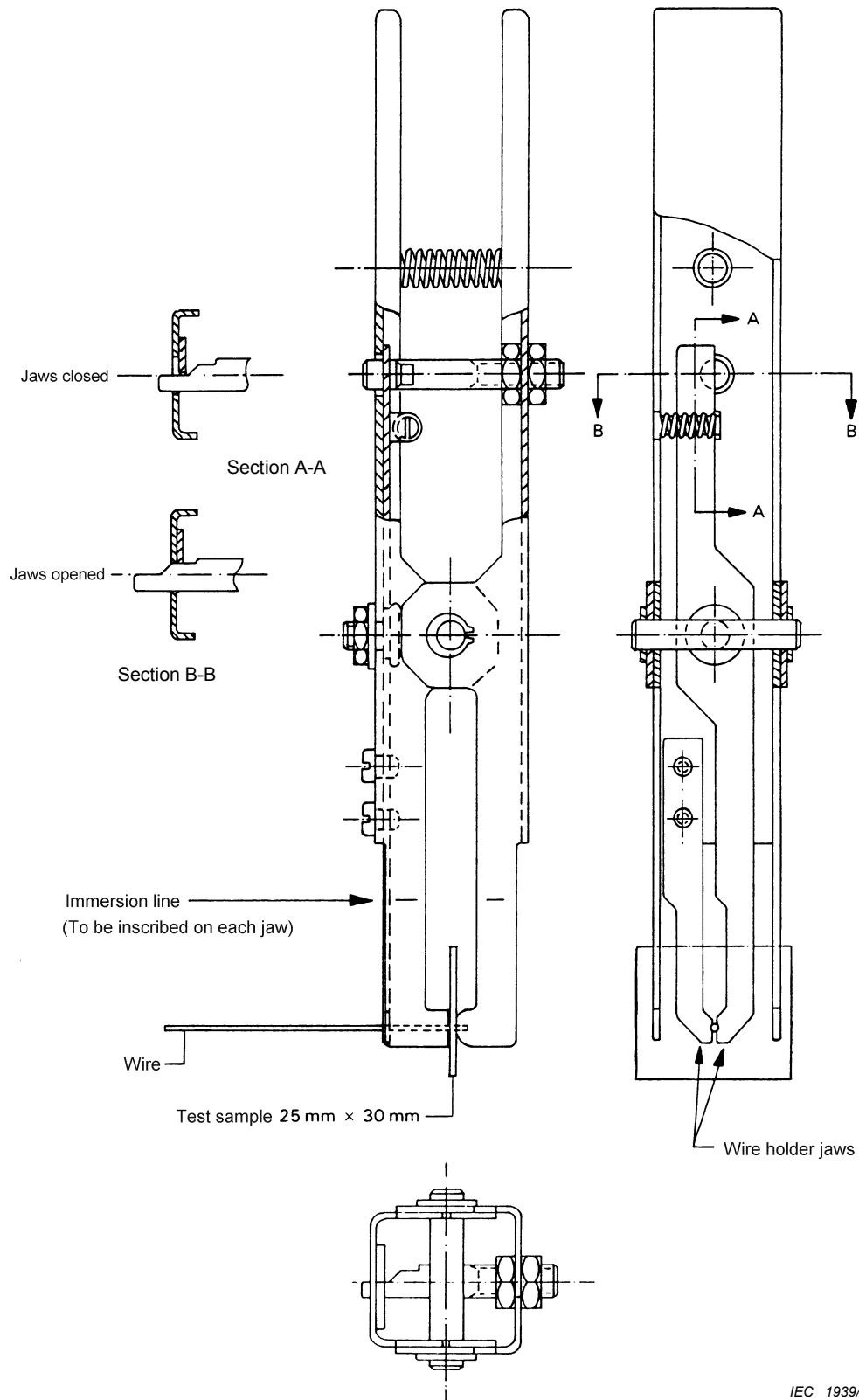


Figure 25 – Plier fixture for thermal shock test, dip soldering

11.5 Test 3N05: Thermal shock, floating, solder bath 288 °C

11.5.1 Object

The purpose of this test is to determine the ability of a printed board with plated through holes (PTH) test specimen to withstand a specified thermal shock by floating the specimen on molten solder. This test is also applicable to holes from which interconnection is achieved with other technologies than the plated through hole process.

11.5.2 Test specimen

Specimens shall be removed from the panel or test coupon by sawing, routing, or an equivalent method. Allow sufficient clearance to prevent damage to the area to be tested.

The specimen shall be removed in such a manner that at least three of the smallest possible adjacent size plated through holes can be viewed in a finished microsection.

As a referee, the specimen shall be preconditioned in an air circulation chamber at 125 °C for a time designated in the relevant specification, then allowed to cool down under standard atmospheric conditions until its temperature is below 35 °C. The recovery time shall not exceed 8 h.

11.5.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A circulating air chamber, capable of maintaining uniform temperature at $(125 \pm 5) ^\circ\text{C}$.
- b) A solder pot of suitable design, electrically heated, thermostatically controlled and containing at least 1 kg of Sn60, Sn62 or Sn63. Test temperature shall be $(288 \pm 3) ^\circ\text{C}$.
- c) A thermometer for measuring the temperature of the solder at a depth of 25 mm below the surface.
- d) A timer capable of resolving to ± 1 s.
- e) Low- or non-activated water soluble flux or rosin of 0,2 % maximum chloride content.

11.5.4 Procedure

The following steps shall be taken:

The surface and plated through holes of the specimen shall be coated with flux.

Any oxide shall be removed from the surface of the solder and the specimen shall be floated on the molten solder in a manner that only one side is in contact with the solder. The specimens shall be floated for the time specified in the relevant specification.

The specimens shall be removed from the solder pot and allowed to cool to ambient temperature.

After cooling, the specimen shall be cleaned with propane 2-OL (isopropyl alcohol) and air dried.

The specimen shall then be microsectioned in accordance with test 3X09 of this standard and examined for conformance.

11.5.5 Report

The report shall include:

- a) test method and revision;
- b) date of analysis;
- c) identification of the material tested;
- d) immersion time;
- e) results of the microsectional evaluation including but not limited to delamination, interconnection, separation, through-hole cracks, lifted lands, and resin recession;
- f) any deviation from this test method.

11.5.6 Additional information

This test is also applicable as a precondition step prior to the evaluation of the plated through hole resistance test method 3E16 of this standard.

Although reliability concerns are normally focused on small holes, large holes may still be a reliability concern even though the smallest holes have passed this test.

11.6 Test 3N06: Damp heat, steady state

11.6.1 Object

The purpose of this test is to determine the suitability of samples of printed board materials, printed boards and printed board assemblies for use and storage under conditions of high relative humidity.

This test is primarily intended to permit the observation of the effects of high humidity at constant temperature over a prescribed period.

11.6.2 Test specimen

Not specified.

11.6.3 Test apparatus and materials

The following test apparatus and materials shall be used:

11.6.3.1 Testing chamber

A testing chamber, so constructed that

- a) the temperature and the humidity of the chamber are monitored by sensing devices located in the working space,
- b) the temperature and relative humidity can be maintained at $(40 \pm 2) ^\circ\text{C}$ and $(93^{+2}_{-3})\%$ RH,
- c) the condensed water is continuously drained from the chamber and is not used again unless it has been re-purified,
- d) when using an injection-type chamber, the water shall have a resistivity of not less than $500 \Omega\text{m}$.

11.6.3.2 Precautions

Precautions shall be taken to ensure that

- a) the conditions prevailing throughout the working space are uniform and as similar as possible to those prevailing in the immediate vicinity of the monitoring devices,
- b) the properties or loading of the specimen under test do not appreciably influence conditions within the chamber,

c) no condensed water from the walls and roof of the test chamber can fall on the specimen.

11.6.4 Procedure

The following steps shall be taken:

The specimen shall be visually inspected and electrically and mechanically checked, as required by the relevant specification.

11.6.4.1 Conditioning

The specimen shall be conditioned as follows:

- a) the specimen shall be put into the chamber in the state specified in the relevant specification. To avoid condensation on the specimen it can be pre-heated to the chamber temperature;
- b) as required by the relevant specification, conditioning time can be
 - 4 days,
 - 10 days,
 - 21 days,
 - 56 days;
- c) note that the relevant specification may call for loading and/or measurements during, or at the end of, conditioning while the specimen is still in the chamber.

11.6.4.2 Recovery

The specimen shall be recovered as follows:

- a) before any measurements the specimen shall recover at standard atmospheric conditions for not less than 1 h and not more than 2 h;
- b) if the standard conditions given above are not appropriate for the specimen to be tested, the relevant specification may call for other recovery conditions;
- c) the relevant specification shall state whether any special precautions shall be taken regarding removal of surface moisture.

11.6.4.3 Inspection

The specimen shall be visually inspected and electrically and mechanically checked, as required by the relevant specification. Unless otherwise specified, measurements shall be completed within 30 min after being removed from the chamber.

11.6.5 Report

The report shall include:

- a) test method number and revision index;
- b) the date of the test;
- c) the identification of the materials;
- d) preconditioning procedure;
- e) conditioning time;
- f) recovery conditions if other than standard.

11.6.6 Additional information

Information required in the relevant specification:

- a) preconditioning procedure;
- b) electrical and mechanical checks to be made prior to the test;
- c) state of the specimen as introduced in the chamber;
- d) conditioning time;
- e) loading during conditioning;
- f) electrical checks to be made during conditioning and the period(s) after which they shall be performed;
- g) special precautions to be taken regarding removal of surface moisture;
- h) recovery conditions if other than standard;
- i) electrical and mechanical checks to be made at the end of the test, the parameters to be measured first and the maximum period allowed for the measurement of these parameters.

11.7 Test method 3N07: Temperature cycling

11.7.1 Object

This test is conducted for the purpose of determining the resistance of a material such as a laminate or multilayer printed circuit board, to the shock of repeated exposures to extremes of high and low temperatures for comparatively short periods of time.

11.7.2 Test specimen

The test specimen for this test shall be a sheet of laminate material at least 150 mm x 150 mm x laminate thickness. In the case of multilayer boards, the test specimen shall be the entire qualification board.

11.7.3 Test apparatus and materials

The following test apparatus and materials shall be used:

A chamber automatic temperature cycling equipment suitable for the temperature extremes specified herein. The air temperature shall be maintained by forced air circulation. The chamber shall have sufficient heating or cooling capacity to maintain the specified air temperature.

11.7.3.1 Conditions

During the exposures, maintain the chamber at the temperatures shown in Table 4.

Table 4 – Chamber temperatures for one cycle

Step	Class A		Class B	
	Temperature °C	Time Min	Temp °C	Time Min
1	125+3/-0	30	85+3/-0	30
2	25+10/-5	10-15	25+10/-5	10-15
3	-65+0/-5	30	-55+0/-5	30
4	25+10/-5	10-15	25+10/-5	10-15

11.7.4 Procedure

The following steps shall be taken:

The test specimen shall be cleaned of dirt, grease, and other contaminants prior to thermal exposure. The test specimen should be cleaned by wiping with a dry, clean lint-free cloth, or wiped with a clean lint-free cloth dampened with acetone or isopropyl alcohol. The specimen shall be allowed to air dry prior to thermal exposure.

11.7.4.1 Test exposure

The test specimen shall be placed in the thermal chamber maintained at the temperature specified in Step 1 for the appropriate test condition. The test specimen shall be maintained at each temperature in the order specified. Throughout the test, the specimen shall be placed in such a position that there is essentially no obstruction to the flow of air around them. The specimen shall be subjected to the specified temperatures in the sequence specified for a total of five cycles performed continuously.

After concluding the five temperature cycles, allow the specimen to return to room temperature. Visually inspect the specimen in accordance with detailed requirements.

11.7.4.2 Report

The report shall include:

- a) test method number and revision;
- b) any deviation from this method;
- c) specimens used in the test;
- d) information regarding type(s) and measurements of plating;
- e) results of evaluation, including failure modes and degree of failure, in case of failure;
- f) the date of the test;
- g) the name of the person that conducted the test.

11.7.4.3 Additional information

None.

11.8 Test 3N08: Thermal shock, immersion, fluidized sand bath at 260 °C

11.8.1 Object

The purpose of this test is to determine the ability of a printed board with plated through holes (PTH) test specimen to withstand a specified thermal shock by immersion in heated sand. This test is also applicable to holes from which interconnection is achieved with other technologies than the plated through hole process.

11.8.2 Test specimen

Specimens shall be removed from the panel or test coupon by sawing, routing or an equivalent method. Allow sufficient clearance to prevent damage to the area to be tested.

The specimen shall be removed in such a manner that at least three of the smallest possible adjacent size plated through holes can be viewed in a finished microsection.

As a referee, the specimen shall be preconditioned in an air circulation chamber at 125 °C for a time designated in the relevant specification, then allowed to cool down under standard atmospheric conditions until its temperature is below 35 °C. The recovery time shall not exceed 8 h.

11.8.3 Apparatus and test materials

The following test apparatus and materials shall be used:

- a) A fluidized sand bath of suitable design electrically heated and thermostatically controlled, kept at (260^{+5}_0) °C throughout the test.
- b) A thermometer for measuring the temperature of the sand in approximately the same location that will be occupied by the specimen.
- c) A timer capable of resolving to ± 1 s.
- d) A holder with low heat capacity such that the temperature of the sand is not brought below 260 °C.
- e) A circulating air chamber, capable of maintaining uniform temperature at (125 ± 5) °C.

11.8.4 Procedure

The following steps shall be taken:

The specimen shall be placed in the holder.

The specimen shall be immersed edgewise in the heated sand for the time given in the relevant specification.

The specimen shall be removed from the sand bath and cooled to ambient temperature.

The specimen shall then be microsectioned in accordance with test 3X09 of this standard and examined for conformance.

11.8.5 Report

The report shall include:

- a) test method and revision index;
- b) date of analysis;
- c) identification of the material tested;
- d) immersion time;
- e) results of the microsectional evaluation including but not limited to delamination, interconnection, separation, through-hole cracks, lifted lands and resin recession;
- f) any deviation from this test method.

11.8.6 Additional information

This test is also applicable as a precondition step prior to the evaluation of plated through hole resistance test method 3E16 of this standard.

Although reliability concerns are normally focused on small holes, large holes may still be a reliability concern even though the smallest holes have passed this test.

11.9 Test method 3N12: Moisture and insulation resistance, rigid, rigid/flex and flexible printed wiring boards

11.9.1 Object

This test method is designed to determine the visual degradation and the changes of the electrical insulation resistance of rigid printed wiring board specimens due to the deleterious effects of the high humidity, and the heat conditions typical of tropical environments. See Figure 26 for a graphical illustration of environmental exposure.

11.9.2 Test specimen

"Y" Pattern

- See Figure 27. There is also a variety of "Y" test patterns in various specifications within the industry, but the specimen preparation and testing are as described in this standard.

Comb patterns

- See Figure 28. Various "comb patterns" such as used in industry that are involved with moisture and insulation resistance requirements. These comb patterns, though different in terms of conductor width and spacing, are all wired and tested at the same terminal lands. These circuits can be properly tested following the procedures in this standard.

Production board testing

- Occasionally, production boards must be tested in lieu of test patterns. When this is required, one must exercise good judgement and select adjacent conductors for wiring terminal lands for testing. In all other respects, this procedure will produce significant results.

11.9.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- A clean test chamber capable of programming and recording an environment of temperature ranging between $(25 \pm 2) ^\circ\text{C}$ and $(65 \pm 2) ^\circ\text{C}$, and 85 % to 93 % relative humidity.
- A power supply capable of producing a standing bias potential of 100 V d.c. with a tolerance of ± 10 %.
- A resistance meter capable of reading high resistance ($10^{12} \Omega$ or greater), with a test voltage of 500 V.
- Two 3 500 ml beakers.
- Electrical hot plate.
- Exhaust ventilation hood.
- Metal tongs.
- Soft bristle brush.
- Deionized water ($2 \times 10^6 \Omega\text{-cm}$ minimum resistivity recommended).
- Isopropyl alcohol.
- Drying oven capable of maintaining at least $80 ^\circ\text{C}$ to $120 ^\circ\text{C}$.
- Conformal coating as agreed upon between user and supplier.

11.9.4 Test conditions

- Class 1 $(35 \pm 5) ^\circ\text{C}$, 90 % to 98 % relative humidity, 4 days (static).
- Class 2 $(50 \pm 5) ^\circ\text{C}$, 90 % to 98 % relative humidity, 7 days (static).
- Class 3 20 Cycles of $(25 \pm 5) ^\circ\text{C}$ to $(65 \pm 2) ^\circ\text{C}$, 90 % to 98 % relative humidity, 160 h total.

11.9.5 Specimen preparation

- Positive, permanent, and non-contaminating identification of test specimen is of paramount importance.
- Visually inspect the test specimens for any obvious defects. If there is any doubt about the overall quality of any test specimen, the test specimen should be discarded.

- c) Solder single stranded (to simulate discrete component axial leads). Polytetrafluoro-ethylene (Teflon) insulated wires in each of the connection points of the specimens. These wires will be used to connect the test patterns to polarization voltage and for insulation resistance testing.

11.9.6 Flux removal

Remove flux from lead lands as follows:

- a) Wet test lead terminals with deionized or distilled water and scrub with a soft bristle brush for a minimum of 30 s. During the remainder of the specimen preparation, handle test specimens by the edges only.
- b) Spray rinse thoroughly the specimen with deionized or distilled water. Hold test specimen at an approximate 30° angle and spray from top to bottom.
- c) Immerse the test lead terminals with clean isopropyl alcohol and agitate for a minimum of 30 s. Scrub with a soft bristle brush to remove flux residue.
- d) Rinse cleaned area thoroughly with fresh isopropyl alcohol.
- e) Dry test specimens in a drying oven for a minimum of 3 h at 49 °C to 60 °C.
- f) If boards are to be stored before coating, place the boards in ionic contaminant free plastic bags, and close bags (do not heat seal).

11.9.7 Application of coating

- a) Apply coating(s) to the appropriate area of the test specimen, in a manner concurrent with user's production techniques or as specified by the supplier.
- b) After the application of coating, the test specimens are to be cured, as specified by the supplier.
- c) After curing, both bare and coated test specimens are to be conditioned at laboratory ambient temperature for 24 h.

11.9.8 Procedure

The following steps shall be taken:

Condition either type of specimen at $(50 \pm 5) ^\circ\text{C}$ with no added humidity for a period of 24 h.

Allow specimens to cool and take the initial insulation resistance measurements at laboratory ambient temperature. Apply 500 V d.c. on the specimen's test points with the resistance meter, and take the reading after 1 min.

The test points for comb patterns are 1 to 2, 2 to 3, 3 to 4, and 4 to 5. Test points 1-3-5 are connected to the positive (+) terminal, and test points 2-4 are connected to the negative (–) terminals of the resistance meter. For quality conformance, test points are at all connection points on the "Y" pattern test coupons (see Figure 34).

Place specimens in chamber, in a vertical position and under a condensation drip shield, and connect the specimen test points to the d.c. voltage source. Apply a $(100 \pm 5) \text{ V d.c.}$ voltage to all specimens.

Expose the class 1 and 2 test boards to the static conditions described under 11.9.4 a) or b). The specimen should be shielded to minimize formation of surface moisture while in the chamber.

Expose the Class 3 test specimens to 20 cycles of temperature and humidity. The specimens shall be maintained at 90 % relative humidity minimum throughout the 20 cycle period, except that when going to low temperature, the humidity may drop to 80 % relative humidity minimum.

- Start test at 25 °C, and raise temperature to 65 °C, over a time span of $(1,75 \pm 0,75) \text{ h}$.
- Maintain temperature at 65 °C, with 90 % to 98 % relative humidity, over a time span of

3 h minimum.

- Lower temperature from 65 °C to 25 °C, maintaining greater than 80 % relative humidity, over a time span of $(1,75 \pm 0,75)$ h.

This constitutes one complete cycle. There shall be no delay between cycles. Specimen shall be exposed to 19 more cycles making a total of 20 cycles and for a total duration of 160 h. See Figure 26.

11.9.9 Measurement (when specified)

- a) Disconnect 100 V d.c. voltage source before taking any insulation resistance measurement. Electrical connections to the specimen or to the “Y” pattern of the test board as illustrated in Figure 27 shall be made so that electrical polarization voltage and the test voltage, of the same polarity, are connected to the same terminal.
- b) For final resistance testing, measurements shall be taken, at (23 ± 2) °C and (50 ± 10) % relative humidity, after disconnecting the bias voltage, removal of specimen from the chamber; stabilize the specimen from 1 h to 2 h at ambient.
- c) Any reason for deleting values, i.e. scratches, condensation, bridged conductors, etc., shall be noted.

11.9.10 Evaluation

- a) Each test specimen shall be evaluated for insulation resistance quality for its class and type, following and/or during the initial, wet and/or dry conditions, as applicable.
- b) After completion of all electrical testing, the test specimens shall be examined for measling, blister, delamination, or other forms of degradation, following 24 h stabilization at laboratory ambient temperature.

11.9.11 Report

The report shall include:

- a) test method number and revision;
- b) any deviation from this method;
- c) specimens used in the test;
- d) results of evaluation, including failure modes and degree of failure, in case of failure;
- e) date of the test;
- f) name of the person that conducted the test.

11.9.12 Additional information

- a) Documented alternative cleaning procedures may be implemented if there is a concern that scrubbing will adversely affect test results, i.e., when the test specimens have very fine spacing and/or are plated with soft metals (tin/lead, gold, etc.).
- b) Initial wire placement must be maintained to ensure reproducible results.
- c) A failure such as measling, blistering, delamination or any other form of degradation, may be due to several factors, and not necessarily inferior coatings. Such system failures observed in solder mask and/or conformal coating on a substrate should not be grounds for coating rejection. In such cases, a retest of the entire system is mandatory.

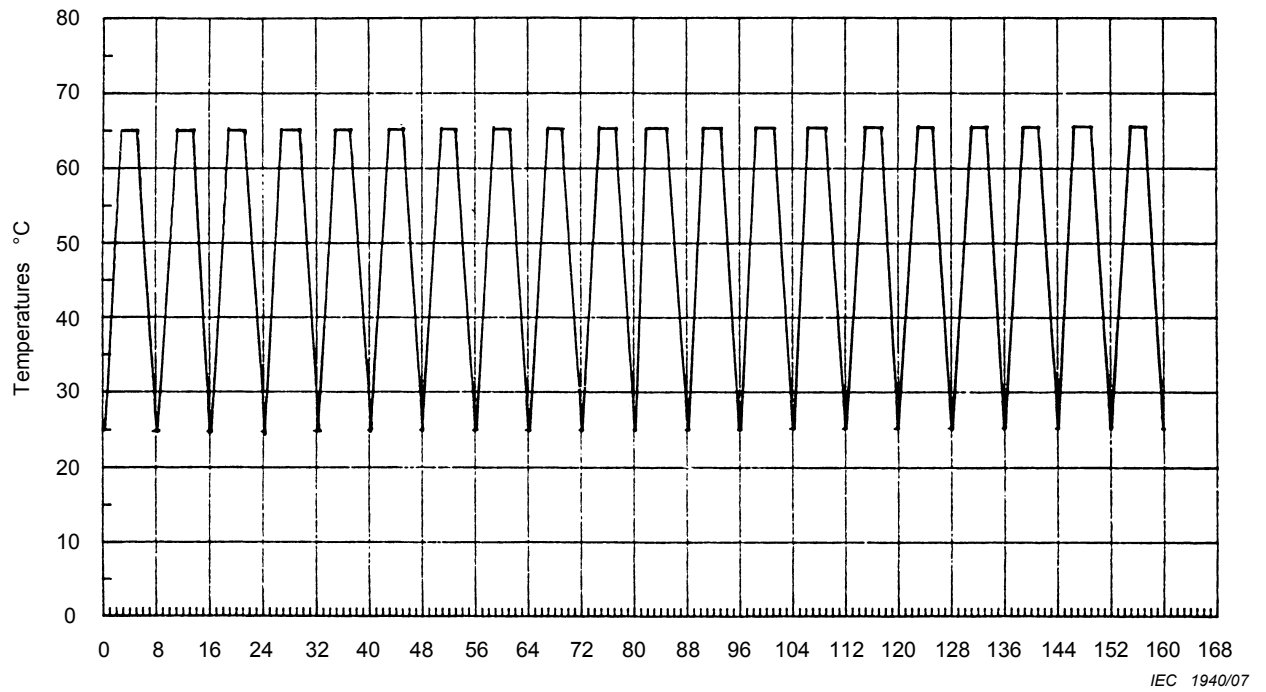


Figure 26 – Temperature cycles for moisture and insulation resistance test graph

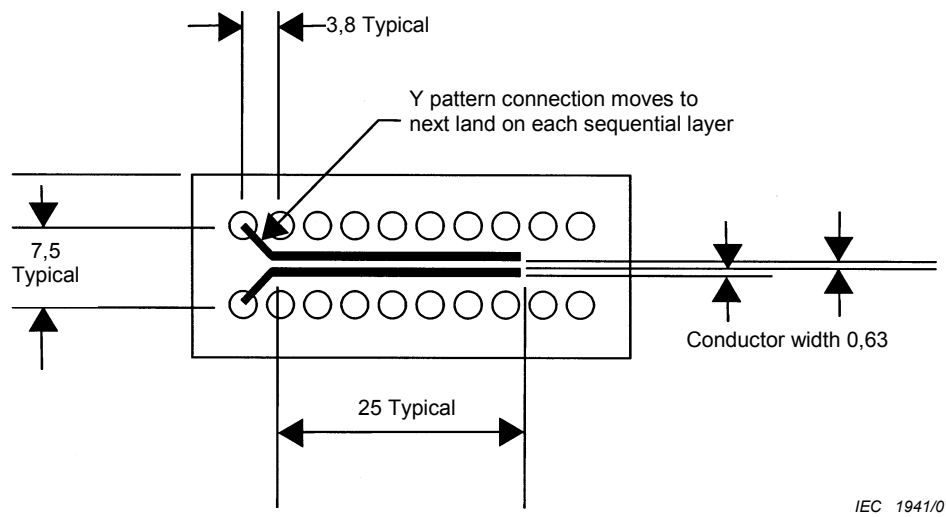


Figure 27 – Insulation resistance coupon (μm)

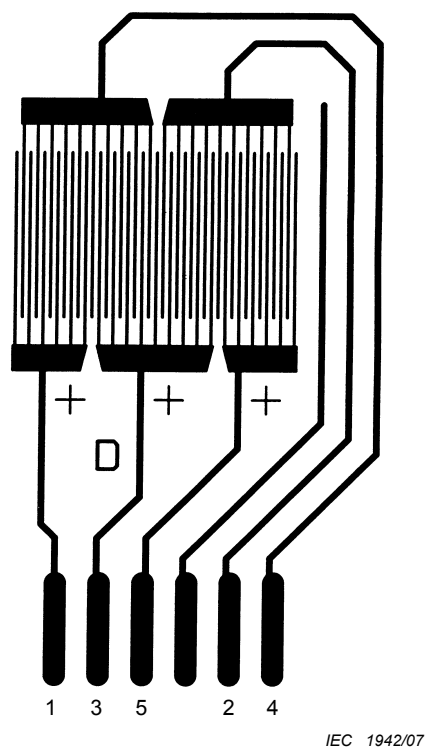


Figure 28 – Typical “comb pattern”

12 X: Miscellaneous test methods

12.1 Test 3X01: Plating adhesion, tape method

12.1.1 Object

The purpose of this test is to determine the adhesion quality of plating on metal clad laminates used in the printed boards.

12.1.2 Test specimen

The adhesion of plating shall be measured on the finished products or a specimen plated with sub-surface metals.

12.1.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- Tape, a roll of pressure sensitive tape, 12 mm wide or similar.

12.1.4 Procedure

The following steps shall be taken:

- a) The surface of test specimen shall be cleaned with alcohol.
- b) Press a strip of pressure sensitive tape 12 mm wide and 50 mm long firmly across the surface of the conductor pattern without air inside. After 10 s, rapidly remove the tape by manual force applied approximately perpendicular to the circuit pattern. Fresh tape shall be used each time.
- c) Visually examine tape and specimen for evidence of any portion of the plating having come off the specimen as evidenced by particles of plating or pattern adhering to the tape.

- d) If overhang breaks off (slivers) and adheres to the tape, it is evidence of overhang but not a plating adhesion failure.

12.1.5 Report

The report shall include:

- a) the test number and revision
- b) any deviation from this method;
- c) the test date;
- d) specimens used in the test;
- e) results of evaluation, including failure modes and degree of failure, in case of failure;
- f) name of the person that conducted the test.

12.1.6 Additional information

None.

12.2 Test 3X02: Plating adhesion, burnish method (under consideration)

12.3 Test 3X03: Porosity, gas exposure (under consideration)

12.4 Test 3X04: Porosity, electrographic test, gold on copper

12.4.1 Object

To detect discontinuities in certain metal plating on printed boards by an electrographic method.

The test is suitable for the examination of gold, palladium, and rhodium coating on copper without an undercoat of nickel.

The test described is known as the electrographic dye transfer method.

12.4.2 Test specimen

A suitable part of a production board with gold, palladium or rhodium coatings on copper.

NOTE Electrical connection will have to be made to coated areas either via the printed circuit conductors or by using a conductive tape which is insulated on one side.

The coated areas should be wiped with a lint-free cloth dipped in a neutral organic solvent (for example acetone) to remove grease, etc.

12.4.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) An electrographic test machine with high-purity aluminium platen of nominal size 150 mm × 25 mm to act as cathode, capable of exerting pressure up to 200 N/cm² and delivering a smooth ripple-free d.c. current from a source not exceeding 12 V.
- b) Dye transfer paper, grade FDW, i.e. high-quality photographic paper coated with a gelatin layer but not photosensitive, cut to a convenient size.
- c) An electrolyte which is 0,01 molar with respect to both sodium chloride and sodium carbonate, i.e. dissolve 0,59 g of NaCl and 1,06 g of Na₂CO₃ together in 1 l of distilled water.
- d) A developing solution of di-thio-oxamide (rubeanic acid), 0,25 % solution in ethanol. Dissolve 0,25 g of di-thio-oxamide in 100 ml of ethanol by warming gently. If necessary, filter when cold, before use.

12.4.4 Procedure

The following steps shall be taken:

Soak the dye transfer paper strips in the sodium chloride/sodium carbonate electrolyte for 30 min.

After soaking, the excess electrolyte is removed from the dye transfer paper by pressing between sheets of blotting paper. The paper is then placed emulsion side down on to the plated specimen, which acts as the anode. These are then placed under the aluminium platen.

The assembly is then compressed to give a pressure of between 140 N/cm² to 170 N/cm².

Whilst under compression a fixed potential of 4 V d.c. is applied for 60 s. The press is then opened, the dye transfer paper removed and developed in the dithio-oxamide solution for 30 s.

NOTE It is advisable to use tweezers for immersing the paper in the dithio-oxamide solution because it can produce persistent black stains on the fingers.

The electrogram produced is washed in running water and allowed to dry. The presence of any defect in the coating (for example porosity) is revealed by a corresponding dark olive-green stain on the paper.

The electrograms shall be visually examined at not greater than 10X linear magnification. Porosity in the gold will show as bright pink spots. An estimate should be made of the number of pores/cm² of the plated area.

12.4.5 Report

The report shall include:

- a) the test method number and revision;
- b) the identification and description of the test specimens;
- c) the date of the test;
- d) porosity in pores/cm²;
- e) any deviation from this test method;
- f) the name of the person conducting the test.

12.4.6 Additional information

Effects at the edges of the plated areas shall be discounted.

12.5 Test 3X05: Porosity, electrographic test, gold on nickel

12.5.1 Object

To detect discontinuities in certain metal platings on printed boards by an electrographic method.

The test is suitable for the examination of gold, palladium and rhodium coatings on a nickel undercoat.

The test described is known as the electrographic dye transfer method.

12.5.2 Test specimen

A suitable part of a production board with gold, palladium or rhodium coatings on nickel.

NOTE Electrical connection will have to be made to coated areas either via the printed circuit conductors or by using a conductive tape which is insulated on one side.

The coated areas should be wiped with a lint-free cloth dipped in a neutral organic solvent (for example acetone) to remove grease etc.

12.5.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) An electrographic test machine with high-purity aluminium platen of nominal size 150 mm × 25 mm to act as cathode, capable of exerting pressure up to 200 N/cm² and delivering a smooth ripple-free d.c. current from a source not exceeding 12 V.
- b) Dye transfer paper grade FDW i.e. high-quality photographic paper coated with a gelatin layer but not photosensitive, cut to a convenient size.
- c) An electrolyte which is 0,01 molar with respect to both sodium chloride and sodium carbonate, i.e. dissolve 0,59 g of NaCl and 1,06 g of Na₂CO₃ together in 1 l of distilled water.
- d) A developing solution of nioxime (cyclohexane-1.2-dionedioxime), 0,5 % solution by weight in ethanol. Dissolve 0,5 g of nioxime in 100 ml of ethanol.

12.5.4 Procedure

The following steps shall be taken:

Soak the dye transfer paper strips in the sodium chloride/sodium carbonate electrolyte for 30 min.

After soaking, the excess electrolyte is removed from the dye transfer by pressing between sheets of blotting paper. The paper is then placed emulsion side down on to the plated specimen, which acts as the anode. These are then placed under the aluminium platen.

The assembly is then compressed to give a pressure of between 140 N/cm² to 170 N/cm².

Whilst under compression, a fixed potential of 4 V d.c. is applied for 60 s. The press is then opened, the dye transfer paper removed and developed in the nioxime solution for 30 s.

The electrogram produced is washed in running water and allowed to dry. The presence of any defect in the coating (for example porosity) is revealed by a corresponding bright pink stain on the paper.

The electrograms shall be visually examined at not greater than 10X linear magnification. Porosity in the gold will show as bright pink spots. An estimate should be made of the number of pores/cm² of the plated area.

12.5.5 Report

The report shall include:

- a) the test method number and revision;
- b) the identification and description of the test specimens;
- c) the date of testing;
- d) porosity in the pores/cm²;

- e) any deviation from this test method;
- f) the name of the person conducting the test.

12.5.6 Additional information

Effects at the edges of the plated areas shall be discounted.

12.6 Test 3X06: Plating thickness, microsection method

12.6.1 Object

The purpose of this test method is to provide a procedure for determining the plating thickness of a printed board.

12.6.2 Test specimen

The specimen shall be cut out from a production board, a test coupon or a specified part of a composite test coupon having plated-through holes (pth) for test.

Where the use of test coupons as specified in IEC 62326-4-1 or 62326-4 is agreed, the measurement shall be carried out on specimen C.

12.6.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A sample cutter, or router or diamond saw to cut out the specimen from printed boards.
- b) A fixture to hold the specimen during encapsulation.
- c) An encapsulating material, such as room temperature curing epoxy resins.
- d) An abrasive paper (for example grit numbers 150-220, 320, 400, 600, 800, and 1 200) to grind roughly and finely.
- e) A diamond polishing paste (for example 6 μm , 3 μm , 1 μm) or alumina polishing paste (for example 5 μm , 3 μm , 0,5 μm) to remove scratches left by the abrasive paper.
- f) A cloth for polishing with polishing paste.
- g) A microscope having the magnification of 100 $\times$, 250 $\times$, 500 $\times$, and 1 000 $\times$.

12.6.3.1 Etching solutions

12.6.3.1.1 Preparation of ammonium hydroxide, peroxide solution

1 ml to 2 ml of B is added to 100 ml of A. Stir for a few seconds and wait 1 min before using.

A is (14 – 20) % ammonium hydroxide (by volume);

B is (30 – 35) % hydrogen peroxide (by volume).

A fresh preparation every day, and the addition of a few drops of 30 % hydrogen peroxide to the etching solution just before use, are recommended.

12.6.3.1.2 Preparation of sulphuric, peroxide solution

1 ml to 2 ml of B is added to 100 ml of C. Stir for a few seconds and wait 1 min before using.

C is (10 – 20) % sulphuric acid (by volume);

B is (30 – 35) % hydrogen peroxide (by volume).

A fresh preparation every day, and the addition of a few drops of 30 % hydrogen peroxide to

the etching solution just before use, are recommended.

12.6.4 Procedure

The following steps shall be taken:

The specimen shall be placed in the fixture and carefully potted using encapsulating material, making sure that the holes are filled with encapsulating material.

There shall be no voids between the encapsulating material and any layer in the area where the thickness of the layer is to be measured. Air bubbles may be eliminated by stirring, manual agitation, or vacuum degassing.

After curing, the specimen shall be carefully ground using successively finer abrasive paper with water flow, culminating with a 1 200 grit minimum abrasive to the plane containing the observation point. Where cross-sections vertical to the plane of the printed board are to be inspected, the polished plane of the microsection shall be inclined within 85° to 95° to the plane of the printed board.

The specimens shall be rinsed after each grinding and polishing step, to prevent particles from contaminating subsequent steps.

The specimen shall be polished by the polishing cloth with paste, using successively finer abrasive to show a clear, sharp image of the observation point.

Where the wall thickness of the plated-through holes is to be measured, the hole diameter appearing in the cross-section shall be not less than 90 % of the actual hole diameter, as measured prior to preparing the microsection.

After polishing and prior to visual and/or dimensional examination, the specimen shall be etched using the etching solution in such a way that plating boundaries are sharply defined.

The specimen shall be visually examined using the microscope. The magnification shall be chosen so as to be suitable for the characteristics to be examined. Where dimensions are to be measured, a calibrated measuring system shall be incorporated.

When measuring dimensions, both boundaries of the detail to be measured shall be in focus simultaneously. When plating thickness is measured, nodules, voids, and cracks shall not be included.

NOTE For examination of multilayer printed boards for inner layer foil cracking and/or inner layer junction pull away, test 3X11 is more sensitive.

The relevant specification may require examination of the interface of the wall of plated-through holes and the conductors on inner layers prior to etching.

12.6.5 Report

The report shall include:

- a) the test method number and revision;
- b) the date of the test;
- c) the identification and description of the specimen(s);
- d) the plating thickness measured;
- e) the number of data;
- f) the maximum and minimum plating thickness observed;

- g) the average thickness measurements;
- h) any deviation from this test method;
- i) the name of the person conducting the test.

12.6.6 Additional information

None.

12.7 Test 3X07: Solderability, edge dip test (under consideration)

12.8 Test 3X08: Delamination, thermal shock

12.8.1 Object

To determine that correct processing and suitable materials have been used, by proving the ability of a printed board to withstand a specified thermal shock without evidence of delamination.

12.8.2 Test specimen

The test shall be carried out on a production board, a test coupon or a specified part of a composite test coupon.

12.8.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) A circulation air chamber, capable of maintaining uniform temperature of $(125 \pm 5) ^\circ\text{C}$.
- b) A solder pot of suitable design, electrically heated, thermostatically controlled and containing at least 1 kg of Sn60, Sn62, or Sn63. Test temperature shall be $(260^{+5}_0) ^\circ\text{C}$.
- c) A thermometer for measuring the temperature of the solder at a depth of 25 mm below the surface.
- d) A timer capable of resolving to ± 1 s or better.
- e) A low- or non-activated water soluble flux or a resin of 0,2 % maximum chloride content.
- f) Optical equipment with approximately $3\times$ linear magnification which, where possible, illuminates the specimen with transmitted light.

12.8.4 Procedure

The following steps shall be taken:

- a) Specimens shall be conditioned in an air circulating chamber at $(125 \pm 5) ^\circ\text{C}$ for a period as specified in the relevant specification.
- b) Let the specimen cool down under atmospheric conditions until the temperature is less than $35 ^\circ\text{C}$. Recovery time shall not exceed 8 h.
- c) Coat the surface with flux.
- d) Remove the oxide from the solder, and float the specimens on the solder in such a manner that only one side is in contact with the solder. The specimens shall be floated for the time specified in the relevant specification.
- e) Remove the specimens from the solder pot and allow to cool to ambient temperature.

- f) After cooling, the specimens shall be cleaned with isopropyl alcohol and air dried.
- g) The specimens shall then be visually examined concerning delamination.

12.8.5 Report

The report shall include:

- a) the test method number and revision;
- b) the detailed description of the test specimen including, but not limited to, preparation, size, quantity, and drying time/temperature;
- c) the result: delamination/no delamination;
- d) the date of the test;
- e) any deviation from this test method;
- f) the name of the person conducting the test.

12.8.6 Additional information

The pre-conditioning is made to dry the specimen to such an extent that the test result will not be influenced by moisture in the material.

12.9 Test 3X09: Microsectioning

12.9.1 Object

The purpose of this test method is to provide a procedure for determining internal conditions of plated through holes, conductive patterns, base materials and coating of a printed board. For examination of multilayer printed boards for inner layer foil cracking and/or inner layer junction pull-away, test 3X11 (under consideration) is more sensitive for these conditions.

12.9.2 Test specimen

The specimen shall be cut out from a printed board or test coupon with great care to avoid any damage to the area to be tested.

In case of soft and/or thin platings, for example gold, tin, or tin-lead, an overplate with a harder plating of the specimen prior to encapsulation may be necessary to avoid smearing.

When organic coatings are to be examined, it may be necessary to overplate or to use a pigmented encapsulating material giving a colour contrast with the coating to be examined.

12.9.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) Sample cutter, or router or diamond saw to cut out the specimen from printed boards.
- b) Fixture to hold specimen during encapsulation.
- c) Encapsulating material, such as room temperature curing epoxy resin.
- d) Abrasive paper (for example grit numbers 150-220, 320, 400, 600, 800, and 1200) to grind roughly and finely.
- e) Diamond polishing paste (for example 6:m, 3:m, 1:m) or alumina polishing paste (for example 5:m, 3:m, 0,5:m) to remove scratches left by abrasive paper.
- f) Cloth for polishing with polishing paste.
- g) Microscope having the magnification of 100 ×, 250 ×, 500 ×, and 1 000 ×.

12.9.3.1 Etching solutions

12.9.3.1.1 Preparation of ammonium hydroxide, peroxide solution

1 ml to 2 ml of B is added in 100 ml of A. Stir for a few seconds and wait 1 min before using.

A is (14 – 20) % ammonium hydroxide (by volume);

B is (30 – 35) % hydrogen peroxide (by volume).

A fresh preparation every day and the addition of a few drops of 30 % hydrogen peroxide to the etching solution just before using are recommended.

12.9.3.1.2 Preparation of sulphuric, peroxide solution

1 ml to 2 ml of B is added in 100 ml of C. Stir for a few seconds and wait one minute before using.

B is (30 – 35) % hydrogen peroxide (by volume);

C is (10 – 20) % sulphuric acid (by volume).

A fresh preparation every day, and the addition of a few drops of 30 % hydrogen peroxide to the etching solution just before using are recommended.

12.9.4 Procedure

The following steps shall be taken:

- a) The specimen shall be set in the fixture and carefully potted using encapsulating material, make sure that the holes are filled with encapsulating material.
- b) There shall be no voids between the encapsulating material and any layer in the area where thickness of the layer is to be measured. Air bubbles may be eliminated by stirring, manual agitation, or vacuum degassing.
- c) After curing, the specimen shall be carefully ground using successively finer abrasive paper with water flow, culminating with a 1 200 grit minimum abrasive to the observation point. Where cross-sections vertical to the plane of the printed board are to be inspected, the polished plane of the microsection shall be inclined within 85° to 95° to the plane of the printed board.
- d) The specimens shall be rinsed after each grinding and polishing step, to prevent particles from contaminating subsequent steps.
- e) The specimen shall be polished by the polishing cloth with paste using successively finer abrasive to show a clear, sharp image of the observation point.
- f) Where wall thickness of the plated through holes are to be measured, the hole diameter appearing in the cross-section shall be not less than 90 % of the actual hole diameter as measured prior to preparing the microsection.
- g) After polishing and prior to visual and/or dimensional examination, the specimen shall be etched using the etching solution in such a way that plating boundaries are sharply defined.
- h) The specimen shall be visually examined using the microscope. The magnification shall be chosen so as to be suitable for the characteristics to be examined. Where dimensions are to be measured, a calibrated measuring system shall be incorporated.
- i) When measuring dimensions, both boundaries of the detail to be measured shall be in

focus simultaneously. When plating thickness is measured, nodules, voids, and cracks shall not be included.

- j) As specified in the relevant specification, one or more of the following characteristics and details shall be examined:
- thickness of the conductor and plating, and of the copper foil of the laminates;
 - voids and cracks in the plating;
 - cracks in the copper foil of the laminates (see note);
 - burrs and nodules;
 - drilling quality (for example: resin smear, nailheading on internal layers);
 - undercut and metallic overhang;
 - interface of the wall of plated through holes and the conductor on inner layers;
 - separation of plating (see note);
 - thickness of organic layers (including base materials);
 - voids in organic layers (including base materials);
 - etchback;
 - glassfibre protrusion;
 - delamination;
 - misregistration between layers;
 - misregistration between conductor and hole patterns;
 - annular ring width.

NOTE For examination of multilayer printed boards for inner layer foil cracking and/or inner layer junction pull-away, test 3X11 of this standard is more sensitive for these conditions.

- k) The relevant specification may require examination of the interface of the wall of plated through holes and the conductors on inner layers prior to etching.

12.9.5 Report

The report shall include:

- a) the test number and revision index;
- b) the test date;
- c) the identification of the material tested;
- d) the features which were examined;
- e) results of examination.

12.9.6 Additional information

Ammonium hydroxide is a pungent, intense alkaline chemical. Hydrogen peroxide is a strong oxidant. Sulphuric acid is a corrosive solution. They shall be handled with care, sulphuric contact with eyes and skin being prevented by the wearing of protective glasses and chemically resistant gloves. Consult local regulations for complete safety instructions.

12.10 Test 3X10: Solderability, rotary dip test for plated through holes, surface mount conductors and attachment lands

12.10.1 Object

This test describes the methods, defect definitions and gives illustrations for assessing solderability of printed board surface conductors, attachment lands, and plated through holes,

using a rotary dip tester. This test is intended for use by both customer and supplier.

This test method shall not be interpreted as a production soldering or tinning procedure for preparation or soldering of printed boards or assemblies.

The solderability determination is made to verify that the printed board fabrication processes and subsequent storage have had no adverse effect on the solderability of those portions of the printed board intended to be soldered. This is determined by evaluation of the solderability of the test specimen portion of a board, or of a representative coupon, which has been processed as part of the panel of boards and subsequently removed for testing, per the method selected.

The objective of the solderability test method described is to determine the ability of the surface conductors and attachment lands to wet easily with the solder, and to withstand the rigours of the assembly processes.

12.10.2 Test specimen

12.10.2.1 Test specimen requirements

The test specimen shall be a representative coupon, a portion of the printed board being tested, or a whole board, if within size limits, such that the immersion depth defined is possible. The test specimen shall be representative of the lot or the production run being tested.

When the test specimen is to be used for testing the printed board acceptance, the number of test specimens shall be defined in the applicable specification.

12.10.2.2 Test specimen details

The test specimen shall be a complete board, a section of a board, or a suggested coupon. Figures 26 and 30 are suggested coupon styles. The specimen shall be of a width such as to allow 13 mm clearance from the solder pot sides.

If plated through holes are to be tested, the minimum number of holes to be tested is 30 per test lot. The minimum number of terminations (plated through holes or attachment lands) per test specimen shall be six. The test specimen shall be representative of the product.

The exposed length of specimen test face in the direction of travel shall be (25 ± 5) mm.

Dimensions in millimetres

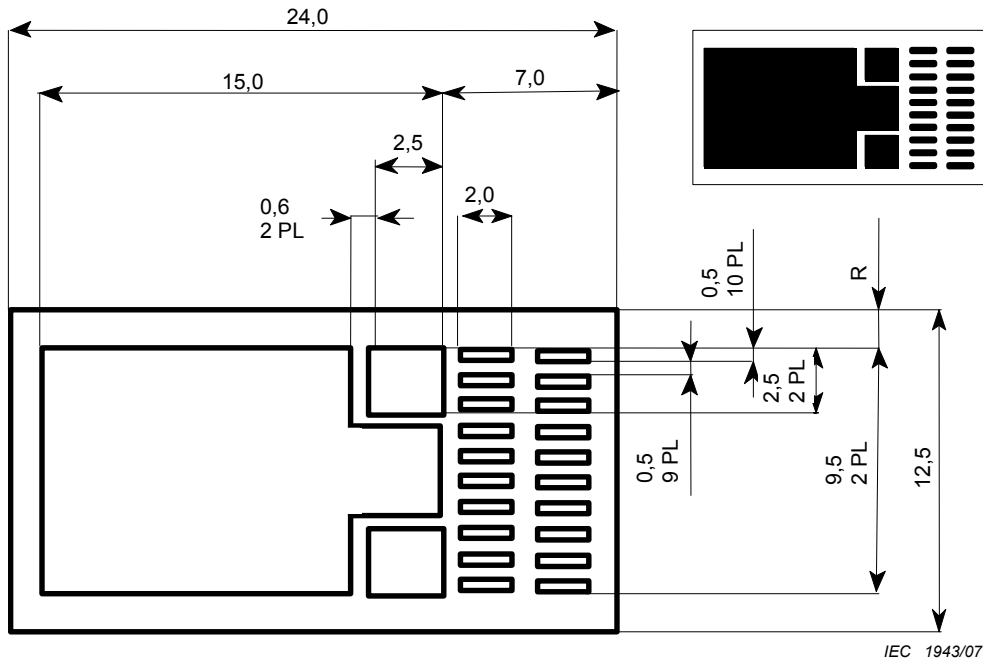


Figure 29 – Suggested test specimen for surface mount features

Dimensions in millimetres

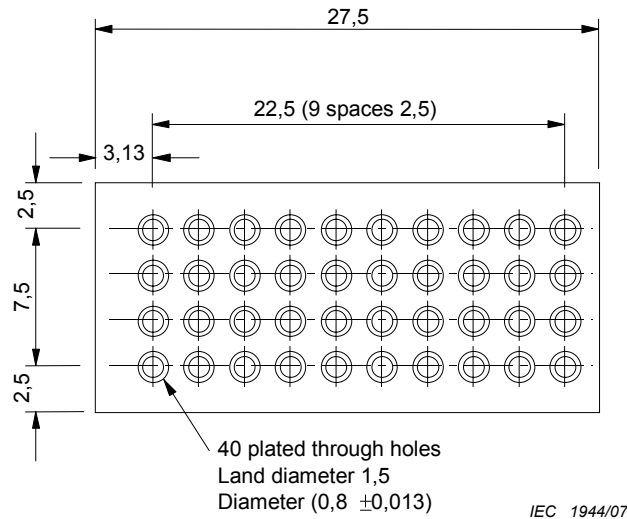


Figure 30 – Suggested test specimen for plated through holes

12.10.2.3 Coating durability

The required coating durability shall be specified. The following are guidelines for determining the needed level of coating durability. Prior to the test, preconditioning of the specimens shall be carried out and the type of flux specified as shown in Table 5.

12.10.2.3.1 Coating durability categories

Category 1: minimum coating durability

Intended for boards which will be soldered within 30 days from time of manufacture, and are likely to experience minimum thermal or solder exposure.

Category 2: average coating durability

Intended for boards which are likely to experience storage up to six months from time of manufacture, and are likely to experience moderate thermal or solder exposures.

Category 3: maximum coating durability

Intended for boards which are likely to experience long storage (over six months) from time of manufacture and/or are likely to experience severe thermal or solder processing steps.

Table 5 – Accelerated ageing and test requirements

Category	Preconditioning	Flux
1 (Minimum)	No ageing	Production type
2 (Average)	No ageing	Rosin (see 12.7.3.1.2)
3 (Maximum)	8 h ageing	Rosin (see 12.7.3.1.2)

12.10.3 Test apparatus and materials

The following test apparatus and materials shall be used:

12.10.3.1 Materials

12.10.3.1.1 Solder

The solder shall be composition Sn60 or Sn63. Other alloys may be used upon agreement between customer and supplier. The composition of the solder shall be maintained within the limits as referenced in Table 6.

Table 6 – Maximum limits of solder bath contaminants

Contaminant ¹⁾	Maximum contaminant weight limit ²⁾ %
Copper	0,300
Gold	0,200
Cadmium	0,005
Zinc	0,005
Aluminium	0,006
Antimony	0,500
Iron	0,020
Arsenic	0,030
Bismuth	0,250
Silver	0,100
Nickel	0,010
¹⁾ The tin content of the solder shall be maintained within ± 1 % of the nominal alloy being used. Tin content shall be tested at the same frequency as testing for copper/gold contamination. The balance of the bath shall be lead and/or the items listed above. ²⁾ The total copper, gold, cadmium, zinc and aluminium contaminants shall not exceed 0,4 %.	

12.10.3.1.2 Flux

A non-active rosin having a nominal composition of 25 % by weight water white gum rosin, for example type LR3CN, in a solvent of 99 % isopropyl alcohol, shall be used. The density shall be $(0,843 \pm 0,005)$ g/cm³ at 25 °C. Other fluxes may be used for solderability testing only upon agreement between customer and supplier (for example, the activated flux as specified in 6.6.2 of IEC 60068-2-20 – see also 12.10.6.1.2 below).

NOTE Caution should be taken in the storage of fluxes used for solderability testing in order to maintain the solids content, and to avoid contamination.

12.10.3.1.3 Flux removal material

Material used for cleaning test specimens prior to solderability evaluations shall be capable of removing visible flux residues.

12.10.3.2 Equipment

12.10.3.2.1 Solder pot/bath

A thermostatically controlled static solder bath shall be used, of adequate dimensions, to accommodate the specimens. The solder bath shall contain sufficient solder to maintain the temperature of (245 ± 2) °C at 25 mm below the surface during testing, unless otherwise agreed to between customer and supplier.

The solder shall be chemically or spectrographically analysed or replaced each 30 operating days as a minimum. An operating day consists of any 8 h period, or any portion thereof, during which the solder is liquified and used. The levels of contamination and Sn content shall be as shown in Table 6.

The interval between analysis may be lengthened if the test results, documented to the user's satisfaction, indicate that the contamination limits are not being approached.

If contamination exceeds the limits specified in Table 6, the solder shall be changed, and the interval between analysis shall be shortened.

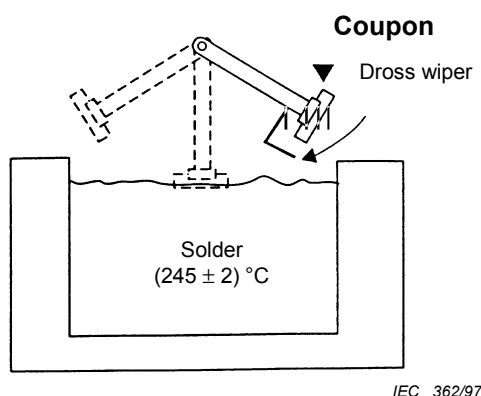
12.10.3.2.2 Optical inspection equipment

Inspection generally shall be carried out by the unaided eye (corrected vision glasses permitted) but, on occasion, either a direct or projection lens system with a maximum of 10× magnification may be used.

12.10.3.2.3 Rotary dip equipment

A device shall be used to move the test specimen in a circular path so that the flat surface of the specimen will contact the solder at a constant speed without stopping. The distance between the centre of rotation and the centre of the test specimen shall be (100 ± 5) mm. An example of rotary dip equipment is shown in Figure 31.

Those parts of the holder, including the retaining spring (if fitted), which come in contact with the specimen and/or the solder should have low thermal capacity and conductivity.



NOTE 1 Dwell timer set at $(3,0 \pm 0,5)$ s.

NOTE 2 Adjustable speed control on 100,0 mm R on solder station.

NOTE 3 Dwell at end of 100,0 mm radius swing to allow solder to solidify.

Figure 31 – Rotary dip solderability test equipment

12.10.3.2.4 Timing equipment

Timing equipment shall be automated, where applicable, and accurate to the limits of the test method.

12.10.4 Procedure

The following steps shall be taken:

12.10.4.1 Limitations

The test procedures are applicable to most printed board constructions, typical of the industry. It is recognized that thick printed boards will not respond in the same way as thin printed

boards, due to their increased thermal mass and number of planes, and weight of the solder column within the hole. These factors greatly reduce the likelihood that all surfaces will display a completely wetted condition.

If it is demonstrated, to the user's satisfaction, that changes in the procedure are necessary, due to the physical characteristics of a specimen and not to the solderability of the specimen, a new procedure shall be documented and used only for that application.

12.10.4.2 Conditioning

12.10.4.2.1 Specimen preparation and conditioning for test

The test specimen(s) in the "as-received" condition shall be prepared for testing in accordance with the user-specified conditioning of 12.10.2.3. Care shall be exercised to prevent contamination (by grease, perspiration, or other) of the surface to be tested.

When agreed upon between customer and supplier, the specimen to be tested may undergo other types of pre-treatment, such as degreasing, aqueous cleaning, copper and soldering brightening, and/or baking.

12.10.4.2.2 Accelerated ageing

Accelerated ageing shall be conducted in accordance with test 1P03.

12.10.4.2.3 Baking

Immediately after steam ageing, and prior to solderability testing, all specimens shall be baked at $(105 \pm 5) ^\circ\text{C}$ for (60^{+15}_0) min to remove surface moisture and other volatiles. Test specimens shall be cooled to room temperature prior to fluxing and testing.

12.10.4.3 Application of flux

The test specimens are to be dipped in the flux for 5 s to 10 s, to the full depth to be soldered. The flux shall be maintained at the prescribed composition defined in 12.10.3.1.2. After withdrawal from the flux, the specimen shall be allowed to drain vertically for a maximum of 60 s and placed on a sheet of absorbent paper. The solderability test shall then be performed in not less than 1 min, and not more than 5 min after blotting.

12.10.4.4 Rotary dip testing

The time of contact between any point of the test face of the test specimen and the molten solder shall be determined by a timer activated by the electrical contact of the sensor with the molten solder. The tip of the sensor shall be located adjacent to the specimen on the specimen holder. The sensor shall be kept clean, and shall be electrically insulated from the specimen holder which carries it.

A strip of 50 mm wide polytetrafluoroethylene (PTFE) or equivalent shall precede the test specimen in the test cycle, in order to remove oxide or flux residue from the solder surface, immediately before the specimen is introduced.

Dross and burned/residual flux shall be completely removed from the surface of the molten solder immediately prior to dipping.

After fluxing and draining, in 12.10.4.3, mount the specimen to be tested in the test equipment specimen holder.

Adjust the test equipment to immerse the specimen 0,75 mm to 1,0 mm into the solder, unless

otherwise specified. Activate the test equipment to expose the specimen to solder. After the specimen has cleared the solder bath, allow all the solder to solidify in the position in which the machine stops before removing the specimen from the holder. Care shall be taken that solder does not flow over the upper face of the specimen. This may be affected by the width of the specimen.

Dwell time at the maximum depth shall be a minimum of $(3 \pm 0,5)$ s.

Prior to examination, specimens shall have the flux removed, using a cleaning agent in accordance with 12.10.3.1.3.

12.10.4.5 Evaluation

Visual inspection shall be carried out with the unaided eye, corrected to 20/20. Lighting shall be suitable for proper inspection. To aid the evaluation of borderline cases, or when a customer and supplier agree that more critical viewing conditions are appropriate, the set-up described in 12.10.3.2.2 is recommended.

An area of 3,2 mm width from the bottom edge of each test specimen shall not be evaluated. Areas contacted by fixtures shall not be evaluated.

12.10.4.6 Report details

12.10.4.7 Plated through hole evaluation

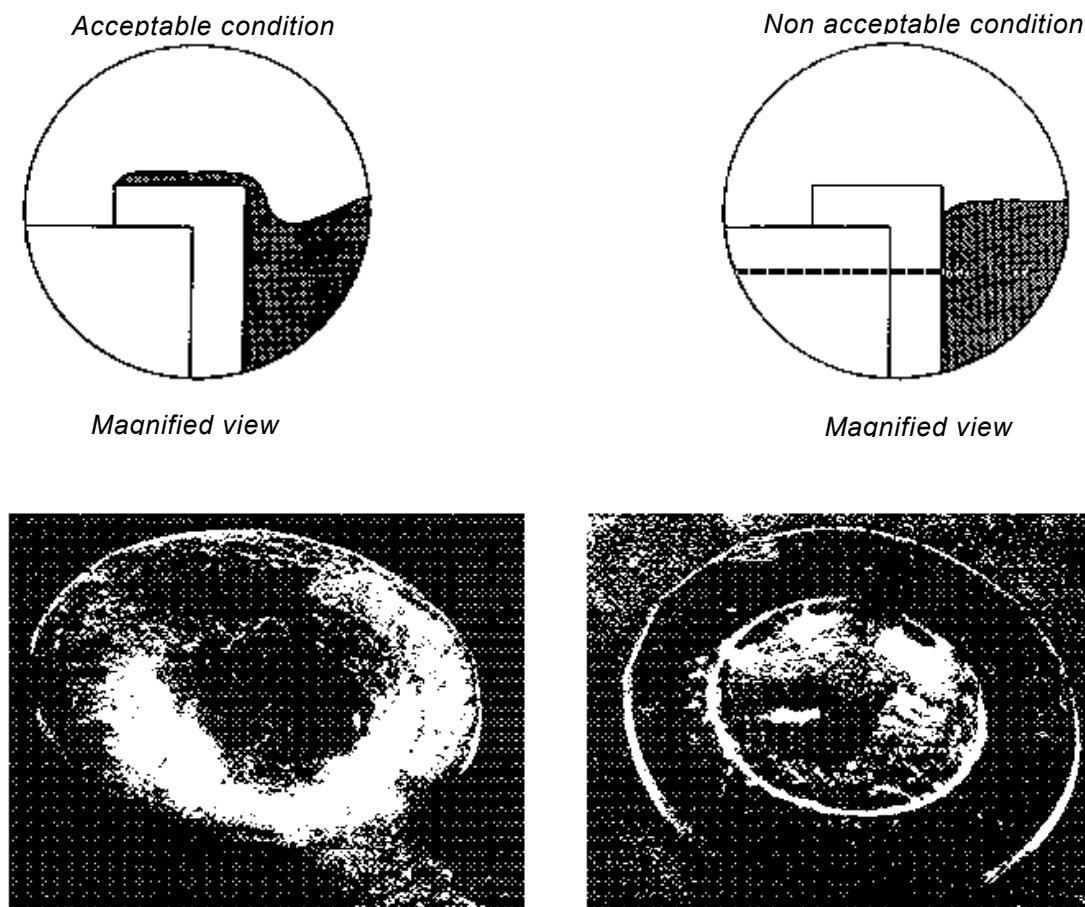
Only plated through holes that are at least 5 mm from any surface or fixturing structure supporting the specimen during the test will be evaluated. An area 3,2 mm wide from the bottom edge of each test specimen shall not be evaluated. Areas contacted by fixtures shall not be evaluated. Figure 32 illustrates effectiveness of solder wetting of plated through holes.

12.10.4.8 Report

The report shall include:

- a) the test number and revision;
- b) the date of the test;
- c) the identification and type/source of specimen(s), (board, portion of board, coupon, etc.);
- d) the number of specimens tested;
- e) the conditioning requirements used;
- f) the accept/reject criteria and results for each specimen tested;
- g) any deviation from this test method.

12.10.5 Additional information



IEC 1945/07

Figure 32 – Effectiveness of solder wetting of plated through holes

12.10.5.1 Flux considerations

12.10.5.1.1 Use of non-activated flux

Non-activated, pure rosin fluxes are specified for solderability testing for two main reasons: to provide maximum sensitivity during the test and to provide a consistent base flux for testing results. Activated rosin materials have different performance characteristics, both within a manufacturer and between manufacturers. Therefore, for testing purposes they do not provide as stable a base material as do the non-activated rosin materials.

12.10.5.1.2 Other fluxes

In certain cases, it may be necessary to use other than non-activated flux, pure rosin flux. It must be remembered that within other generic types, different formulations may give different results. Therefore, it is important that if testing is being performed at both customer and supplier locations, the same formulation shall be used by both.

12.10.5.2 Baking considerations

12.10.5.2.1 Baking/testing time delay

The time between baking and solderability testing should be kept to a minimum, in order to prevent re-absorption of water vapour into the laminate structure. The actual time is dependant

upon ambient temperature and humidity levels.

12.10.5.2.2 Prebaking

The occurrence of outgassing, which may result in blow holes, measling, blisters, or delamination, may be reduced by baking the printed boards and/or test specimens, prior to soldering and/or solderability testing, to eliminate moisture or solvents. Other factors, such as conveyor speed (for wave solder testing), solder temperature, contamination content, may also be involved in producing defects and therefore should be analysed if problems occur. Specimens should be baked in a suitable oven to remove any absorbed moisture. Time between bake and testing should not exceed 24 h. Temperature and baking time is to be determined on an individual basis.

Printed boards should be prebaked only if prebaking is normally used as a production procedure. Baking should be kept to a minimum, but at least be equal to the production procedure, to prevent excessive oxidation and intermetallic growth.

12.11 Test 3X11: Assessment of multilayer printed boards (after thermal shock) for inner layer junction failure and micro-cracking of inner layer foils

12.11.1 Object

This test method details the technique for the metallographic preparation, grinding and polishing of printed board specimens, and the method of assessment for two distinct failure modes.

a) Inner layer junction failure (junction pullaway)

Inner layer junction pullaway failure is the separation of, or failure to join (in whole, or in part), of the electroless or electrolytic plated deposits to inner layer foils or sheets, within plated-through holes of a multilayer printed board.

b) Inner layer foil cracking

Where present, cracks in inner layer foils or sheets are normally found in close proximity to the join-up region of electroless or electrolytically plated deposits within plated-through holes of a multilayer printed board.

This test method is supplementary to test method 3X09 which is the general metallographic (microsection) technique to be used in the assessment of many printed board parameters.

This test method does not seek to detail all the possible failure modes that may be identified using a thermal shock method of conditioning and subsequent microscopic examination.

12.11.2 Test specimen

The sample shall normally comprise specimens A or B in accordance with IEC 62326-4-1. A minimum of four specimens shall be taken, of which at least three shall be microsectioned vertical to the specimen surface and at least one horizontal to the specimen surface.

The sample may also comprise a suitable multilayer production board. A minimum of five holes with connections to internal layers, distributed on one or more specimens shall be examined. The number of microsections should reflect the batch size, and at least 75 % should be taken vertical to the specimen surface and 25 % horizontal to the specimen surface.

In all cases, the area of the board selected as the specimen shall be removed from the board by a technique that will not cause internal or external damage to the holes to be examined. Cut edges shall be at least 5 mm from the holes to be examined.

12.11.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- a) An air-circulating oven capable of maintaining a temperature of at least $(125 \pm 5) ^\circ\text{C}$ is required.
- b) A solder pot for thermal shock capable of maintaining a temperature of between $260 ^\circ\text{C}$ and $265 ^\circ\text{C}$ (measured at least 25 mm below the surface of the solder) shall be used. The pot dimensions shall be at least 25 mm in length, width and depth. The solder may be either 60:40 or 63:37 tin:lead.
- c) Proprietary rotary grinding and polishing equipment with a range of abrasive papers may be used.
- d) A suitable lubricant to assist in polishing may be made up as follows:
 - 65 % distilled or de-ionized water;
 - 25 % methanol;
 - 10 % glycerol;
 - a drop of detergent.
- e) Etchant for assessment of inner layer foil cracks should be made-up fresh for each assessment or daily as follows:
 - 50 ml distilled or de-ionized water;
 - 25 ml ammonia solution (specific gravity 0,88);
 - 20 ml hydrogen peroxide (% by volume).
- f) Etchant for assessment for inner layer junction failure. It should be made-up fresh for each assessment or daily as follows:
 - 65 ml distilled or de-ionized water;
 - 25 ml ammonia solution;
 - 10 mg ammonium persulphate.

NOTE This solution must be allowed to stand for a minimum of 2 h prior to use.

- g) A warm air drier for blast drying of microsections.
- h) A microscope with a magnification range of at least $\times 400$ to $\times 1\,000$.

12.11.4 Procedure

The following steps shall be taken:

12.11.4.1 Pre-conditioning

The specimen shall be stored in an air-circulating oven at a temperature of $(125 \pm 5) ^\circ\text{C}$ for 1 h, and then returned to ambient laboratory conditions for a period not exceeding 8 h.

12.11.4.2 Conditioning (thermal shock)

The sample shall be fluxed using a material as specified in 6.6.2 of IEC 60068-2-20. The sample shall be subjected to a solder float for a period of (10 ± 1) s. After cooling to room temperature, flux deposits shall be removed from the sample using a suitable solvent.

12.11.4.3 Specimen preparation

Specimens for vertical microsection shall be ground, using 180 - 320 grit abrasive paper with water as a lubricant, at right angles to the board edge until the barrels of the plated-through hole are just visible through the laminate and appear parallel. Ensure that ground edges are square to the axis of the drilled hole. Specimens for horizontal microsection shall be ground to remove any surface solder meniscus. All samples should be dried prior to encapsulation.

12.11.4.4 Specimen encapsulation

Specimens shall be mounted using a cold curing epoxide or acrylic mounting compound. Forced curing, increased pressure, or hot press mounting methods must not be used.

Vertical microsection samples should be mounted using a suitable sectioning clip to hold the sample in a vertical position, and centrally in the mould.

Horizontal microsection samples shall be positioned flat within the mould. Metal pins may be placed on either side of the specimen to limit the rate of grinding, if required.

Individual microsections should be identified in a manner that maintains full traceability to the test/inspection lot. Care should be taken to ensure that any identification is not rendered illegible by encapsulation or grinding.

The mounting compound shall be poured into the mould in a manner that ensures that all holes in the specimen are filled, and removes any entrapped air bubbles, prior to completely covering the specimen. After curing, the encapsulated specimen shall be completely removed from the mould.

12.11.4.5 Microsection grinding

Grinding of the mounted specimens shall be initially undertaken with 180 grit abrasive paper with water as a lubricant. Vertical microsections should be ground to a level just short of the centre line of the plated-through hole, ensuring that all holes are ground to the same level. Horizontal microsections should be ground to just short of the first internal layer.

The microsections shall then be lightly ground using a sequence of 320, 600, and 1 200 grit abrasive papers. If required, 2 400 abrasive paper may subsequently be used. Water shall be used as a lubricant. After each grinding stage, the specimen shall be rinsed and rotated through 90° for the next stage. Grinding at each stage should continue until all evidence of previous grinding marks are no longer visible.

12.11.4.6 Microsection polishing

The microsections shall then be polished using suitable polishing cloths and light hand pressure whilst rotating the microsection contrary to the direction of the polish wheel. Progressively finer polishing pads and compounds shall be used as follows:

≤6 µm diamond compound wheel – 30 s

≤3 µm diamond compound wheel – 20 s

≤1 µm diamond compound wheel – 10 s.

A suitable lubricant for polishing is described in 12.11.3.c) of this test method.

After each polishing stage, the microsection shall be washed using a suitable proprietary detergent ensuring that all traces of diamond paste and/or lubricant are removed. After drying, the microsection shall be given a final wash in alcohol.

12.11.4.7 Examination for quality of preparation

After polishing, but prior to etching, the microsection shall be examined at ×400 magnification. The surface of the microsection shall be free from any surface scratches, stains or other blemishes that may obscure features of the specimens.

If necessary, the microsection may be reground from the 1 200 or 2 400 grit stage and subsequently repolished as described in 12.11.4.5 of this test method.

NOTE Examination of the internal junctions prior to etching the specimen may reveal evidence of poor join-up of the deposited metals to the internal layers and other defects which may be masked by subsequent etching.

12.11.4.8 Etching the microsection

After initial examination, the microsection shall be etched to reveal any separation or failure to join of the deposited metals to internal layers or foil cracking of internal layers. Two separate etchings are required.

NOTE Both specified etches are for plated copper deposits and copper inner layer foils. Both etchants have limited shelf life and should be freshly made. Specimens containing plated deposits and internal layer foil materials other than copper may require different etchants. Etchants should only be applied to freshly polished, cleaned and degreased microsections.

12.11.4.8.1 Internal layer foil cracks

The etchant which is described in 12.11.3 point d) of this test method shall be used. A few drops of the etchant shall be applied to a slurry of Gamma or Alpha alumina deposited on a rotary polishing cloth. Using light hand pressure, the microsection shall then be rotated contrary to the direction of the rotation of the cloth for between 10 s and 15 s. The microsections shall then be washed with water to remove any residual etchant, rinsed with alcohol, and dried.

Alternatively, the etchant may be applied to the microsection using a cotton bud impregnated with the alumina/etchant slurry.

It is recommended to begin with a light etch, which can be sequentially repeated to increase the depth of the etch. Overetching should be avoided but may be redressed by repolishing the specimen.

12.11.4.8.2 Inner layer junction failure

The etchant which is described in 12.11.3 of this test method shall be used. After polishing and washing, the microsection shall be immersed in the etchant for approximately 1 s (dependant upon the activity of the etchant), and subsequently washed in water, rinsed in alcohol and dried.

Alternatively, the etchant may be applied to the microsection using a cotton bud impregnated with an alumina/etchant slurry.

It is recommended to begin with a light etch which can be sequentially repeated to increase the depth of the etch. Overetching shall be avoided but may be redressed by repolishing the specimen.

12.11.4.9 Examination

The etched microsection should be examined at a selection of magnifications between $\times 400$ and $\times 1\,000$.

12.11.5 Report

The report shall include (for each sample):

- a) the test method and revision;
- b) the date of the test;
- c) the identification and description of the samples;
- d) the number of specimens tested;
- e) the type of failure if failures are found;
- f) given the visual nature of the examination, photomicrographs shall be included in the report;

- g) any deviation from this test method;
- h) the name of the person conducting the test.

12.11.6 Additional information

The level of experience/competence of the operator is critical to the process of microsection preparation and examination. Operators should be familiar with general microsectioning techniques before attempting to use this method.

Grinding pads should be renewed regularly to maintain an even and efficient level of grinding. Other laboratory consumables should be replenished at regular intervals.

Colour photomicrographs should be taken.

12.12 Test 3X12: Adhesion of organic surface finish to printed board (tape method)

12.12.1 Object

This test method details the procedure for assessing the adhesion of an organic surface finish material to the underlying materials of a printed board.

Materials which may be assessed using this technique include permanent polymeric solder resist, conductive inks, and legend inks.

Underlying printed board materials to be tested in combination with the material being assessed include the base material (substrate) and any inorganic surface finish.

12.12.2 Test specimen

The chequer board test pattern which is detailed as specimen G in IEC 62326-4-1 shall be used. The entire specimen G shall be coated with the organic surface finish to be tested. Figure 33 shows the test specimen.

The underlying materials shall be those to which the material under test is to be applied. These combinations may have to be agreed between the printed board supplier and customer.

The minimum number of assessments shall be five.

12.12.3 Test apparatus and materials

The following test apparatus and materials shall be used:

- Non-transferable transparent pressure sensitive tape, in accordance with IEC 60454-1 shall be used. The tape shall be a minimum of 13 mm wide. A suitable tape is designated IEC 60454-3-1-5/F-PVCP/90x. A fresh piece of tape must be used for each test.

12.12.4 Procedure

The following steps shall be taken:

- a) The specimen shall be cleaned with a suitable organic solvent and allowed to dry.
- b) A 50 mm length of tape shall be taken. The adhesive side of the tape shall be applied to the sample under test in a manner which excludes air bubbles. This may be done by finger pressure, by the use of a hand roller, or by an eraser. The tape shall fully cover the test specimen.

- c) After a minimum interval of 10 s, the tape shall be removed by a snap pull applied approximately perpendicular to the surface of the test specimen.
- d) The tape may then be transferred to a clean white sheet of paper or card. This enables evaluation of the degree of material transfer and facilitates record-keeping. Where a white or light-coloured notation ink is under test, a suitably contrasting coloured paper or card should be used.

12.12.5 Report

The report shall include:

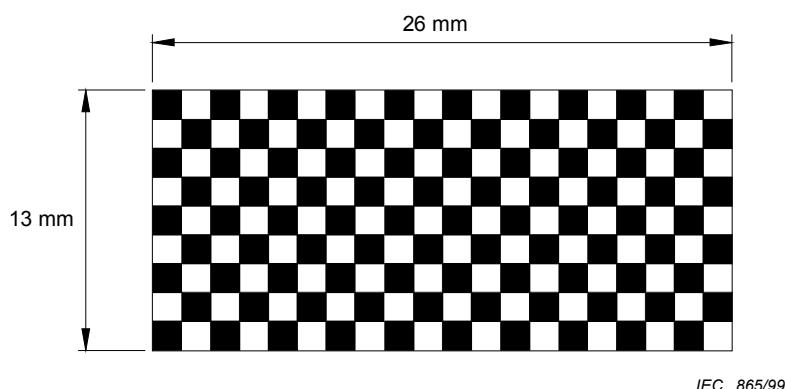
- a) the test method and revision;
- b) the date of the test;
- c) the identification and description of the specimens;
- d) the number of specimens tested;
- e) the underlying surface finish(es);
- f) a description of the degree of material transferred to the tape. Alternatively, the tape itself may be used for reporting where transferred to paper or card, as described in 12.12.4;
- g) a description of the condition of the underlying printed board materials;
- h) the cleaning solvent;
- i) any deviation from this test method;
- j) the name of the person conducting the test.

12.12.6 Additional information

Some standards specify adhesion of printed board materials using the cross-cutting technique. The method described herein should be considered as the preferred technique for printed board organic surface coatings.

The outcome of this test is an attribute. Uncertainty of measurement is not a consideration.

Results are only valid for the combinations of materials and/or finishes actually tested.



NOTE The white squares indicate the presence of the underlying surface finish and/or metallization. The black squares indicate the absence of the underlying surface finish and/or metallization. The material under test is applied to the entire specimen.

Figure 33 – Test specimen

Annex A (informative)

Worked examples

The following are worked examples of the uncertainty estimate for various printed board related tests:

Example 1:

Method: Repeated measurement of a 200 Newton (nominal mass) deadweight with a tensometer. (Comparable with QTH pull-out test)

Raw data:	198,50	198,50	198,75	198,75	198,75
	198,00	198,50	198,50	198,50	198,75
	198,20	198,50	198,50	198,50	198,75
	198,20	198,50	198,50	198,50	198,50
	198,00	198,50	198,75	198,50	198,75

Table "t" value:	t
Sample size (n):	25
Graduation of dial:	1,0 N
Mean:	198,50 N
Standard deviation (σ_{n-1}):	0,215 N

$$U_r = \pm t(\sigma_{n-1}) / \sqrt{n} = \pm 0,08858 \text{ N}$$

$$U_s = \pm 1 \% \text{ (as shown on calibration certificate)} \\ = \pm 1,985 \text{ N}$$

$$U_i = \pm 20 \% \text{ of graduation} = \pm 0,2 \text{ N}$$

$$U_t = \pm \sqrt{U_s^2 + U_r^2} + U_i = \pm 2,187 \text{ N} = \pm 1,1 \%$$

Thus, uncertainty of measurement is estimated as $\pm 1,1 \%$

Example 2:

Method: Repeated measurement of the level of ionic contamination in a test solution, as represented by the introduction of a metered quantity of the calibration solution. Measured in a commercial ionic contamination test instrument.

Raw data:	1,00	1,03	1,11	1,03	1,05
	1,10	0,92	1,06	0,91	0,92
	(measurement unit $\mu\text{g}/\text{cm EQ. NaCl}$)				

Table "t" value:	t
Sample size (n):	10
Mean:	1,013 μg
Standard deviation (σ_{n-1}):	0,074 μg
Graduation of dial:	not applicable

$$U_r = \pm t(\sigma_{n-1}) / \sqrt{n} = 0,053 \text{ mg}/\text{cm}^2$$

$$U_s = \pm 5 \% \text{ (calibration of Gilman pipette/test solution)} = \pm 0,05 \text{ mg}$$

$$U_t = \pm \sqrt{(U_s^2 + U_r^2)} \pm 0,0729 \text{ mg} = \pm 7,29 \%$$

Thus, uncertainty of measurement is estimated at $\pm 7,29 \%$.

Example 3:

Method: Repeated measurement of a surface insulation resistance (SIR) test specimen (nominal value 100 M Ω) using a megohmmeter.

Raw data:	68,00	102,50	105,00	108,75
	81,25	102,50	107,50	113,75
	90,00	102,50	107,50	

Table "t" value: t

Sample size (n): 11

Mean: 99,02 M Ω

Standard deviation (σ_{n-1}): 13,73 M Ω

Graduation of dial: 1,25 M Ω

$$U_r = \pm t (\sigma_{n-1}) / \sqrt{n} = 9,2223 \text{ M}\Omega$$

$$U_s = \pm 5 \% \text{ (as shown on calibration certificate)} \\ \pm 0,5 \% \text{ of } 99,02 = \pm 0,4951 \text{ M}\Omega$$

$$U_i = \pm 20 \% \text{ of graduation (1,25 M}\Omega) = \pm 0,25 \text{ M}\Omega$$

$$U_t = \pm \sqrt{(U_s^2 + U_r^2)} + U_i = \pm 9,456 \text{ M}\Omega = \pm 9,5 \%$$

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